

Policy Spaces as Organizational Schemes: A Semantic Theory of Electoral Competition

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September 7, 2026

Abstract

The policy space in which voters compare alternatives depends on how they understand political life. Campaign communication can change the distinctions they draw and the meanings they attach to them, changing the terms of electoral choice. This manuscript develops a semantic model of electoral competition in which a lattice of understandings records what voters can access and distinguish, together with the meanings attached to those representations. Reception and updating allow the same speech to affirm an existing understanding, introduce valued content, or trigger removal of conflicting prior content. Electoral responses combine recognition of an appeal with valued cognitive change.

With fixed reception parameters and acquisition depending only on epistemic stances, repeated exposure to a fixed message and source eventually stabilizes or enters a two-cycle, even with active conflict triggers. Internally consistent messages stabilize within two encounters under maximal acquisition. The analysis also identifies conditions for reducing the feasible message domain while preserving electoral incentives and, under source neutrality, majority reachability between retained alternatives. The content needed to represent competition depends on what voters value, what they admit, and what can trigger conflict with their priors.

Strategic results identify conditions under which candidates can combine appeals to audiences that interpret the same speech differently. A worked election shows how observing an opposing audience's objection can strengthen supporters' responses and change the equilibrium speech. Another has pure equilibria with active conflict and no common single-peaked policy order.

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1 Introduction.

A spatial benchmark for electoral competition places voters and candidates in a common policy space, assigns each platform a publicly known location, and ranks platforms by their proximity to voters' ideal points. Downs's (1957, pp. 115–116) spatial construction assumes an agreed left–right ordering and single-peaked voter preferences. The benchmark connects electoral choice to the geometry of a space whose interpretation is already specified. Downs also examines uncertainty, ideological shortcuts, and strategic ambiguity: parties can appeal to different constituencies whose members weight policies differently or interpret the same policy in different ways (Downs, 1957, pp. 98–100, 135–136).

This interpretive variation is the starting point for the present account of mass politics. Affective polarization between American party identifiers shows a long-run rise through 2016 (Iyengar et al., 2019). Lakoff (1996) proposes an account of political disagreement organized around contrasting models of family morality. Entman (1993, pp. 52–53) distinguishes the interpretations promoted by a text from those a receiver adopts. A campaign appeal therefore raises separate questions about the meanings it offers and the meanings different voters take from it. These phenomena raise a question about how a platform acquires its meaning for a voter. In the benchmark, assigning the platform a common location settles its interpretation before voter preferences are applied. Here, the encounter through which a voter receives and interprets political content is itself an object of analysis.

The organization of electoral choice was central to Stokes's (1963) critique of spatial models. He questioned four assumptions: a single dimension, a stable evaluative structure, ordered policy alternatives, and a common frame of reference among political actors. His distinction between position and valence issues separates disagreement over policies from competition over parties' association with outcomes voters broadly value (Stokes, 1963, pp. 372–374). He also emphasized that voters and party leaders may perceive political alternatives differently, and proposed treating the relevant cognitive features as variables (Stokes, 1963, pp. 374–376).

Subsequent behavioral work has identified specific problem points. Converse (1964) reported weaker associations among issue positions in an electorate sample than among congressional candidates, instability in repeated responses to several policy questions, and greater stability among respondents with particular interest in the issue. His account of issue publics directs attention to variation in what citizens engage with and organize politically. Zaller (1992, chap. 3) develops a model linking political information acquisition to the formation of opinion statements. Bartels (2002) finds partisan differences in retrospective economic perceptions. Taber and Lodge (2006) find selective argument evaluation and search in two student studies of affirmative action and gun control. These findings motivate distinguishing the reception of arguments from changes in commitment to a position.

These lines of inquiry motivate two questions for a semantic account of electoral competition. Which political distinctions and associations do voters bring to an election, and how does communication change them? The framework below makes representational differences explicit and specifies how an encounter can supply distinctions, admit meanings, or remove conflicting prior content. Its empirical application requires measuring those changes separately from reported attitudes and candidate evaluations. Section 3.2 develops this distinction for source-sensitive reception.

Formal electoral theory has made both communication and adaptation objects of analysis. Hinich and Munger (1994) model ideological communication through symbols and analogies alongside political commitment. Kollman, Miller and Page (1992) study spatial competition with parties that search adaptively using limited polling information. Their later analysis connects the distribution of voter preferences to the difficulty of that search (Kollman, Miller and Page, 1998). Bendor et al. (2011)

develop models of party competition, turnout, and voter choice using adaptation governed by aspiration levels.

The contribution here centers on an explicit state for each voter’s understanding: the representations they can access and distinguish, together with the meanings attached to those representations. Reception maps describe how a speech reaches a voter who uses different categories; the update specifies how the encounter changes distinctions and content. This makes it possible to follow the same broadcast through different voters’ understandings before comparing their electoral responses. The analysis then identifies conditions under which those responses admit a common policy order or permit a smaller set of feasible messages to represent the electoral game.

Consider three voters in the same election, all with opinions about immigration. The first favors enforcement, regards paths to citizenship as unworkable, and sees immigration’s economic impact as harmful. The second regards enforcement as effective but cruel, and paths to citizenship as viable and desirable. The third attaches evaluative meanings to enforcement and citizenship—enforcement feels cruel, and a welcoming posture feels right—without attaching factual meanings to either or any meaning to economic impact. All three can distinguish these topics; their differences here concern what the topics mean to them.

These differences raise concrete questions for a campaign. What does each voter recognize in a factual argument, an evaluative appeal, or a speech combining both? When does an appeal affirm an existing understanding, introduce valued content, or put prior commitments at risk? An answer requires specifying what voters already understand, what they value, and how they receive and respond to the available speeches. Those responses determine their rankings.

A common policy axis is then a possible pattern in the rankings: an ordering of the speeches along which each voter’s response is single-peaked. The worked immigration election in Example 5.25 follows the three voters through a specified five-speech menu and shows how this pattern can fail. Among three of its speeches, each voter strictly ranks a different one last, so every possible axis creates a valley in someone’s response. The same election has pure equilibria with active conflict triggers. It thus supplies both an obstruction to a common single-peaked order and an example of strategic stability.

The framework models voters and candidates as holding *understandings*: pairs combining an epistemic stance (which political representations an agent can access and discriminate) with a semantics (what those representations mean to them). Understandings are ordered by a complete lattice whose meet and join operations formalize the notions of semantic common ground and pooled representational capacity. A *semantic update operator* governs how understandings change when agents encounter each other’s framings. Credibility filters determine which incoming meanings are admitted on the forward branch; a conflict trigger instead selects removal of incompatible prior meanings. These operations specify changes in semantic content, while their connection to reported attitudes requires an explicit measurement model. Candidates choose from a feasible menu of messages represented in this lattice; the resulting voter rankings determine whether that menu admits a common policy order.

The electoral response distinguishes recognition of a message from the valued cognitive change it produces. A familiar appeal can be affirming without changing the voter’s state; another appeal can introduce a valued distinction or meaning. The evaluative anchor is a reduced-form representation of what the voter can recognize as valuable, while the best available appeal depends on the current understanding, the source, and the feasible messages. This link from cognition to evaluation is an explicit modeling assumption.

Repeated communication can stabilize or sustain a periodic response. For a fixed source and message, every orbit eventually reaches a fixed understanding or alternates between two understandings, according to whether absorption regenerates enough conflict to activate the trigger

(Theorem 5.4). Internally consistent messages stabilize within two encounters under maximal acquisition, even with active triggers. This supplies a sufficient condition for acyclic chains of public reactions to settle and permits an observed objection to add to a supporter’s appreciation of the original appeal.

A common policy order must organize the resulting voter rankings. The conditional median result (Proposition 4.15) assumes strict single-peakedness of those actual rankings, feasible peaks, source neutrality, and a specified treatment of voting ties. The coalition mountain condition is equivalent to the existence of a strict Condorcet winner; it does not by itself recover a voter-level axis. Likewise, maximality of a winning broadcast requires the direct-broadcast ranking restriction stated in Theorem 4.14. The semantic framework makes the conditions visible without deriving them from lattice structure alone.

A second question concerns the conditions under which a smaller set of messages can represent electoral competition. Two ingredients matter: the content needed to preserve voter responses and the availability of messages expressing that content. In the direct-broadcast benchmark, taking a broadcast’s meet with the join of the voter anchors preserves every score (Proposition 5.13). For the full encounter response at a common stance, a sufficient reduction retains both credibly valued meanings and meanings capable of triggering conflict with a voter’s prior (Proposition 5.14). When the feasible menu is closed under the relevant reduction, replacing platforms by their reduced representatives preserves expected vote shares. Every full-game pure equilibrium projects to a reduced-game equilibrium, and every reduced-game equilibrium is also one of the full game (Corollary 5.15). Under source neutrality, strict majority-improvement paths can likewise be projected into the reduced menu. The reductions identify conditions under which the strategic analysis can be conducted on a smaller message domain.

The scope of the framework is correspondingly broad. Arbitrary finite weak-ranking profiles can be represented even with fixed disjoint anchors, empty priors, full credibility, and no conflict (Proposition 5.11). An explanation of observed political order therefore requires restrictions on the messages available, how voters receive them, or what they value. The strategic analysis connects these restrictions to actual vote shares. It gives sufficient conditions for independently choosing audience-specific components with active triggers, and a worked election in which an observed objection makes a provocative combined speech the equilibrium choice. When a strict majority has the same maximizing set, symmetric equilibrium must lie within it; minority voters can select among alternatives that leave that majority indifferent. The running immigration election has pure equilibria with active triggers despite a three-speech obstruction to any common single-peaked axis. Field comparisons distinguish a common restriction, which preserves comparability, from different restrictions that can make initially comparable understandings incomparable. The sufficient conditions identify either resources retained exclusively by each voter, or a loss of access by the richer voter combined with an advantage surviving on the retained field.

Section 2 defines the representational framework, with immigration as a running example. Section 3 specifies the update operator. Section 4 connects encounters to voting and states the conditional policy-ordering result. Section 5 examines communication, electoral reduction, and candidate incentives. Sections 6 and 7 discuss empirical interpretation, legislative bargaining, and variation in the representational field.

2 The representational field.

The framework describes how voters can use shared representations while differing in what they access, distinguish, and mean by them. It has three interlocking layers—an ontic layer of representations,

an epistemic layer of agents’ stances toward those representations, and a semantic layer of meanings agents attach to them. Together these generate a complete lattice of *understandings* whose meet and join operations formalize semantic common ground and pooled representational capacity.

2.1 What representations are—and aren’t.

One way to model political knowledge begins with a set Ω of mutually exclusive and exhaustive states of the world, exactly one of which is true, and represents agents’ epistemic states as probability distributions or partitions over Ω . Here the primitive objects are the representational resources agents use to conceive of political life. A voter might model Congress as a hierarchy in which a decisive leader controls legislative outcomes, as a network of committees and agenda-setting procedures, or as a narrative of heroes and villains contesting legitimacy. Each schema can organize the voter’s understanding, including when it is simplified or mistaken. The questions concern which schemas an agent can access, which distinctions they draw, and which meanings they attach.

We collect these resources in a shared *representational field*. This common formal universe lets us compare different ways of understanding the same domain. An ordering of electoral alternatives, when one exists, will be a property of the responses generated from those understandings and the available messages.

Definition 2.1. *The representational field $\mathcal{R}$ is a nonempty set whose elements are representational resources: models, schemas, symbols, narratives, formal structures, and other devices through which domains of political life can be conceived, described, or contested.*¹

Elements of $\mathcal{R}$ are not claims about how things actually are. They are the *tools available for making such claims*—or for framing, contesting, or refusing them. A Petri net model of congressional procedure, a conspiracy theory about legislative capture, and a spatial model of ideological competition are all elements of $\mathcal{R}$, regardless of their epistemic status. Nothing in the field is privileged as the correct or canonical representation.

Throughout, $\mathcal{R}$ and the meaning space M (definition 2.5) are assumed to be finite. This ensures that the understanding lattice $\mathcal{U}_{\mathcal{R}}$ is finite—a standing assumption used in the convergence results of section 5 and the extensions of section 7.

The field $\mathcal{R}$ is *shared* in a specific and limited sense: it is the common ambient space within which all agents operate. Shared means neither uniformly used nor uniformly interpreted. Different agents access different portions of $\mathcal{R}$, and even where two agents access the same element, they may assign it entirely different meanings. For example, the word “immigration” can be represented as a shared token whose policy implications, evaluative associations, and social identifications are supplied by each agent’s semantic mapping. What the field provides is a common formal reference space: a single universe of representational possibility within which differences in access *and* in interpretation can be located and compared.²

To fix ideas, consider immigration as a representational domain. The field $\mathcal{R}$ includes, among many other elements, three representations that will recur as a running example: r_1 (border enforcement mechanisms), r_2 (paths to citizenship), and r_3 (the economic impact of immigration). These are not policy positions; they are representational resources available for making, contesting,

¹The field is fixed within a given model. Field variation can be modeled by comparing different finite sets of resources; Section 7.2 supplies restriction and embedding maps for those comparisons.

²Sniderman and Bullock (2004, pp. 337–339) hypothesize *menu dependence*: consistency of political choice depends jointly on citizens’ orientations and the organization of the alternatives they face. Here the field specifies representational resources, while the feasible message menu specifies the choices available to candidates. Their distinction between the properties of choosers and of choices motivates attending to both voter understandings and the electoral menu.

or refusing claims about immigration. A voter may engage with all three, with a subset, or—as the epistemic layer will formalize—may be unable to distinguish some of them from one another. Different voters may assign categorically different meanings to the same representation: r_1 can carry the meaning “enforcement works” or the meaning “enforcement is cruel,” and a single voter can hold both meanings simultaneously. The next three subsections build the apparatus to make these differences precise.

2.2 Epistemic stances.

The representational field is shared, but agents do not engage with it uniformly. One voter might distinguish several legal statuses while another distinguishes forms of economic contribution, with neither drawing all the distinctions available to the other. The *epistemic stance* formalizes this variation: it captures which parts of $\mathcal{R}$ are available to an agent and how finely they can discriminate among those parts.

Definition 2.2. Fix a marker $\star \notin \mathcal{R}$ and write $\mathcal{R}_\star = \mathcal{R} \cup \{\star\}$. An epistemic stance is a pair $e = (\mathcal{R}_e, \sim_e)$ where $\sim_e$ is an equivalence relation on $\mathcal{R}_\star$, its invisible class is $I_e = [\star]_e$, and the agent’s visible scope is $\mathcal{R}_e = \mathcal{R} \setminus I_e$. Thus visibility is determined by the marked partition: the invisible resources are $I_e \setminus \{\star\}$. The perceived representations $\mathcal{R}_e / \sim_e$ are the classes other than I_e in $\mathcal{R}_\star / \sim_e$ —the finest distinctions the agent can draw within their visible scope. The set of all epistemic stances is denoted $E_{\mathcal{R}}$.

The marker is a bookkeeping device, not a representational resource or a primitive meaning. It identifies the invisible class even when every resource is visible, in which case $I_e = \{\star\}$. Without the marker, two identical partitions could have different classes designated invisible; partition refinement would then fail to distinguish the stances.

To illustrate variation in epistemic stance, consider a variant of the immigration example on the field $\{r_1, r_2, r_3\}$. The enforcement voter and the integration voter both access all three representations and draw fine distinctions among them: their visible scopes are $\mathcal{R}_{e_A} = \mathcal{R}_{e_B} = \{r_1, r_2, r_3\}$, and their equivalence relations place each r_i in its own class. Suppose voter C does not distinguish enforcement mechanisms from economic impact—both register as “the harsh side of the debate”—while distinguishing the path-to-citizenship representation from both. Her epistemic stance collapses r_1 and r_3 into a single equivalence class while keeping r_2 distinct: $\mathcal{R}_{e_C} = \{r_1, r_2, r_3\}$ with $r_1 \sim_{e_C} r_3$. She can access every representation the other voters can, but she draws fewer distinctions—her stance is representationally poorer in the sense formalized next. This coarsened stance is a separate assumption; C’s evaluative semantics in the introduction and the worked election are compatible with distinguishing all three resources.

Representational richness. It is natural to compare epistemic stances by the representational capacity they confer. Say that e_2 is *representationally richer* than e_1 , written $e_1 \preceq_E e_2$, when $\sim_{e_2}$ refines $\sim_{e_1}$ as partitions of $\mathcal{R}_\star$:

$$e_1 \preceq_E e_2 \iff \sim_{e_2} \subseteq \sim_{e_1} \text{ as binary relations on } \mathcal{R}_\star,$$

i.e., every $\sim_{e_2}$ -class is contained in a $\sim_{e_1}$ -class. The marker makes access and discrimination part of the same comparison. Since the fine class containing $\star$ must lie inside the coarse class containing $\star$, $I_{e_2} \subseteq I_{e_1}$ and hence $\mathcal{R}_{e_1} \subseteq \mathcal{R}_{e_2}$. A richer stance preserves existing access and distinctions; it may split an invisible block to make resources accessible or split visible blocks to distinguish them. Gaining access while losing an existing distinction can instead produce an incomparable stance.

A terminological remark is necessary. The relation $\preceq_E$ is *not* an order of epistemic accuracy, calibration, or rationality. Being higher in $\preceq_E$ means having more representational resources and finer discriminations—nothing more. A more elaborated representational toolkit may be epistemically harmful (finer but misleading distinctions), epistemically neutral (finer distinctions in domains irrelevant to the decision at hand), or both. For this reason we use the phrase *representationally richer* rather than “more informative,” which carries connotations of Blackwell dominance and truth-localization that are inappropriate here.

Lemma 2.3. $(E_{\mathcal{R}}, \preceq_E)$ is a partially ordered set.

Incomparability. Two epistemic stances e_1 and e_2 are *incomparable* in $\preceq_E$ when neither $e_1 \preceq_E e_2$ nor $e_2 \preceq_E e_1$: each agent can represent something the other cannot.

As an illustration, consider two agents reasoning about the legislative process through disjoint, nonempty sets of resources. Agent i accesses precisely a set $\mathcal{R}_P \subset \mathcal{R}$ of Petri net representations, modeling procedure as a directed flow of tokens through transitions. Agent j accesses precisely a disjoint set $\mathcal{R}_N \subset \mathcal{R}$ of narrative representations, understanding the same process through accounts of protagonists, obstacles, and institutional drama. Since $\mathcal{R}_P \cap \mathcal{R}_N = \emptyset$, the stances $e_i = (\mathcal{R}_P, \sim_P)$ and $e_j = (\mathcal{R}_N, \sim_N)$ are incomparable: each can represent things the other cannot, and neither is richer than the other in the sense of $\preceq_E$.

Applying this comparison requires specifying what each agent accesses and distinguishes. Lakoff (1996), for example, proposes contrasting Strict Father and Nurturant Parent models of political morality. Such a contrast can be represented through different meanings attached to common resources, through different epistemic stances, or through both. At a common stance, agents may attach discipline and care associations differently while distinguishing exactly the same political categories. Epistemic incomparability follows only when their specified partitions fail to refine one another.

Incomparable stances can also have overlapping visible scopes. A shared resource can carry communication between agents whose surrounding schemas differ. If a recipient groups together representations that a sender distinguishes, pooling their meanings can preserve the associations while losing which source category carried each one. When the recipient preserves the relevant distinctions, this particular loss need not occur. The reception rule below specifies how such overlap supports communication.

The space of epistemic stances also admits operations for extracting shared categorical ground (the meet) and for combining access and distinctions (the join). These operations apply whether or not the stances are comparable. Their semantic counterparts will define common ground between understandings, while the epistemic join bounds the acquisition available in an encounter.

The epistemic lattice. Formally, the set $E_{\mathcal{R}}$ admits the following meet and join operations. For a nonempty family $\{e_j\}_{j \in J}$ with $e_j = (\mathcal{R}_j, \sim_j)$:

1. *Meet* $\bigwedge_{j \in J} e_j$: the most representationally rich stance lying below every e_j . Its partition of $\mathcal{R}_*$ is $\sim^\wedge = \text{EqCl}(\bigcup_j \sim_j)$ —merge any elements that some e_j identifies, then close transitively. Its invisible class is $[\star]_{\sim^\wedge}$ and its visible scope is $\mathcal{R} \setminus [\star]_{\sim^\wedge}$, which is contained in, but need not equal, $\bigcap_j \mathcal{R}_{e_j}$. The meet is the representational *common ground*.
2. *Join* $\bigvee_{j \in J} e_j$: the least representationally rich stance lying above every e_j . Its partition of $\mathcal{R}_*$ is $\sim^\vee = \bigcap_j \sim_j$ —two elements are identified only when every stance already treats them as indistinguishable. The visible scope is $\bigcup_j \mathcal{R}_{e_j}$. The join is the representational *horizon*: what an agent fluent in all the schemas could access.

In the legislative example, the meet of e_i and e_j is $e_\perp$ —they share no representational ground at all. Their join is $(\mathcal{R}_P \cup \mathcal{R}_N, \sim_{P \cup N})$ —the stance of an agent at home in both Petri nets and narrative. In this disjoint-scope example, reception at the initial stances has no overlap through which to pass meanings. Acquiring access to the other’s resources can create such overlap using elements already in $\mathcal{R}$; the acquisition rule in Section 3 specifies how far an encounter may extend the receiver’s stance. In the immigration case, the meet of the enforcement voter’s stance and the coarsened stance e_C is e_C itself: it preserves only distinctions both agents make and is strictly below the enforcement voter’s stance.

Overlapping access alone need not supply a shared category. On $\mathcal{R} = \{a, b\}$, compare the partitions $\{a\} \mid \{b, \star\}$ and $\{a, b\} \mid \{\star\}$. The first agent sees only a ; the second sees a and b as one inseparable category. Their meet merges a with b and b with $\star$, leaving nothing visible, although both initially access a . Communication through overlapping resources can nevertheless occur under the reception rule below; shared categorical ground and the possibility of partial uptake are different requirements.

Lemma 2.4. *$(E_{\mathcal{R}}, \preceq_E)$ is a complete lattice with bottom element $e_\perp$ (the single class $\mathcal{R}_\star$, with empty visible scope) and top element $e_\top = (\mathcal{R}, \text{id}_{\mathcal{R}_\star})$ (full visible scope, each resource its own class). The meet of the empty family is $e_\top$; its join is $e_\perp$.*

The bottom $e_\perp$ represents total representational blindness relative to the chosen field; the top $e_\top$ represents full access with maximal discrimination. For a small field such as the three-resource immigration example, full access and discrimination can be an appropriate assumption about a voter.

2.3 Semantics.

The epistemic layer specifies which representations an agent can access and distinguish. It says nothing about what those representations *mean*. Two agents with identical epistemic stances may attach entirely different meanings to the same representations: one reads congressional procedure as a mechanism for collective deliberation; another reads it as a site of elite capture and manufactured consent. This variation—*independent of representational scope and discrimination*—is the province of the semantic layer.

At a fixed stance, meanings can vary independently of access and discrimination. We therefore assign a separate semantic mapping to each stance and later order the resulting understandings by both their epistemic distinctions and their semantic content. Reception, updating, and electoral evaluation can then depend on either layer.

The meaning space.

Definition 2.5. *The meaning space M is a nonempty set of primitive meanings.*

Like $\mathcal{R}$, the space M is left unstructured for generality. In applications, its elements might include:

- *Policy implications:* “entails higher taxes,” “requires redistribution,” “commits troops”
- *Evaluative content:* “just,” “threatening,” “wasteful,” “legitimate”
- *Social associations:* “for people like me,” “an elite agenda,” “a working-class cause”
- *Factual associations:* “correlated with crime,” “associated with economic growth”
- *Affective content:* dread, resentment, solidarity, hope

The diversity of this list is intentional. A representation of the Affordable Care Act might carry associations concerning health-care access, government intervention, personal responsibility, and partisan identity. The set-valued range in the definition below allows several associations to be attached at once. Their contribution to electoral response is specified separately in Section 4.

To see the semantic layer in isolation, suppose all three immigration voters share the top epistemic stance $e_\top$ —full scope, each representation in its own class—so that the action is purely in what meanings they attach. Let $M = \{T, F, G, B\}$, where T and F are factual primitives (“the policy works,” “the policy fails”) and G and B are evaluative primitives (“desirable,” “undesirable”). A representation can carry any subset of M : assigning $\{T, G\}$ to r_1 says “enforcement works and is good”; assigning $\{T, B\}$ says “enforcement works but is bad.” The enforcement voter assigns $\mu_A(r_1) = \{T, G\}$, $\mu_A(r_2) = \{F, B\}$, $\mu_A(r_3) = \{T, B\}$: enforcement is effective and good, paths to citizenship are based on false premises and bad, and immigration’s economic impact is real but harmful. The integration voter assigns $\mu_B(r_1) = \{T, B\}$, $\mu_B(r_2) = \{T, G\}$, $\mu_B(r_3) = \{T, G\}$: enforcement is real but cruel, paths to citizenship are viable and good, and immigration is economically beneficial. The affect voter assigns $\mu_C(r_1) = \{B\}$, $\mu_C(r_2) = \{G\}$, $\mu_C(r_3) = \emptyset$: enforcement is bad and citizenship is good, with no factual meanings and no meanings attached to economic impact.

These three semantics illustrate pure semantic disagreement. Voters A and B agree on the factual status of enforcement ($T \in \mu_A(r_1) \cap \mu_B(r_1)$) but disagree evaluatively (G versus B). Voter C’s displayed semantics attach only evaluative meanings. Her stance nevertheless distinguishes enforcement, citizenship, and economic impact, just as A’s and B’s do. When used as an evaluative anchor, this content gives a factual-only receipt zero semantic recognition value. Her response to a campaign also depends on her prior, credibility, and updating; Example 5.25 specifies these inputs and computes the resulting rankings.

The semantic layer captures this variation independently of the epistemic layer—all three voters see the same representations and draw the same distinctions.

Semantic mappings. We now link epistemic stances to meanings via semantics.

Definition 2.6. For an epistemic stance $e = (\mathcal{R}_e, \sim_e)$, a semantics is a function

$$\mu_e : \mathcal{R}_\star / \sim_e \longrightarrow 2^M,$$

assigning to each class of the marked partition a set of meanings, subject to $\mu_e(I_e) = \emptyset$. The perceived representations $\mathcal{R}_e / \sim_e$ of Definition 2.2 are the visible classes of this partition. Even at full scope the marker’s singleton class receives empty semantics. A semantics μ_e is e -consistent when it satisfies the condition $\mu_e(I_e) = \emptyset$. Let M_e denote the set of all e -consistent semantics. Write O_e for the semantics assigning $\emptyset$ to every class.

Invisible resources carry no meaning not because they are intrinsically meaningless, but because the agent cannot access them. If an agent later encounters such a resource—through a politician’s platform or a conversation—it may enter scope and acquire meaning; but until then $\mu_e(I_e) = \emptyset$ by definition.

Two agents sharing the same epistemic stance e but holding different semantics $\mu_e \neq \mu'_e$ exhibit *pure semantic disagreement*: they access the same representations and draw the same distinctions, but interpret those representations differently. This is a distinct kind of political disagreement from epistemic incomparability. Both forms of disagreement can coexist and interact, as the reframing discussion below illustrates.

Before we move on, one design choice about M itself deserves to be made explicit.

Remark 2.7 (Relations between meanings). *Distinct meanings can occur independently: a voter may assign factual effectiveness T without evaluative approval G , or both together. A systematic implication from T to G would require an additional restriction on permitted semantics, such as closure under a specified implication rule. The conflict relation Cf in Section 3.1 instead records which meaning pairs count as incompatible during an encounter. An individual understanding or evaluative anchor may associate T and G without imposing that association on every voter. Restrictions on these associations are possible sources of structure in electoral responses; whether they support a common single-peaked order also depends on reception, updating, evaluation, and the feasible menu.*

2.4 Understandings.

An agent’s *understanding* combines both layers: which representations the agent can access and distinguish (the epistemic stance) and what meanings the agent attaches to those representations (the semantics). Two agents can differ in either layer or both. The immigration voters illustrate the full range: voters A and B share the same epistemic stance but differ semantically; voter C (in her coarsened version from Section 2.2) differs from both in the epistemic layer as well. The definition below packages both layers into a single object and organizes the space of all such objects into a structure—a complete lattice—that supports the comparison, aggregation, and updating operations that drive the rest of the manuscript.

Definition 2.8. *An understanding is a pair $u = (e, \mu_e)$ consisting of an epistemic stance $e \in E_{\mathcal{R}}$ and an e -consistent semantics $\mu_e \in M_e$. The space of understandings is*

$$\mathcal{U}_{\mathcal{R}} = \bigsqcup_{e \in E_{\mathcal{R}}} M_e = \{(e, \mu_e) \mid e \in E_{\mathcal{R}}, \mu_e \in M_e\},$$

where $\bigsqcup$ denotes disjoint union: each epistemic stance defines a fiber of compatible semantics, and $\mathcal{U}_{\mathcal{R}}$ assembles all fibers.

Semantic transport. The fiber structure requires us to distinguish the pooling of associations during reception from the extraction of content common to finer distinctions. For a fixed stance e , write $\mu \preceq_M \nu$ for pointwise inclusion on its visible classes. Suppose $e_1 \preceq_E e_2$. Every fine class D lies in a unique coarse class C_D , which may be invisible. The following maps compare semantics at these levels:

$$\begin{aligned} \text{Ref}_{e_1 \rightarrow e_2}(\mu)(D) &= \mu(C_D), \\ \text{Crs}_{e_2 \rightarrow e_1}(\nu)(C) &= \bigcup_{\substack{D \in \mathcal{R}_{e_2} / \sim_{e_2} \\ D \subseteq C}} \nu(D), \\ \text{All}_{e_2 \rightarrow e_1}(\nu)(C) &= \bigcap_{\substack{D \in \mathcal{R}_{e_2} / \sim_{e_2} \\ D \subseteq C}} \nu(D), \end{aligned}$$

where the last two formulas apply to visible coarse classes C ; both maps assign $\emptyset$ to I_{e_1} . Every visible coarse class contains at least one fine class. Refinement also respects invisible semantics: the fine invisible class has an invisible coarse parent. Newly visible classes inherit $\mu(I_{e_1}) = \emptyset$, so refinement alone creates no meanings at newly accessible resources.

Union-pooling Crs collects all associations attached to any finer subclass and will drive reception in the updating model. Intersection transport All retains only associations attached to every finer subclass; it supplies the common content needed for the understanding meet.³

Proposition 2.9. *For $e_1 \preceq_E e_2$, the maps Ref, Crs, and All above are monotone maps between the indicated semantic fibers, and*

$$\text{Ref}_{e_1 \rightarrow e_2}(\mu) \preceq_M \nu \iff \mu \preceq_M \text{All}_{e_2 \rightarrow e_1}(\nu).$$

Thus $\text{Ref} \dashv \text{All}$ is a Galois connection. If $\mathcal{R}_{e_1} = \mathcal{R}_{e_2}$, there is also the adjunction $\text{Crs} \dashv \text{Ref}$; this second claim does not in general hold when visible scope expands. Refinement has identity maps and composes: for $e_1 \preceq_E e_2 \preceq_E e_3$, $\text{Ref}_{e_2 \rightarrow e_3} \circ \text{Ref}_{e_1 \rightarrow e_2} = \text{Ref}_{e_1 \rightarrow e_3}$.

The first equivalence says that a coarse meaning can be retained below a finer understanding exactly when every finer subclass carries it. The restriction on the second adjunction matters: coarsening may erase meanings at resources outside the coarse scope, while refinement assigns empty semantics to those resources when they become visible.

Reception across incomparable stances. Neither agent’s vocabulary need refine the other’s. For arbitrary stances e_i and e_j , define the *overlap reception map* from the sender’s semantics to the receiver’s vocabulary by

$$\text{Rec}_{e_j \rightarrow e_i}(\nu)(C) = \bigcup_{\substack{D \in \mathcal{R}_{e_j} / \sim_{e_j} \\ D \cap C \neq \emptyset}} \nu(D), \quad C \in \mathcal{R}_{e_i} / \sim_{e_i},$$

and set $\text{Rec}_{e_j \rightarrow e_i}(\nu)(I_{e_i}) = \emptyset$. A receiver category collects associations from each sender category with which it shares a resource. The common ambient field supplies this correspondence; a sender’s association with a bundle is available at its overlapping parts. This is a rule of semantic assimilation, not a claim that either agent acquires the other’s distinctions.

Proposition 2.10. *For arbitrary e_i, e_j , $\text{Rec}_{e_j \rightarrow e_i} : M_{e_j} \rightarrow M_{e_i}$ is monotone and, writing $k = e_i \vee e_j$, satisfies*

$$\text{Rec}_{e_j \rightarrow e_i} = \text{Crs}_{k \rightarrow e_i} \circ \text{Ref}_{e_j \rightarrow k}.$$

If $e_i \preceq_E e_j$, it reduces to $\text{Crs}_{e_j \rightarrow e_i}$; if $e_j \preceq_E e_i$, it reduces to $\text{Ref}_{e_j \rightarrow e_i}$.

For example, a sender may attach “good” to the fused category $\{a, b\}$ while a receiver sees only a . Their epistemic meet is empty, but overlap reception delivers “good” at the receiver’s a -class, leaving b invisible. Partial uptake can therefore occur without shared categorical ground.⁴

³These downward transports answer different questions. If a source attaches “good” to a and “bad” to b , union-pooling into the category $\{a, b\}$ retains both associations, representing the ambivalence that coarse reception can produce. Intersection transport retains neither: there is no association shared across the two subclasses. More generally, union asks whether a meaning occurs anywhere in the bundle, while intersection asks whether it occurs throughout the bundle. The latter may also be useful for cautious summarization or a communication rule restricted to shared associations. Neither operation assesses the truth of a meaning. The choice between them is substantive for reception, although intersection has a necessary formal role in the richness order adopted here. Both erase content that becomes invisible; pooling such content into the invisible class would violate semantic consistency.

⁴An alternative route through $h = e_i \wedge e_j$ is $\text{Ref}_{h \rightarrow e_i} \circ \text{Crs}_{e_j \rightarrow h}$. This rule first expresses the message in common coarse categories and transfers nothing when $h = e_\perp$. It agrees with overlap reception when the stances are comparable, but can differ otherwise. It may also blur more associations: if the sender distinguishes $\{a, b\}$ from $\{c\}$ and the receiver distinguishes $\{a\}$ from $\{b, c\}$, their meet fuses all three resources and pools the sender’s associations across both receiver classes. We use overlap reception to allow uptake through shared resources while preserving the receiver’s current categories. Neither route transfers content between disjoint visible scopes; doing so would require a bridge resource, expanded access, or an additional translation relation.

The order on understandings. Using the refinement operator to compare meanings across epistemic levels, we extend the order on epistemic stances to the full space of understandings:

$$(e_1, \mu_{e_1}) \preceq_{\mathcal{U}} (e_2, \mu_{e_2}) \iff e_1 \preceq_E e_2 \text{ and } \text{Ref}_{e_1 \rightarrow e_2}(\mu_{e_1}) \preceq_M \mu_{e_2},$$

where $\preceq_M$ is the pointwise inclusion order on M_{e_2} : $\nu \preceq_M \nu'$ iff $\nu(D) \subseteq \nu'(D)$ for all $D \in \mathcal{R}_{e_2}/\sim_{e_2}$. Unpacking the refinement: the condition requires $\mu_{e_1}(C) \subseteq \mu_{e_2}(D)$ for every fine class D at e_2 's level that is contained in the coarse class C at e_1 's level. An understanding (e_2, μ_{e_2}) lies above (e_1, μ_{e_1}) when it is epistemically richer *and* each fine distinction D in e_2 's partition carries at least all the meanings e_1 attributes to the encompassing coarse class C . In the immigration example, $u_A \preceq_{\mathcal{U}} u_B$ would require $\mu_A(r_i) \subseteq \mu_B(r_i)$ for all i (since both voters share $e_{\top}$). This fails at r_1 : $\mu_A(r_1) = \{T, G\}$ is not contained in $\mu_B(r_1) = \{T, B\}$, because $G \notin \mu_B(r_1)$. The same check in reverse fails at r_2 . The two voters are incomparable: neither understanding contains the other in the richness order. Whether their electoral responses nevertheless admit a common single-peaked order is a separate question.

The same terminological caution applies: the order $\preceq_{\mathcal{U}}$ measures *representational richness*, not epistemic virtue. A higher understanding has more representational resources and richer meanings. It does not have more accurate, more rational, or more politically appropriate representations.

Theorem 2.11. *($\mathcal{U}_{\mathcal{R}}, \preceq_{\mathcal{U}}$) is a complete lattice. For any nonempty family $\{(e_j, \mu_{e_j})\}_{j \in J} \subseteq \mathcal{U}_{\mathcal{R}}$, writing $e^{\wedge} = \bigwedge_j e_j$ and $e^{\vee} = \bigvee_j e_j$, the meet and join are*

$$\bigwedge_{j \in J} (e_j, \mu_{e_j}) = (e^{\wedge}, \mu_{\wedge}), \quad \bigvee_{j \in J} (e_j, \mu_{e_j}) = (e^{\vee}, \mu_{\vee}),$$

where

$$\mu_{\wedge}(C) = \bigcap_{j \in J} \text{All}_{e_j \rightarrow e^{\wedge}}(\mu_{e_j})(C), \quad \mu_{\vee}(D) = \bigcup_{j \in J} \text{Ref}_{e_j \rightarrow e^{\vee}}(\mu_{e_j})(D),$$

for $C \in \mathcal{R}_{e^{\wedge}}/\sim_{e^{\wedge}}$ and $D \in \mathcal{R}_{e^{\vee}}/\sim_{e^{\vee}}$ respectively. Both semantics assign $\emptyset$ to their invisible class. The global bottom is $u_{\perp} = (e_{\perp}, 0_{e_{\perp}})$, and the global top has stance $e_{\top}$ and meaning set M at every visible singleton. The empty-family join and meet are these bottom and top elements, respectively.

The lattice operations have direct political interpretations. The meet $u_i \wedge u_j$ is their *semantic common ground*: the greatest understanding below both in the understanding order. The join $u_i \vee u_j$ is their *pooled representational toolkit*: the least understanding above both, combining their distinctions and meanings. Electoral evaluation in Section 4 uses meets with a fixed evaluative anchor to value received content and cognitive change. Under full credibility and maximal acquisition, forward absorption takes the join of the prior and the broadcast (Proposition 3.4).

The immigration voters make the lattice operations concrete. Consider the meet of voters A and B (at the shared stance $e_{\top}$, the meet reduces to pointwise intersection of meaning sets):

$$u_A \wedge u_B : \quad r_1 \mapsto \{T\}, \quad r_2 \mapsto \emptyset, \quad r_3 \mapsto \{T\}.$$

Their common semantics is $(\{T\}, \emptyset, \{T\})$: the two voters share the factual judgment that enforcement has consequences and that immigration has economic impact, but they share *nothing* evaluative and have zero semantic overlap on paths to citizenship. This is the formal content of the familiar observation that “they agree on the facts but not on the values”—the meet strips away everything contested and leaves only what survives intersection.

Now consider the meet of voters A and C:

$$u_A \wedge u_C : \quad r_1 \mapsto \emptyset, \quad r_2 \mapsto \emptyset, \quad r_3 \mapsto \emptyset.$$

Write 0_e for empty semantics at every class and $u_{e,0} = (e, 0_e)$ for the bottom of the fixed semantic fiber. The result is $u_{e_\top,0}$: the voters share no meanings, but both still see and distinguish all three representations. It is strictly above the global empty-scope bottom $u_\perp$. Empty common semantics does not preclude reception: a message can introduce an association absent from the receiver’s prior.

The example also exhibits semantic incomparability. Voters A and B are incomparable in $\preceq_{\mathcal{U}}$: A assigns G to r_1 where B does not ($u_B \not\preceq_{\mathcal{U}} u_A$), and B assigns $\{T, G\}$ to r_2 where A assigns $\{F, B\}$ ($u_A \not\preceq_{\mathcal{U}} u_B$). Neither understanding is richer than the other; each carries semantic content the other lacks. Voters A and C are also incomparable, and their meet has empty semantics at the shared stance. This distinguishes the absence of shared meanings from the absence of shared representations.

The join $u_A \vee u_B = (\{T, G, B\}, \{T, F, G, B\}, \{T, G, B\})$ pools both meaning systems, producing an understanding that is internally conflicted—at r_2 it simultaneously carries T and F , G and B —and is the least understanding containing both voters’ semantic systems. It need not be maximal in the full understanding lattice. The tension at r_2 is informative, not paradoxical. It says that the full semantic picture of paths to citizenship includes both “viable and good” and “false and bad”—the pooled understanding does not resolve this conflict but rather *registers* it as a feature of the electorate’s semantic landscape.

These computations distinguish semantic richness from electoral orderability. Incomparable understandings cannot be ranked by inclusion, but voters’ electoral responses might still be single-peaked on a common ordering of feasible messages. The *semantic dimension* defined later records antichain width; it does not by itself settle whether an election admits a policy axis. That question also depends on reception, initial understandings, credibility, updating, evaluation, and the feasible message set. Example 5.25 supplies these inputs and establishes the axis obstruction from the resulting rankings.

Figure 1 summarizes the lattice structure of the three immigration voters.

The evaluative anchor. The lattice operations also supply objects for evaluating encounters. An agent’s *evaluative anchor* u_v^* , introduced in Definition 4.1, represents dispositions through which political content can be recognized as valuable. It need not be identical to the agent’s current understanding or consciously articulated in that understanding. Section 4 uses meets with the anchor to value both a received message and the cognitive change produced by the encounter. The best available appeal is then derived from the agent’s prior, reception, and response rule; it is not identified with the anchor itself.

In the immigration example, voter C’s anchor contains evaluative meanings but no factual meanings. At the common fixed stance, a received message containing only factual meanings has the same meet with this anchor as an empty message at that stance, so its normalized recognition term is zero. This recognition calculation alone does not determine the response to general encounters: their changes in valued prior content or acquired distinctions must also be evaluated. These distinctions are reasons to keep the received message and the resulting understanding separate.

Epistemic regression. The state space can represent a loss of access or distinctions. For example, an agent who coarsens an epistemic partition while retaining the same visible scope moves downward in $\preceq_E$. To move downward in the understanding order as well, the resulting coarse meanings must be included in every corresponding prior fine class; intersection transport supplies the greatest

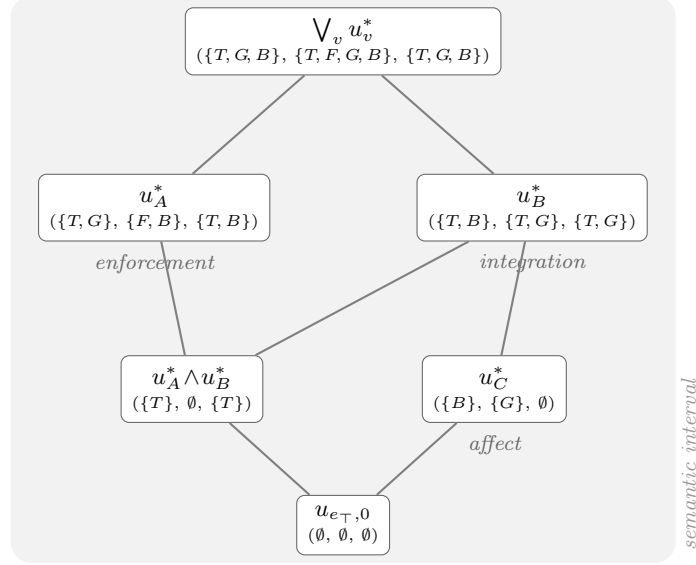


Figure 1: Order relations among selected understandings at the common stance e_T . Nodes show semantic content $(\mu(r_1), \mu(r_2), \mu(r_3))$; edges are covering relations among the displayed nodes, not all covers in the full lattice. The displayed understandings, used below as evaluative anchors, have u_A^* and u_B^* incomparable (neither lies above the other); u_C^* lies below u_B^* . The three-way meet $\bigwedge_v u_v^* = u_{e_T, 0}$ has empty meanings but full epistemic scope. The displayed nodes lie in the interval between that meet and the join; the diagram alone makes no claim about feasible majority-preference paths.

semantics satisfying that condition. If coarsening instead pools new associations, the old and new understandings can be incomparable. Similarly, losing distinctions while making previously invisible resources accessible does not give a downward epistemic move: an increase in visible scope precludes it. The order does not assign a common numerical weight to gains in access and losses of discrimination.

The state space’s ability to describe these changes must be distinguished from the transitions generated by a particular update rule. The rule in Section 3 expands or preserves the epistemic stance on its forward branch and preserves it on contraction. Actual loss of epistemic distinctions would require a further transition rule.

2.5 Reframing: the paradigm interaction.

Maintaining separate epistemic and semantic layers lets us distinguish several ways a political appeal can reshape understanding. Entman (1993, pp. 52–53) describes framing through selection and salience, directed toward interpretations of problems, causes, evaluations, or remedies. He distinguishes the communicator’s framing intention, the frame in the text, and the receiver’s thinking; a frame’s presence in a text does not guarantee its adoption. This distinction motivates separating the broadcast from the understanding produced by an encounter.

Within the present state space, a voter can acquire a new interpretation of a familiar category while retaining the same epistemic stance. For example, a procedural category can acquire a moral association alongside its existing meanings. A reframe can also involve new access or distinctions: a voter may come to distinguish cases previously treated alike and attach different meanings to

them. The framework represents semantic change at a fixed stance, epistemic change, and their combination; the update rule determines which changes an encounter generates.⁵

In Section 4, a candidate broadcasts an understanding via a map $\phi : X \rightarrow \mathcal{U}_{\mathcal{R}}$ (Definition 4.1). The feasible image $\phi(X)$ specifies which representations and meanings the candidate can offer. The update operator and credibility filter α_{vc} in Section 3 determine the resulting voter state, and the electoral response evaluates the encounter separately. Adopting an interpretation need not make the voter favor its speaker.

The immigration case. Suppose a politician seeks to shift how voters understand immigration. Voters begin with an epistemic stance e_0 distinguishing several immigration categories—high-skill workers, low-skill workers, asylum seekers, temporary labor visa holders, undocumented entrants—and a semantics μ_{e_0} that assigns meanings including “economic contributor,” “humanitarian obligation,” and “procedural status.”

Suppose the politician offers a statistical display organized by immigration status, represented by $r_{\text{crime}} \in \mathcal{R} \setminus \mathcal{R}_{e_0}$. The broadcast attaches a “threat” meaning both to that display and to specified existing categories, such as undocumented entry. If the encounter takes the forward branch, acquisition can make the display visible, and filtered absorption can add the offered associations where they are admitted. For a voter who acquires both, this is joint epistemic and semantic change. The new association on an existing category is carried by the broadcast itself; assigning a meaning to the display alone does not propagate it automatically to other categories. If raw conflict activates the contraction branch, the encounter instead removes conflicting prior meanings at the voter’s existing stance.

If voters also collapse their distinctions among immigration categories into a “legal” versus “illegal” binary, the reframe combines gained access with lost discrimination. With r_{crime} newly visible, the new stance cannot lie below e_0 ; if a previous distinction is lost, it cannot lie above e_0 either. The stances are then incomparable, and so are the associated understandings. The state space can represent this combination of changes, but the update operator below would need an additional coarsening mechanism to generate it. Attaching a threat association to an already visible category is a simpler possible reframe: overlap reception and filtered absorption can add it on the forward branch without changing the stance or erasing existing meanings.

3 Semantic updating.

The framework developed in Section 2 describes the static geometry of understandings: how representational resources are accessed, how meanings are assigned, and how understandings are ordered.

The update model distinguishes admission of incoming meanings from removal of prior meanings. It uses a conflict relation, source-specific credibility, a tolerance threshold, and a rule for epistemic expansion. On the forward branch, the receiver first gains any access or distinctions supplied by the expansion rule, then pools admitted associations from overlapping sender categories at that expanded stance. On the contraction branch, the receiver deletes prior meanings that conflict with the translated message while retaining the same epistemic stance. Both branches are specified on the full understanding space.

⁵A separate salience variable could represent changes in the relative prominence of associations while access, distinctions, and attached meaning sets remain fixed. The present understanding (e, μ_e) records those sets and distinctions, so a pure prominence change would require an additional state variable or an explicitly specified effect on reception or evaluation.

Festinger’s (1957, chap. 1) theory links cognitive inconsistency to pressure for change and avoidance of additional dissonance. Harmon-Jones and Mills (2019, pp. 3–4) distinguish inconsistent cognitions from experienced discomfort and describe several routes to reduction, including removing cognitions, adding consonant ones, and changing their importance. Here the operator specifies which prior meanings survive and which incoming meanings enter. Its conflict count measures the number of receiver categories containing incompatible prior and message content; the threshold selects the update branch. An empirical application can investigate how felt discomfort relates to this count and to observed updating. Following content across encounters then tests whether exposure preserves prior meanings, removes them, or leads to later uptake.

The understanding lattice makes the branches comparable: the forward update is extensive, the baseline is the greatest admissible contraction below the prior, and the parameter comparisons below hold for the explicitly fixed expansion rule. The existence and behavior of equilibria still require the additional hypotheses stated in their respective models.

3.1 Conflict structure.

Before specifying how understandings change, we need a formal notion of when two understandings are in tension. The conflict relation captures incompatibility at the level of individual meanings; the conflict set aggregates this to the level of representation classes, measuring how broadly one understanding clashes with another.

Definition 3.1. *A conflict relation is a symmetric, irreflexive relation $\text{Cf} \subseteq M \times M$. A set of meanings $A \subseteq M$ is conflict-free if no pair $(m, m') \in \text{Cf}$ has both $m, m' \in A$. A semantics μ_e is coherent if $\mu_e([r])$ is conflict-free for every $[r] \in \mathcal{R}_e / \sim_e$.*

The conflict relation is left as a parameter of the model. Its natural content comes from two sources. For truth-apt meanings $M_T \subseteq M$ equipped with a negation map $\neg : M_T \rightarrow M_T$, the pairs $(m, \neg m)$ are the canonical conflicts: an agent should not simultaneously hold a truth-apt claim and its negation for the same perceived representation. For evaluative, affective, and social-association meanings, conflicts may be present or absent depending on the application; the framework imposes no requirement. Incoherence of the latter kind—holding incommensurable values, carrying tensions between social associations—is representationally permissible.

The conflict relation is an analytical choice. For fixed understandings, adding conflict pairs can enlarge the set of receiver classes at which conflict is detected. How often contraction occurs also depends on the actual message, the prior meanings, and the receiver’s threshold; the number of possible conflict pairs alone does not determine the frequency of reversion.

Given two understandings $u_i = (e_i, \mu_i)$ and $u_j = (e_j, \mu_j)$, write

$$F_{ij}([r]) := \{m \in M : \exists m' \in \text{Rec}_{e_j \rightarrow e_i}(\mu_j)([r]), (m, m') \in \text{Cf}\}$$

for the set of meanings at i ’s class $[r]$ that conflict with j ’s translated content. The reception map of Proposition 2.10 is defined even when the stances are incomparable. The *conflict set* of u_j against u_i is then

$$\text{Conf}(u_i, u_j) = \{[r] \in \mathcal{R}_{e_i} / \sim_{e_i} : \mu_i([r]) \cap F_{ij}([r]) \neq \emptyset\}.$$

This is i ’s perspective: it asks whether what j is saying, translated into i ’s representational language, would require i to hold conflicting meanings simultaneously. Conflict is assessed on the translated message before credibility filtering. Thus a meaning can trigger a reaction even if the receiver would not admit it into the forward update. Filtering before conflict detection would define a different

model; here credibility controls absorption while the raw message determines conflict exposure. An empty conflict set means that no prior meaning conflicts with the translated message; it does not require the prior or message to be internally coherent.

In the immigration example, the natural conflict relation on $M = \{T, F, G, B\}$ is $\text{Cf} = \{(T, F), (F, T), (G, B), (B, G)\}$: factual values conflict with each other and evaluative values conflict with each other. Consider voter A encountering an integration understanding u_L with semantics $\mu_L = (\{T, B\}, \{T, G\}, \{T, G\})$ —the integration voter’s understanding from Section 2. At the shared epistemic level $e_\top$, reception is the identity, and the conflict meanings are:

$$F_{AL}(r_1) = \{F, G\}, \quad F_{AL}(r_2) = \{F, B\}, \quad F_{AL}(r_3) = \{F, B\}.$$

The meanings in voter A’s semantics that fall within these conflict sets are $\mu_A(r_1) \cap F_{AL}(r_1) = \{G\}$, $\mu_A(r_2) \cap F_{AL}(r_2) = \{F, B\}$, and $\mu_A(r_3) \cap F_{AL}(r_3) = \{B\}$ —all nonempty. Thus $\text{Conf}(u_A, u_L) = \{r_1, r_2, r_3\}$: every representation class is in conflict. The enforcement voter’s entire semantic system is under pressure from the integration understanding.

3.2 Source credibility.

The conflict structure determines when an encounter is contentious; the credibility structure determines what content passes through. Together they govern the two branches of the update operator: forward absorption (filtered by credibility) and semantic contraction (triggered by conflict).

The degree to which agent i absorbs meanings from agent j is governed by a *source credibility* structure. Credibility is representation-specific: what matters is which *types* of meaning an agent is willing to absorb from a given source—policy implications, evaluative associations, affective signals, social-group content. The credibility function encodes this type-specific receptivity as a *meaning filter*: a subset $\alpha \subseteq M$ specifying which categories of meaning i is willing to receive. The filtering operation is intersection: on the forward branch, i admits from j exactly $\alpha \cap A$, the portion of j ’s content A that falls within i ’s receptivity set. An empty filter excludes all incoming meanings from that sender class. The raw message can still trigger contraction of prior content, and a forward encounter can still supply distinctions through the separately specified expansion rule.

Definition 3.2. A source credibility function for agent i toward agent j is a map

$$\alpha_{ij} : \mathcal{R}_{e_j} / \sim_{e_j} \rightarrow 2^M,$$

assigning to each of j ’s perceived representations a meaning filter $\alpha_{ij}([r]) \subseteq M$. When the source stance can vary, a credibility specification is a family $\{\alpha_{ij}^e\}_{e \in E_{\mathcal{R}}}$ with $\alpha_{ij}^e : \mathcal{R}_e / \sim_e \rightarrow 2^M$; at source stance e_j , write α_{ij} for $\alpha_{ij}^{e_j}$.

Credibility is representation-specific: agent i may be highly receptive to j ’s affective associations while entirely filtering out j ’s policy implications, or vice versa. Full receptivity at $[r]$ is $\alpha_{ij}([r]) = M$; complete exclusion of incoming meanings is $\alpha_{ij}([r]) = \emptyset$. Filtering acts on each sender class before overlap-pooling: define $\nu_{ij}(D) = \alpha_{ij}(D) \cap \mu_j(D)$ on visible sender classes and $\nu_{ij}(I_{e_j}) = \emptyset$. This permits a meaning admitted from one sender category to reach an overlapping receiver category even if another sender category would block that meaning.

The empirically interesting case is when credibility filters are *heterogeneous across voters and sources*. The parameter α_{ij} specifies receptivity in an encounter, separately from the receiver’s current understanding $u_i = (e_i, \mu_i)$. Its formation is a further modeling question: a rule using source cues, for example, could generate different admission sets for otherwise identical content.⁶

⁶For discussion of political heuristics, see [Lau and Redlawsk \(2006, chap. 11\)](#). A heuristic using source cues could be one process generating a credibility filter. The set-intersection specification describes which meanings pass; explaining how that set is formed requires an account of the receiver’s decision process.

Bartels (2002) reports partisan differences in retrospective economic perceptions in the 1988 NES. Iyengar et al. (2019) review evidence on partisan affect, trust, and favoritism. These findings motivate studying how political attachments shape encounters with political content. Within this framework, such attachments could enter prior meanings, evaluative anchors, or source-specific admission sets; identifying their role requires distinguishing these objects empirically.

An empirical study could present the same speech under randomly assigned source labels and measure the meanings listeners associate with its representations before and after exposure, alongside their feelings about the source. Connecting those observations to α_{ij} requires an account of perceived distinctions, prior content, and which update branch occurred. At a common fixed stance on the forward branch, an offered meaning absent from the prior enters exactly when the filter admits it. Already-held meanings do not reveal admission through a change in content, while contraction and acquisition of distinctions require separate treatment. This design would therefore test restrictions on reception with an explicit measurement model; source-label effects on warmth alone would leave the admission sets undetermined.

When voter types admit different portions of a broadcast—for example, one admits factual content and another evaluative associations—a candidate can supply different receipts through a single speech. These receipts can produce different post-update understandings and supply the reception mechanism for polysemy. Its electoral value depends on feasibility and the full response, as specified in Section 5.3.

3.3 The update operator.

With conflict and credibility in hand, we can specify the full update. The operator has two branches: a *forward update* that absorbs filtered content from the source and enriches the receiver’s understanding, and a *baseline* that contracts the receiver’s understanding by stripping conflicting content. Which branch fires depends on a threshold parameter that measures the receiver’s tolerance for conflict.

Let $u_i = (e_i, \mu_i)$ and $u_j = (e_j, \mu_j)$. The update of u_i upon encountering u_j is governed by a *trigger threshold* $\tau_i \in \mathbb{N} \cup \{\infty\}$ and source credibility α_{ij} . The trigger threshold is a tolerance for conflict: the forward update fires when the number of conflicting representation classes is at most τ_i ; when it exceeds τ_i , the agent reverts. $\tau_i = 0$ means any conflict triggers reversion; $\tau_i = \infty$ means the agent never reverts regardless of conflict size.

Fix an *epistemic expansion rule* $\Gamma : E_{\mathcal{R}} \times E_{\mathcal{R}} \rightarrow E_{\mathcal{R}}$ satisfying

$$e_i \preceq_E \Gamma(e_i, e_j) \preceq_E e_i \vee e_j.$$

This rule depends on the two stances, not on credibility, the threshold, or their semantic contents. It is a model parameter held fixed throughout the analysis; minimal and maximal expansion choose e_i and $e_i \vee e_j$ respectively. The source understanding describes the distinctions and meanings offered in this encounter; in the electoral application it is the broadcast understanding, rather than the speaker’s entire private understanding. Maximal expansion assumes that every offered distinction is acquired while the receiver’s existing distinctions are preserved. Partial acquisition is permitted by other choices of Γ .

The conflict test is applied to the raw message at the receiver’s initial stance, as specified above. If it passes, epistemic expansion precedes semantic reception; otherwise the receiver contracts at the initial stance. Thus the same successful forward encounter can make a resource distinguishable and attach meanings to it. Testing conflict after expansion would define a different operator: splitting a category can change the number of conflicting classes against which the threshold is compared.

Definition 3.3. For the fixed conflict relation and expansion rule Γ , the update of u_i by u_j is

$$T_{\alpha_{ij}, \tau_i}(u_i, u_j) = \begin{cases} T_+(u_i, u_j) & \text{if } |\text{Conf}(u_i, u_j)| \leq \tau_i, \\ \beta(u_i, u_j) & \text{otherwise,} \end{cases}$$

where T_+ is the forward update and β is the baseline, defined below. The notation suppresses the fixed parameters Cf and Γ .

The forward update. Set $e'_i = \Gamma(e_i, e_j)$ and transport the prior semantics to that stance. The forward update then adds the filtered source content there:

$$\begin{aligned} \mu_i^+(C) &= \text{Ref}_{e_i \rightarrow e'_i}(\mu_i)(C) \cup \text{Rec}_{e_j \rightarrow e'_i}(\nu_{ij})(C) \\ &= \text{Ref}_{e_i \rightarrow e'_i}(\mu_i)(C) \cup \bigcup_{\substack{D \in \mathcal{R}_{e_j} / \sim_{e_j} \\ D \cap C \neq \emptyset}} (\alpha_{ij}(D) \cap \mu_j(D)), \end{aligned}$$

for every visible class C at e'_i , with $\mu_i^+(I_{e'_i}) = \emptyset$. The resulting understanding is

$$T_+(u_i, u_j) = (e'_i, \mu_i^+).$$

Refinement preserves the prior and assigns empty inherited semantics to newly visible classes; reception can supply meanings to those classes within this encounter. When a receiver class overlaps several visible source classes, their admitted associations are pooled. When it overlaps none, it retains only its inherited meanings. At a common stance the expansion interval is a singleton, reception is the identity, and the formula reduces to pointwise filtered union.⁷

Proposition 3.4 (Maximal expansion). *Let $u_j^\alpha = (e_j, \nu_{ij})$ be the source understanding with its semantics filtered by α_{ij} . If $\Gamma(e_i, e_j) = e_i \vee e_j$, then*

$$T_+(u_i, u_j) = u_i \vee u_j^\alpha.$$

Under full credibility, $T_+(u_i, u_j) = u_i \vee u_j$. These identities give the actual update T whenever its conflict test selects the forward branch.

Example 3.5 (Articulating an invisible resource). Take $\mathcal{R} = \{r_p\}$, where r_p is a representational resource. Initially the receiver has the single block $\{r_p, \star\}$, with empty semantics. The source distinguishes $\{r_p\}$ from $\{\star\}$ and attaches a meaning m to $\{r_p\}$. The receiver has no visible class at which conflict can arise. With maximal expansion and full credibility, the encounter produces the partition $\{\{r_p\}, \{\star\}\}$ and semantics $\{m\}$ at the newly visible resource. The same encounter therefore supplies both a distinction and an association.

The initial state represents inability to distinguish r_p , not belief in the negation of a proposition p . The resource already belongs to the ambient field; its availability to the receiver changes. This is one precise sense in which an appeal can articulate something previously unavailable to thought. The construction does not by itself imply that the acquired distinction is accurate or that the receiver responds favorably to it.

⁷An alternative timing absorbs at the old stance and then expands: $\tilde{T}_+(u_i, u_j) = (e'_i, \text{Ref}_{e_i \rightarrow e'_i}(\mu_i \cup \text{Rec}_{e_j \rightarrow e'_i}(\nu_{ij})))$. It models an encounter interpreted entirely through the receiver's existing categories, with any new distinctions becoming usable for semantic reception only later. The two rules agree when there is no expansion. They need not otherwise coincide: the alternative leaves newly visible classes empty, and it can spread associations across distinctions learned during the encounter. For example, if the receiver groups two visible resources together while the sender distinguishes them and assigns $\{g\}$ and $\{h\}$, absorption before a full split copies $\{g, h\}$ into both new classes when the prior semantics are empty. Expansion before reception instead delivers $\{g\}$ and $\{h\}$ separately. Consequently neither timing produces a uniformly richer understanding across all encounters.

The forward update in the immigration example. Since voter A and the integration understanding u_L share $e_\top$, the formula reduces to $\mu_A^+(r_k) = \mu_A(r_k) \cup (\alpha_{AL}(r_k) \cap \mu_L(r_k))$. Under *full credibility* ($\alpha_{AL} = M$ everywhere), the filter is transparent:

$$\begin{aligned}\mu_A^+(r_1) &= \{T, G\} \cup \{T, B\} = \{T, G, B\}, \\ \mu_A^+(r_2) &= \{F, B\} \cup \{T, G\} = \{T, F, G, B\}, \\ \mu_A^+(r_3) &= \{T, B\} \cup \{T, G\} = \{T, G, B\}.\end{aligned}$$

At r_1 and r_3 , voter A now holds evaluative ambivalence: enforcement is simultaneously good (G) and bad (B). At r_2 , the full meaning space is active—factual and evaluative tension coexisting. This is the augmentation interpretation made concrete: holding G and B simultaneously represents attitudinal ambivalence. Under the stated conflict relation it is incoherent in the sense of Definition 3.1; such semantics are nevertheless allowed in the understanding lattice.

Under *factual-only credibility* ($\alpha_{AL}(r_k) = \{T, F\}$ for all k), the filter admits only factual content:

$$\begin{aligned}\mu_A^+(r_1) &= \{T, G\} \cup (\{T, F\} \cap \{T, B\}) = \{T, G\}, \\ \mu_A^+(r_2) &= \{F, B\} \cup (\{T, F\} \cap \{T, G\}) = \{T, F, B\}, \\ \mu_A^+(r_3) &= \{T, B\} \cup (\{T, F\} \cap \{T, G\}) = \{T, B\}.\end{aligned}$$

Voter A absorbs the factual claim that paths to citizenship are viable (T enters at r_2) but does not absorb the evaluative endorsement (G is blocked by the filter). The result at r_2 is $\{T, F, B\}$: factual ambivalence (T and F simultaneously active) alongside the retained negative evaluation B . Here selective admission preserves the old negative evaluation while adding a factual meaning on the forward branch. Its empirical interpretation requires measuring which meanings are admitted separately from the strength of the voter’s evaluations.

The contrast between the two cases illuminates what is at stake in the credibility parameter. Under full credibility, voter A’s post-update understanding at r_2 is $\{T, F, G, B\}$ —the entire meaning space, carrying both factual claims and both evaluative assessments simultaneously. Under factual-only credibility, r_2 carries $\{T, F, B\}$: the voter has acquired factual ambivalence (“paths to citizenship might work *and* might fail”) but the evaluative frame remains undisturbed (B persists, G is excluded). The full-credibility output contains more meanings and more conflicting pairs. Under the same subsequent message and threshold, its conflict count is weakly larger than that of the factual-only output. At a class holding $\{T, F, G, B\}$, every received meaning in this four-meaning universe conflicts with a meaning already present. Whether the next encounter triggers contraction depends on which classes receive content and the threshold. The factual-only filter prevents the absorption of new evaluative meanings, but does not shield prior evaluations from the raw-message conflict test. Credibility controls which meanings accumulate; the trigger still responds to conflicting content in the translated message.

The baseline. When the conflict in $\text{Conf}(u_i, u_j)$ exceeds the threshold τ_i , the update reverts to the *baseline*—the largest understanding below u_i that avoids the conflicting content:

$$\begin{aligned}\beta(u_i, u_j) &= \bigvee S_{ij}, \quad \text{where} \\ S_{ij} &= \{u \preceq_{\mathcal{U}} u_i : \forall [r] \in \text{Conf}(u_i, u_j), \text{Ref}_{e_u \rightarrow e_i}(\mu_u)([r]) \cap F_{ij}([r]) = \emptyset\}.\end{aligned}$$

The construction retains as much prior content as possible while avoiding meanings that conflict with the source.⁸ Several features are immediate. First, $\beta(u_i, u_j) \preceq_{\mathcal{U}} u_i$: the baseline is always weakly below the current understanding. Second, the baseline depends on both arguments: what you revert to depends on what triggered the reversion. Third, (e_i, μ_i) itself is in S_{ij} iff $\text{Conf}(u_i, u_j) = \emptyset$ —that is, iff u_i has no conflicting content with u_j at any representation class. When $\tau_i > 0$, this is strictly stronger than the trigger-passing condition $|\text{Conf}(u_i, u_j)| \leq \tau_i$: the trigger can pass even when $u_i \notin S_{ij}$. Fourth, and most consequentially:

Proposition 3.6. *If $\text{Conf}(u_i, u_j) \neq \emptyset$, then $\beta(u_i, u_j) \prec_{\mathcal{U}} u_i$. More explicitly, S_{ij} has a greatest element and*

$$\beta(u_i, u_j) = (e_i, \mu_i^-), \quad \mu_i^-(C) = \mu_i(C) \setminus F_{ij}(C)$$

for every visible receiver class C , with $\mu_i^-(I_{e_i}) = \emptyset$. In particular, contraction preserves the receiver’s epistemic stance.

The inequality is strict because every element of S_{ij} has its refinement to e_i ’s level disjoint from $F_{ij}([r])$ at each conflicting class $[r]$; their join therefore also avoids $F_{ij}([r])$; but $\mu_i([r]) \cap F_{ij}([r]) \neq \emptyset$ by definition of conflict, so the join cannot recover to u_i .

In the immigration example, with $|\text{Conf}(u_A, u_L)| = 3$ and threshold $\tau_A = 2$, the trigger fires. The baseline strips the conflicting meanings at each class: at r_1 , removing G (which is in $F_{AL}(r_1) \cap \mu_A(r_1)$) leaves $\{T\}$; at r_2 , removing F and B (both in $F_{AL}(r_2) \cap \mu_A(r_2) = \{F, B\}$) leaves $\emptyset$; at r_3 , removing B leaves $\{T\}$. The baseline is $\beta(u_A, u_L) = (\{T\}, \emptyset, \{T\})$: bare factual acknowledgments at r_1 and r_3 , and total semantic disengagement at r_2 . Voter A initially attached evaluative meanings to all three immigration representations; only the factual acknowledgments at r_1 and r_3 survive this contraction. The citizenship representation remains accessible and distinct, but has no attached meanings after this encounter. Evaluative meanings at the other two classes are also removed. At a subsequent encounter with this same internally consistent integration message, the remaining factual meanings no longer conflict, so the forward branch admits the content allowed by the credibility filter.

Lemma 3.7. *For any understandings u_i, u_j with $|\text{Conf}(u_i, u_j)| \leq \tau_i$ (the trigger does not fire):*

$$u_i \preceq_{\mathcal{U}} T_+(u_i, u_j).$$

The forward update is weakly above the input: encountering another understanding can only enrich, never impoverish, when the trigger does not fire.

3.4 Parameter dependence.

The previous subsection defined the update operator for fixed credibility and trigger threshold. In practice, these parameters vary across voters and across voter-source pairs: some voters are more receptive than others, some more tolerant of conflict. How does the update vary as the parameters change? For the fixed expansion rule and raw-message conflict test, more permissive credibility enriches the forward branch, and a higher threshold weakly favors expansion over contraction. The prior-content comparison below additionally holds the epistemic stance fixed.

⁸Retention subject to removal also motivates belief contraction. In partial meet contraction, [Alchourrón, Gärdenfors and Makinson \(1985, pp. 511–513\)](#) intersect selected maximal subsets of a belief set that fail to entail the proposition to be removed, when such subsets exist. Here each receiver class has a greatest subset of prior meanings avoiding the specified conflicting content. A formal connection would require specifying logical consequence for encoded meanings and checking the contraction postulates.

Lemma 3.8. *For fixed u_i, u_j and expansion rule Γ , if $\alpha_{ij}(D) \subseteq \alpha'_{ij}(D)$ for every visible sender class D , then $T_+(u_i, u_j; \alpha_{ij}) \preceq_{\mathcal{U}} T_+(u_i, u_j; \alpha'_{ij})$.*

Higher credibility produces richer forward updates: more of j 's meanings pass the filter and are absorbed.

Lemma 3.9. *For fixed α_{ij} , the output $T_{\alpha_{ij}, \tau}(u_i, u_j)$ is weakly increasing in τ .*

A lower threshold makes reversion more likely. Since $\beta(u_i, u_j) \preceq_{\mathcal{U}} u_i \preceq_{\mathcal{U}} T_+(u_i, u_j)$ (Proposition 3.6 and lemma 3.7), a more hair-trigger agent ends up at a weakly lower understanding than a more tolerant one.

Content removed and retained. When the trigger fires, the baseline removes exactly the prior meanings in conflict with the translated message. At a fixed stance, a larger prior can contain both more content exposed to removal and more content that survives. The following comparison identifies these two sets separately.

Lemma 3.10 (Prior content and conflict). *Fix a source understanding u_j and two priors $u_i = (e, \mu_i) \preceq_{\mathcal{U}} u_{i'} = (e, \mu_{i'})$ at the same stance. For each visible class C , put $L_i(C) = \mu_i(C) \cap F_{ij}(C)$, the content removed by $\beta(u_i, u_j)$, and define $L_{i'}$ analogously. Then*

$$\text{Conf}(u_i, u_j) \subseteq \text{Conf}(u_{i'}, u_j), \quad L_i(C) \subseteq L_{i'}(C), \quad \beta(u_i, u_j) \preceq_{\mathcal{U}} \beta(u_{i'}, u_j).$$

At a common threshold τ , if the trigger fires for u_i , it also fires for $u_{i'}$, and both contractions are strictly below their respective priors.

The trigger counts affected representation classes. Adding another conflicting meaning at an already affected class increases the content removed without increasing that count; adding only nonconflicting meanings leaves conflict exposure unchanged. In the immigration example, voter A holds $\{F, B\}$ at r_2 , and both meanings conflict with the integration message. A prior with only $\{B\}$ there loses one meaning, while A loses two, with the same one-class contribution to the trigger. The empirical comparison therefore requires distinguishing semantic content, affected classes, and the meanings retained after exposure.

Remark 3.11 (Source-dependent articulation). *The fixed expansion rule can be extended to model a source's ability to make distinctions available to a particular receiver. For example, let $\mathcal{D}(e_j)$ be a finite family of marked partitions $d \preceq_E e_j$ representing distinctions offered by the source. Assign each a receiver-specific acquisition threshold $\theta_i(d)$ and let κ_{ij} describe the source's effective articulatory influence. One possible rule is*

$$\Gamma_{ij}(e_i, e_j; \kappa_{ij}) = e_i \vee \bigvee \{d \in \mathcal{D}(e_j) : \kappa_{ij} \geq \theta_i(d)\}.$$

The empty join is $e_{\perp}$. Every output lies between e_i and $e_i \vee e_j$. If the offered partitions join to e_j and all pass the threshold, maximal expansion is recovered. Increasing κ_{ij} with the family and thresholds fixed refines the resulting epistemic stance. It need not increase the resulting understanding in the full order: finer reception can prevent associations from being pooled across categories.

This is a possible formalization of charisma through articulation, rather than an assumption of the base model or an established psychological law. It differs from admitting meanings through α_{ij} , tolerating conflict through τ_i , or evaluating the candidate favorably. Source-dependent tolerance and a direct candidate-evaluation term would be further, distinct extensions. Any results using this extension must specify its source parameters and the expansion held fixed in a comparison; candidate-independent credibility alone would no longer remove candidate identity from the update.

3.5 Special cases.

The general update operator has many moving parts: a conflict relation, a credibility filter, a trigger threshold, an epistemic expansion rule. To build intuition for what each part does—and to connect the operator to familiar models from the literature—it helps to examine what happens when specific parameters take extreme values. The cases below distinguish affect-only reception, full credibility, and an additional interpretation in terms of Bayesian conditioning. The first two restrict the operator’s parameters; the Bayesian comparison additionally replaces its output.

We partition M into semantically typed subsets: M_{policy} (factual and policy-implication meanings), M_{affect} (affective and evaluative meanings), and M_{social} (social-association and identity meanings). The partition is an analyst’s choice, not a structural feature of the framework; different partitions yield different special cases below and different Downsification conditions in Section 4.7.

Warm glow. Suppose $\alpha_{ij}([r]) \subseteq M_{\text{affect}} \cup M_{\text{social}}$ for all $[r]$ —agent i filters j ’s meanings down to affective and social-association content only—and suppose the epistemic expansion is minimal ($e'_i = e_i$). The forward update then adds only admitted affective and social-association meanings. On this branch, the voter can absorb emotional and social associations while retaining the prior factual content. The restriction is a modeling assumption that isolates one form of selective uptake.⁹ Consider a voter whose credibility filter admits only affective content encountering a policy-dense broadcast about net neutrality—bandwidth regulation, interconnection fees, common-carrier obligations. Suppose the broadcast also offers “this is about fairness” or “this is about freedom” as affective meanings at categories the voter already distinguishes. The technical content is filtered out; admitted affective meanings can attach to those categories. On the forward branch, the voter acquires admitted affective content about net neutrality but no new policy content, regardless of how much policy the broadcast contained. The raw-message conflict test can still select contraction when the broadcast’s factual claims conflict with the prior.

Policy campaign. With full credibility, every sender meaning at an overlapping category is admitted on the forward branch. At a common fixed stance this gives $\mu_i^+ = \mu_i \cup \mu_j$ pointwise. With maximal expansion, $e'_i = e_i \vee e_j$, Proposition 3.4 gives the join of the prior and the broadcast on the forward branch. Both newly acquired distinctions and admitted meanings can therefore enter in one encounter. This absorption identity does not equate an evaluation of the resulting understanding with an evaluation of the broadcast alone. If only policy content is to be admitted, replace full credibility by $\alpha_{ij}(D) = M_{\text{policy}}$ at every visible sender class. The electoral analysis must evaluate the resulting post-update understanding or supply additional assumptions justifying a reduction to direct evaluation of the broadcast.

A Bayesian comparison. A connection to conditioning requires an additional probability layer and a replacement rule. For example, at a common fixed epistemic stance, equip the visible classes with a probability distribution p_i and interpret a message as an event E with $p_i(E) > 0$. One may then stipulate

$$p_i^+(C) = \begin{cases} p_i(C)/p_i(E), & C \in E, \\ 0, & C \notin E. \end{cases}$$

⁹Lodge and Taber (2013, chap. 5) examine how implicit and explicit information shapes impressions of political candidates. Their processing question motivates studying affective influence; the content-exclusion rule here is a separate specification. Stokes (1963, pp. 373–374) distinguishes valence issues, which associate political actors with positively or negatively valued conditions, from competition over positions. A model of such competition also needs assumptions about common valuation and the assignment of credit or blame to candidates.

This is conditioning in the additional probability layer. It is not a consequence of the set-valued forward operator: union accumulates meanings and does not normalize a probability distribution or remove probability mass outside an event. The event interpretation and normalization must therefore be supplied separately before claiming a Bayesian special case.

Further comparisons with AGM contraction and public announcement logic require explicit encodings of their objects and operations. Set deletion in the baseline does not alone establish AGM postulates, and union of Boolean meanings does not alone implement world elimination. These comparisons remain questions for a separate formal treatment. A separate comparison concerns electoral behavior. [Little, Schnakenberg and Turner \(2022\)](#) show how their motivated-reasoning accountability model can be mapped to models with Bayesian voters and different preferences or information. For the present framework, a behavioral equivalence would need to specify the alternative model and match voting behavior and candidate incentives over the same feasible speeches and encounter protocol.

4 Semantic electoral competition and Downsification.

Political communication can give a voter words for an existing commitment, introduce a distinction, or alter the meanings she attaches to an issue. These are different effects, and approval need not track the amount of cognitive change. A familiar message can be affirming even when it leaves the voter’s understanding unchanged. This section specifies an electoral response that distinguishes recognition of a message from the valued change produced by the encounter. It then asks when the resulting voter rankings can be organized on a common policy axis.

4.1 The electoral setup.

The strategy space X is a nonempty finite set of *platforms*, without a pre-given order, topology, or metric. A broadcast specifies the distinctions and meanings offered in the particular encounter, rather than the candidate’s entire private understanding.

Definition 4.1. *A semantic election consists of*

$$\mathcal{E} = (C, V, X, \phi, \text{Cf}, \Gamma, \mathcal{W}, \{u_v^0, u_v^*, \lambda_v, \tau_v\}_{v \in V}, \{\alpha_{vc}\}_{(v,c) \in V \times C}).$$

Here C and V are nonempty finite sets of candidates and voters; $\phi : X \rightarrow \mathcal{U}_{\mathcal{R}}$ is the broadcast map; Cf and Γ are the conflict relation and fixed stance-only expansion rule of Section 3; u_v^0 is voter v ’s initial understanding and u_v^* her evaluative anchor; $\lambda_v \geq 0$ weights valued cognitive change; $\tau_v \in \mathbb{N} \cup \{\infty\}$ is her trigger threshold; and α_{vc} is her family of source credibility filters toward c . The common content valuation $\mathcal{W}$ is specified in Definition 4.3. Voting ties are treated as in Definition 4.8.

The evaluative anchor represents dispositions through which an encounter can be recognized as expressing something the voter values. It may include distinctions or meanings that she does not yet articulate in u_v^0 . It is not asserted to be a second consciously held belief state, an objectively correct understanding, or the best message a politician could deliver. The anchor is held fixed during the encounter; a theory of how evaluative dispositions themselves change would require another transition rule.

The evaluative anchor u_v^* is distinct from the appeal that best addresses a particular voter. That appeal will depend on her current understanding, the source, the available messages, and the

response rule. The phrase *that's what I've been trying to say* motivates this distinction: the voter can recognize an expression of something she values, whether the encounter supplies new content or articulates content already present.

The anchor and the response rule are reduced-form assumptions connecting cognition to evaluation. The lattice does not derive what a voter values, nor does it guarantee that a richer understanding is more accurate or politically desirable. Different anchors can be incomparable, but that fact alone does not establish the absence of a common policy axis. Their electoral implications depend on reception, evaluation, and the feasible broadcasts.

4.2 Campaign updating and voter response.

A *campaign* is a profile $(x_c)_{c \in C}$. Each candidate's broadcast is processed from the same initial state:

$$\hat{u}_v^c(x_c) = T_{\alpha_{vc}, \tau_v}(u_v^0, \phi(x_c)).$$

Write $\bar{e}_v^c(x_c)$ for its epistemic stance. On the forward branch this is $\Gamma(e_v^0, e_{x_c})$; on contraction it remains e_v^0 . The raw conflict test, source-class credibility filter, and expansion-before-reception timing are exactly those of Definition 3.3.

We also need the message as available at the stance the encounter actually leaves the voter able to use. For $\phi(x) = (e_x, \mu_x)$, let $\nu_{vc,x} = \alpha_{vc}^{e_x} \cap \mu_x$ on visible sender classes, with empty invisible semantics, and set

$$r_v^c(x) = (\bar{e}_v^c(x), \text{Rec}_{e_x \rightarrow \bar{e}_v^c(x)}(\nu_{vc,x})).$$

This *received message* contains the filtered message at that stance, without adjoining the voter's prior meanings. On the forward branch its meanings are absorbed into the refined prior. On contraction it represents accessible, credibility-filtered content that need not be retained. This distinction allows recognition of part of an encounter to coexist with a negative cognitive effect.

Remark 4.2 (Campaign priming). *The base election compares separate encounters from a fixed prior. Sequential exposure would instead apply the response rule at the state reached after the preceding message. That extension requires an exposure order and a rule for retaining or aggregating encounter evaluations. It can produce rankings dependent on the opponent's platform, so the platform-only comparisons below do not automatically apply to a primed campaign. For separately candidate-attributed public reactions, Section 5.1 specifies evaluation after a fixed number of rounds.*

4.3 Electoral evaluation and voting.

Definition 4.3. A content valuation is a common map $\mathcal{W} : \mathcal{U}_{\mathcal{R}} \rightarrow \mathbb{R}$ such that $u \prec_{\mathcal{U}} u'$ implies $\mathcal{W}(u) < \mathcal{W}(u')$. For voter v , write

$$F_v(z) = \mathcal{W}(z \wedge u_v^*).$$

The meet measures common ground under the understanding order. The requirement that richer common ground receive a higher value is a substantive restriction. It does not assign a penalty to every meaning outside the anchor. Such content can matter through conflict and updating, but the formula does not supply a general theory of aversion, truth, or message quality. The common $\mathcal{W}$ fixes a valuation scale; priors, anchors, filters, thresholds, and λ_v also contribute to voter heterogeneity. The differences in $\mathcal{W}$ used below have cardinal significance: the relative sizes of recognition and valued change can affect a voter's ranking of messages. A common positive affine transformation

of $\mathcal{W}$ scales every response by the same positive factor and preserves electoral comparisons; a general strictly increasing transformation need not. Empirical use of the response therefore requires information or assumptions about value differences as well as ordinal content preferences.

Lemma 4.4. *If $f(u, u')$ is a lower bound of u, u' and lies above every common lower bound, then $f(u, u') = u \wedge u'$.*

This is the defining property of the meet. It identifies the greatest shared understanding once that object is chosen as the basis of evaluation. It does not uniquely derive a scalar response function from general requirements of common ground and monotonicity.

For a stance e , write $u_{e,0} = (e, 0_e)$ for the understanding with empty semantics at that stance, as in Section 2. This is the bottom of its semantic fiber, not generally the bottom of the entire understanding lattice.

Definition 4.5. *The recognition and valued change from candidate c 's platform x are*

$$\begin{aligned} R_v^c(x) &= F_v(r_v^c(x)) - F_v(u_{\bar{e}_v^c(x),0}), \\ G_v^c(x) &= F_v(\hat{u}_v^c(x)) - F_v(u_v^0). \end{aligned}$$

The voter's encounter response is

$$s_v^c(x) = R_v^c(x) + \lambda_v G_v^c(x). \quad (1)$$

In the base campaign, $w_v(c, (x_{c'})_{c' \in C}) = s_v^c(x_c)$.

Recognition credits the meanings expressed by the received message, including meanings already held by the voter. Its reference point is an empty message at the same stance. Thus access to a distinction alone is not counted as semantic recognition; its value can enter through G_v^c . Valued change compares the resulting understanding with the prior on the same fixed evaluative scale. Subtracting prior value alone would not solve the recognition problem: whenever two encounters produce the same final understanding, their change terms still tie.

The coefficient on recognition is normalized to one, while λ_v determines the relative weight of change. Additive combination is a behavioral assumption, not a consequence of lattice theory. It permits recognition and contraction to offset one another; a rule giving the trigger an absolute evaluative veto would be a different specification.

Proposition 4.6. *Recognition is nonnegative and is zero for an empty received message. Valued change is nonnegative on the forward branch and nonpositive on contraction. Consequently the forward-branch response is nonnegative. A broadcast at the global bottom produces no update and has response zero. If $u_v^0 \succeq_{\mathcal{U}} u_v^*$, every forward encounter has $G_v^c(x) = 0$, but recognition can still distinguish the messages.*

Example 4.7 (Affirmation and learning). *Fix a stance with one visible representation and two possible meanings p, q . Let the prior contain $\{p\}$ and the anchor $\{p, q\}$. Assume full credibility, no conflicts, and $\lambda_v = 1$. Use cardinality valuation within this fiber, up to a constant. Then:*

Message	Resulting meanings	Recognition	Change	Response
$\emptyset$	$\{p\}$	0	0	0
$\{p\}$	$\{p\}$	1	0	1
$\{q\}$	$\{p, q\}$	1	1	2

The second message affirms without changing the state. The third adds valued content. Its higher score depends on the specified valuation and positive weight on change; learning is not assumed to outrank affirmation in every comparison. If the prior instead contains $\{q\}$, with the same anchor and other parameters, the rankings of messages $\{p\}$ and $\{q\}$ reverse. Unlike Example 3.5, this example adds a meaning at an already visible representation. In that earlier example, a newly accessible resource can support recognition in the same encounter because r_v^c is evaluated at the acquired stance.

For a given source, the best available appeal is a derived object:

$$A_v(c \mid u_v^0) = \arg \max_{x \in X} s_v^c(x).$$

It need not be unique, and the broadcast of a maximizing platform need not equal u_v^* . A change in the prior or in reception can change this set even when the anchor is fixed. This is the sense in which the best way to address a voter is conditional on her current understanding.

Charisma and direct candidate evaluation. The source-dependent acquisition gate of Remark 3.11 can affect both the accessible message and the resulting state. Its effect on response must be computed through those objects; more acquired distinctions do not imply more approval. An additional term z_{vc} could represent a direct evaluation of candidate c beyond the message’s cognitive effects. That would be a further extension of eq. (1), rather than an identification of charisma, credibility, and valence. Neither the acquisition extension nor z_{vc} is assumed in the base election.

Definition 4.8. Let $K_v = \arg \max_{c \in C} s_v^c(x_c)$. Voter v assigns probability $1/|K_v|$ to each candidate in K_v and zero to other candidates. Candidate c ’s expected vote share is

$$\mathcal{V}_c((x_{c'})_{c' \in C}) = \frac{1}{|V|} \sum_{v \in V} \frac{\mathbf{1}\{c \in K_v\}}{|K_v|}.$$

Uniform treatment of ties is an assumption about electoral incentives. It matters even with an odd electorate: voters can be indifferent between distinct platforms, and identical candidates tie in their evaluations. Candidates maximize expected vote share, which is a different objective from maximizing their probability of winning the election.

4.4 Candidate strategy and symmetric equilibrium.

Definition 4.9. A campaign is a symmetric semantic equilibrium at $x^* \in X$ if every candidate chooses x^* and no unilateral platform change raises that candidate’s expected vote share:

$$\mathcal{V}_c(x^*, (x^*)_{c' \neq c}) \geq \mathcal{V}_c(x, (x^*)_{c' \neq c}) \quad \text{for all } c \in C, x \in X.$$

The definition permits source-dependent responses. Even at identical platforms, candidates need not receive equal evaluations if their credibility differs. The more specific majority comparisons below require source neutrality.

4.5 Policy as organizational scheme.

Under candidate-independent credibility, $\alpha_{vc} = \alpha_v$, the base election gives a platform-only response $s_v(x)$. This follows because its other response parameters and its expansion rule do not depend on candidate identity. It removes the candidate label while retaining the prior, reception, and update. A source-dependent acquisition gate, tolerance, or direct candidate evaluation would require additional equal-source assumptions.

For comparison, define the *direct-broadcast benchmark*

$$D_v(x) = \mathcal{W}(\phi(x) \wedge u_v^*).$$

It evaluates the broadcast itself. The following gives precise cases in which it also supplies the encounter rankings.

Proposition 4.10. *Suppose the prior and every feasible broadcast have the same stance e , credibility is full, and every feasible encounter takes the forward branch. Then $s_v(x) = D_v(x) - F_v(u_{e,0})$ if $\lambda_v = 0$. If instead $u_v^0 = u_{e,0}$, then for any $\lambda_v \geq 0$,*

$$s_v(x) = (1 + \lambda_v)[D_v(x) - F_v(u_{e,0})].$$

In either case s_v and D_v induce exactly the same rankings and ties. The conditions may be imposed voter by voter.

Full credibility and maximal expansion alone give the forward join identity of Proposition 3.4; they do not establish either reduction above. At a fixed stance the recognition term already ranks direct broadcasts under full credibility, but a nonconstant change term can alter that ranking. With expansion, the received message is also translated to a possibly different stance. Results for D_v therefore require a stated bridge before they can be applied to s_v .

A *policy ordering* is a total order $\leq_X$ organizing the actual rankings s_v . For the strict single-peaked version used below, each voter has a unique feasible peak $p_v \in X$, with

$$\begin{aligned} x <_X y \leq_X p_v &\implies s_v(x) < s_v(y), \\ p_v \leq_X x <_X y &\implies s_v(x) > s_v(y). \end{aligned}$$

No restriction is imposed on comparisons across opposite sides of the peak. Neither a shared broadcast field nor incomparable anchors settles whether such an order exists.

Write $N(x, y) = |\{v : s_v(x) > s_v(y)\}|$. Platform x *strictly majority-defeats* y when $N(x, y) > |V|/2$, and let

$$W(x) = \{y \in X : N(x, y) > |V|/2\}.$$

A strict Condorcet winner defeats every distinct platform in this sense. With indifference, some pairs have no strict-majority winner, even when $|V|$ is odd.

Proposition 4.11. *In a source-neutral two-candidate election, (x^*, x^*) is an equilibrium if and only if*

$$N(y, x^*) \leq N(x^*, y) \quad \text{for every } y \in X.$$

Every strict Condorcet winner satisfies this condition, but the converse need not hold.

At identical platforms each candidate receives expected share $1/2$. After a deviation, indifferent voters split while strict supporters choose their preferred platform. This compares the two strict-support counts, not the deviator's support with half of the entire electorate. It is the relevant condition for the stipulated expected-vote-share objective.

Finite illustration. Suppose five voters all rank 3 first, 2 and 4 tied next, and 1 and 5 tied last. These rankings are strictly single-peaked on $1 <_X 2 <_X 3 <_X 4 <_X 5$. Their majority-wins counts are 0, 2, 4, 2, 0, as in Figure 2. The example displays a possible aggregate shape; inferring a common voter-level axis from that shape alone requires more information.

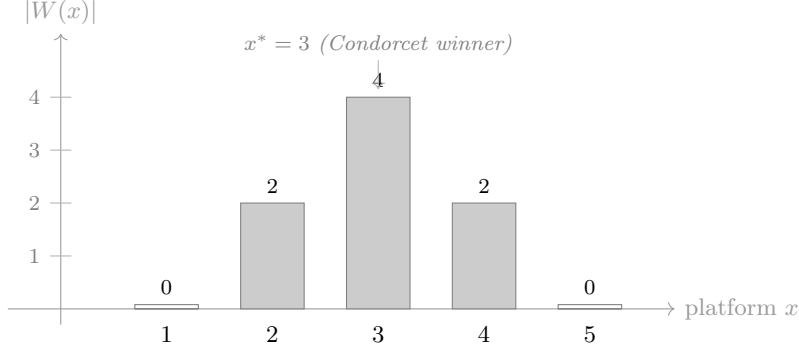


Figure 2: An illustrative majority-score profile. Each bar shows $|W(x)|$, the number of platforms that x defeats by strict majority. The displayed scores arise when every voter ranks 3 above 2 and 4, tied with one another, and both above 1 and 5, also tied. A mountain in these aggregate counts does not by itself establish a voter-level policy ordering.

4.6 The coalition mountain condition.

The coalition mountain condition concerns aggregate win counts. Its logical strength must be separated from the voter-level ordering sought above.

Definition 4.12. The coalition mountain condition (CMC) holds at x^* if some total order of X makes $|W(x)|$ weakly nondecreasing up to x^* and weakly nonincreasing thereafter, with $|W(x^*)| = |X| - 1$. The peak is permitted to be an endpoint of the order.

Definition 4.13. In a source-neutral two-candidate election with an odd number of voters, an equilibrium at x^* is Downsifiable if some total order makes the responses s_v strictly single-peaked and x^* is the median of their feasible peaks.

The definition uses encounter responses. Direct-broadcast scores may substitute only when their rankings and ties agree with those responses, for example under Proposition 4.10.

Theorem 4.14.

- (i) CMC holds at x^* if and only if x^* is a strict Condorcet winner. It is therefore necessary for Downsifiability but does not establish a common single-peaked ordering.
- (ii) If all voter responses have the same rankings as the direct-broadcast benchmark and x^* is a strict Condorcet winner, then $\phi(x^*)$ is maximal in $\phi(X)$.

For part (i), any strict Condorcet winner can be placed last after sorting the other platforms by their win counts. The score-order requirement thus adds no restriction to strict Condorcet existence. It cannot identify the policy dimension. For example, voters with rankings

$$d > a > b > c, \quad d > b > c > a, \quad d > c > a > b$$

all prefer d , yet their rankings restricted to $\{a, b, c\}$ form a majority cycle. No common single-peaked axis exists, whereas CMC holds at d .

Part (ii) states an order-theoretic conclusion under an additional ranking assumption. Maximal means that no available broadcast is strictly richer than the winner; it does not mean that the winner lies above every available broadcast. Incomparable alternatives remain possible. The conclusion

relies on the direct-broadcast rankings, under which increasing a broadcast cannot decrease its meet with any anchor. It is not a general consequence of the encounter rule.

For a counterexample, take a common stance with one visible representation, prior and anchor $\{p\}$, full credibility, $\lambda_v = 1$, and threshold zero. Let p and q conflict, and use cardinality valuation within this fiber, up to a constant. A broadcast $\{p\}$ earns recognition 1 and change 0, while $\{p, q\}$ earns recognition 1 but triggers loss of p , giving change -1 . The thinner broadcast is strictly preferred. If every voter has these parameters and these are the two feasible platforms, it is a strict Condorcet winner despite the richer alternative. The example also shows why candidate-independent credibility cannot justify dropping the update.

4.7 Conditional recovery of a policy ordering.

The following sufficient condition is stated directly on the encounter response profile. A policy projection is a map $q : X \rightarrow \mathbb{R}$ that separates feasible platforms. If a projection π on understandings is available, $q = \pi \circ \phi$ is one possibility. The existence of an arbitrary real-valued label is not itself a behavioral restriction on a finite menu; compatibility with voter rankings does the work.

Proposition 4.15. *Suppose the base election is source-neutral, has two candidates and an odd number of voters, and uses the tie rule of Definition 4.8. If an injective $q : X \rightarrow \mathbb{R}$ orders every s_v strictly single-peaked with a feasible peak $p_v \in X$, then the median peak x_{med} :*

- (i) *is a strict Condorcet winner and satisfies CMC;*
- (ii) *supports the symmetric equilibrium $(x_{\text{med}}, x_{\text{med}})$;*
- (iii) *is Downsifiable on the ordering induced by q .*

Feasibility is explicit: x_{med} is one of the voters' peak platforms. A median numerical projection of anchors need not be the image of any available platform, and injectivity alone cannot supply one. Here single-peakedness is an assumption on the actual encounter rankings, not a consequence of full credibility, maximal acquisition, or inactive triggers.

A policy-focused benchmark. A policy-focused application imposes four conditions:

1. All broadcasts carry only policy meanings, all voters admit those meanings from every candidate, and acquisition is maximal.
2. An injective policy projection q orders the feasible platforms.
3. The encounter responses s_v are strictly single-peaked on that order, with feasible peaks.
4. All triggers are inactive.

Together with the electoral and tie assumptions of Proposition 4.15, these conditions imply its conclusion. Conditions 1 and 4 specify a reception regime; Conditions 2 and 3 supply the common ordering and its behavioral compatibility. The theorem itself also applies to other reception regimes producing the required rankings. It does not derive single-peakedness from the semantic primitives.

Failure of a reception condition therefore does not establish failure of a policy ordering. Conversely, incomparable anchors, selective credibility, or a large meaning vocabulary do not on their own demonstrate multiple peaks. Example 5.25 specifies a complete immigration campaign with active triggers and compatible combined appeals. Its actual response rankings admit no common single-peaked axis, even allowing flat portions, although the campaign has pure equilibria. A general characterization of which semantic inputs generate a common single-peaked order remains open.

5 Semantic equilibrium beyond Downsification.

Section 4 states a conditional median result for the actual encounter responses. Its single-peakedness condition need not hold for an arbitrary broadcast menu. This section separates three questions: stability of communication, exact reduction of electoral responses, and the strategic consequences of heterogeneous reception.

The communication results below characterize fixed-message orbits with active triggers and give a sufficient condition for public reactions to settle. Section 5.2 identifies conditions under which a smaller message domain preserves electoral responses and strategic comparisons. The subsequent subsections consider how heterogeneous reception affects candidate incentives.

5.1 Communication and public reactions.

A political message can affect a voter directly and through the reactions it elicits from others. To study these effects, distinguish the circulation of a fixed stock of meanings from repeated encounters that can activate conflict, and from communication that reports an audience's response. A communication equilibrium is a profile left unchanged by the specified communication process. Such stability is compatible with disagreement.

Pooling a fixed stock of meanings. As a benchmark, suppose N participants share a stance e , triggers are inactive, and each receiver has a fixed filter α_i independent of the sender. For a semantic profile $\mu \in M_e^N$, set

$$G(C) = \bigcup_j \mu_j(C), \quad \mathbf{T}(\mu)_i(C) = \mu_i(C) \cup \left(\alpha_i(C) \cap \bigcup_{j \neq i} \mu_j(C) \right) = \mu_i(C) \cup (\alpha_i(C) \cap G(C)).$$

All operations are on visible classes; invisible semantics remain empty.

Proposition 5.1. *The pooling map $\mathbf{T}$ is monotone, extensive, and idempotent. It preserves G . Starting from μ , the profile $\mathbf{T}(\mu)$ is the least communication equilibrium above μ , reached after one simultaneous update.*

Proposition 5.2. *The fixed points of $\mathbf{T}$ form a complete lattice. Their least and greatest elements are respectively the all-empty and all-full visible semantic profiles, for every filter profile.*

The reached profile records which parts of the initial stock each participant can absorb. In the immigration example, A and B add each other's factual content, while C also acquires evaluative content from A and B. All eligible additions occur in the first simultaneous round. Disjoint receiver filters separate the kinds of content admitted; a sender may still possess meanings admitted by a receiver of another type. The initial profile and the filters therefore jointly determine the transmission pattern.

Repeated exposure with active triggers. Fix a broadcast $b = (f, \nu)$, a source-credibility family α , a threshold τ , and the expansion rule Γ of Definition 3.3. Repeated exposure follows

$$u^{t+1} = T_{\alpha, \tau}(u^t, b).$$

The same source and broadcast are used at every encounter. Both contraction and acquisition remain available.

First consider a stance e with $\Gamma(e, f) = e$. Write

$$B = \text{Rec}_{f \rightarrow e}(\nu), \quad A = \text{Rec}_{f \rightarrow e}(\alpha^f \cap \nu).$$

Thus B is raw reception, $A \subseteq B$ is admitted reception, and both remain fixed during repetition at e . At each visible class C , put

$$F(C) = \{m \in M : \text{some } n \in B(C) \text{ satisfies } (m, n) \in \text{Cf}\}, \quad d(P) = |\{C : P(C) \cap F(C) \neq \emptyset\}|.$$

For semantics P , the update is exactly

$$H(P) = \begin{cases} P \cup A, & d(P) \leq \tau, \\ P \setminus F, & d(P) > \tau. \end{cases}$$

Set $\kappa = d(A)$: the number of classes at which admitted content itself conflicts with the raw broadcast.

Lemma 5.3. *At the fixed stance, if $\kappa \leq \tau$, every orbit of H reaches a fixed point within three updates. If $\kappa > \tau$, H has no fixed point and every orbit enters a two-cycle within three updates. From initial semantics P , that cycle alternates between*

$$C_0 = (P \cup A) \setminus F \quad \text{and} \quad C_1 = C_0 \cup A.$$

The distinction is between conflict that contraction can remove permanently under this broadcast and conflict that later absorption restores. A trigger can fire on the way to a stable understanding. A periodic orbit occurs when the admitted content renews conflict above the receiver's tolerance.

Acquisition determines the stance at which this distinction eventually operates. Starting at $e^{(0)} = e^0$, iterate $e^{(k+1)} = \Gamma(e^{(k)}, f)$ until the sequence reaches a fixed stance e^∞ . Let h be the number of strict steps. The sequence is increasing in a finite epistemic lattice, so h is finite. Evaluate A, B, F , and κ at e^∞ .

Theorem 5.4. *Under repeated exposure to b , the receiver reaches stance e^∞ within $2h$ encounters. If $\kappa \leq \tau$, the full understanding reaches a fixed point within $2h + 3$ encounters. If $\kappa > \tau$, it enters a two-cycle within $2h + 3$ encounters. Thus every fixed-message orbit is eventually constant or has period two.*

The theorem permits arbitrary source filters, active thresholds, and every fixed stance-only expansion rule satisfying the bounds in Definition 3.3. The eventual stance depends on the expansion rule and the offered distinctions; triggers can delay acquisition. The semantic orbit is then governed by whether absorption renews excessive raw conflict.

Example 5.5 (A stable response and a periodic response). *At one visible class, let p and q conflict and set $\tau = 0$. Full credibility and the fixed broadcast $\{q\}$ give the orbit*

$$\{p\} \longrightarrow \emptyset \longrightarrow \{q\} \longrightarrow \{q\}.$$

The initial trigger fires, after which the message is absorbed and the understanding stabilizes. With the fixed broadcast $\{p, q\}$, the orbit instead contains

$$\emptyset \longrightarrow \{p, q\} \longrightarrow \emptyset \longrightarrow \{p, q\}.$$

Here $\kappa = 1 > \tau$. Such mixed reception can also arise when a receiver pools two sender categories, one carrying p and the other q , while acquiring no distinction between them.

Call a broadcast *internally consistent* if its meanings contain no conflicting pair within any visible sender class. This condition permits conflict with a receiver's prior. Maximal acquisition gives a stronger convergence bound for such broadcasts.

Corollary 5.6. *Suppose b is internally consistent and $\Gamma(e, f) = e \vee f$. For every initial understanding, credibility filter, and threshold,*

$$T_{\alpha, \tau}(T_{\alpha, \tau}(T_{\alpha, \tau}(u, b), b), b) = T_{\alpha, \tau}(T_{\alpha, \tau}(u, b), b).$$

Repeated exposure therefore stabilizes within two encounters, including when a trigger fires.

At the acquired stance, each received category lies within a sender category. The admitted content can consequently remain without regenerating conflict with the same raw broadcast. Distinguishing the sender's categories can therefore change a periodic response under coarse reception into a stable response.

Public reactions as further messages. A report about an audience's reaction supplies content beyond the initial speech. Specify a finite nonempty directed acyclic graph of observation dependencies. For each participant i , let $\text{pa}(i)$ be the participants whose current understandings determine the message available to i . Given a fixed speech b , a deterministic observation rule produces a broadcast

$$b_i^t = \Psi_i((u_j^t)_{j \in \text{pa}(i)}; b), \quad u_i^{t+1} = T_{\alpha_i, \tau_i}(u_i^t, b_i^t).$$

All participants update simultaneously from their current understandings. For a participant with no parents, Ψ_i supplies a fixed broadcast. An observation rule can encode a publicly expressed judgment and its attribution to the speaker; it is an additional primitive, not a property of union-pooling. Its outputs may depend nonmonotonically on the observed states.

Proposition 5.7. *Suppose every output of each Ψ_i is internally consistent and acquisition is maximal. Let d be the largest number of participants on a directed path in the observation graph. For arbitrary initial understandings, fixed credibility families, and thresholds, the communication profile reaches a fixed point within $2d$ simultaneous rounds.*

The proposition follows the causal order of observations. Once a participant's observed predecessors stabilize, her input is fixed and Corollary 5.6 gives stabilization within two further encounters. The terminal profile can depend on initial understandings. A graph containing recurrent observation loops requires its own orbit analysis; the acyclic result supplies a sufficient condition for settling.

Example 5.8 (An objection becomes part of an appeal). *Use one visible representation and meanings p, r, h , with only p and r conflicting. A speech offers p . An opposing listener starts with $\{r\}$, has anchor $\{r\}$, admits none of p , and has threshold zero. The first encounter contracts her semantics to $\emptyset$, which remains stable under repetition. A supporter starts with empty semantics, has anchor $\{p, h\}$, admits both p and h , and also has threshold zero.*

For this illustration, the observation rule reports h when the opposing listener no longer holds r . The meaning of h is a candidate-attributed public objection to this speech. The supporter receives p together with h when that report is present. The rule mapping the opponent's state to a public objection is a behavioral assumption. It does not identify every contraction with anger.

The synchronous profile evolves as

$$(\{r\}, \emptyset) \longrightarrow (\emptyset, \{p\}) \longrightarrow (\emptyset, \{p, h\}) \longrightarrow (\emptyset, \{p, h\}).$$

With cardinality valuation in the common fiber, the supporter gains one unit of valued content from the speech and another from the observed objection. Her anchor values the report of opposition to this speaker; she need not accept the opposing listener's judgment.

Evaluation at election time. To connect communication to electoral choice, fix an integer horizon $H \geq 1$, common to the candidates and specified by the electoral environment. For each candidate c , run a separately attributed communication process from the original priors. The first round is the direct speech encounter, $u_v^{c,1} = T_{\alpha_{vc}, \tau_v}(u_v^0, \phi(x_c))$; later rounds follow a specified protocol for observing and attributing reactions to that speech. Let $u_v^{c,H}$ be the understanding reached when voting occurs. The voter’s evaluation is

$$S_v^c(H) = R_v^c(x_c) + \lambda_v(F_v(u_v^{c,H}) - F_v(u_v^0)). \quad (2)$$

The initial-speech recognition term is held fixed; subsequent communication is evaluated through the net valued change present at the election date. At $H = 1$, this is the encounter response of Section 4.2. If the process stabilizes before voting, it evaluates the stable understanding; if the process cycles, its phase at H can affect the score. No convergence assumption is needed to evaluate a specified process at a finite horizon. In Example 5.8, taking $\lambda = 1$ gives the supporter score 2 at $H = 1$ and 3 at every $H \geq 2$. Voters compare these scores using the same uniform treatment of voting ties as in Definition 4.8. This defines electoral evaluation for a specified communication protocol; the base campaign is its single-encounter benchmark.¹⁰

5.2 Electoral relevance and feasible compression.

In the model of McKelvey (1976), where preferences depend on Euclidean distance, absence of a total median allows majority-improvement paths between any two alternatives. His more general analysis identifies restrictive conditions on the frontiers of reachable sets (McKelvey, 1979). These results make the conditions that restrict majority reachability a central question for models of competition.

The semantic framework also asks which features of voters’ understandings determine the domain needed to represent that competition. Voter anchors identify valued content, reception determines how a message engages it, and the feasible menu determines which messages candidates can offer. The results below give sufficient conditions for reducing the feasible menu while preserving electoral incentives and, under source neutrality, majority reachability between retained endpoints. Shared content supplies separate restrictions on symmetric equilibria.

Anchor width and electoral rankings.

Definition 5.9. *The anchor width of V , also called its semantic dimension, is the width of $(\{u_v^* : v \in V\}, \preceq_{\mathcal{U}})$: the largest number of pairwise incomparable anchors.*

Width one means that the anchors form a chain. Width is not a Euclidean dimension or a measure of how many axes are needed to represent voter rankings. The following algebraic obstruction concerns a particular kind of map on understandings.

Lemma 5.10. *If the anchor poset contains incomparable elements, there is no strictly increasing, meet-preserving map from $\mathcal{U}_{\mathcal{R}}$ into a real chain.*

For incomparable a, b , their meet is strictly below both, whereas meet preservation would give $\pi(a \wedge b) = \min\{\pi(a), \pi(b)\}$. This argument applies to any incomparable pair in the entire

¹⁰The update can already depend on the order of encounters, so evaluation at a fixed date does not imply path-independent dynamics. One could additionally make evaluation sensitive to the intervening history by replacing the endpoint change with $\sum_{t=1}^H w_{v,t}[F_v(u_v^{c,t}) - F_v(u_v^0)]$, where the weights are fixed across candidates, nonnegative, and sum to one. Such weights can represent duration or recency and distinguish histories reaching the same final understanding. Putting all weight on $t = H$ recovers (2).

understanding lattice, whether or not the pair consists of voter anchors. The global obstruction can therefore hold even when anchor width is one: two resources provide incomparable visibility states, and two meanings provide incomparable singleton-content states at a common visible stance. It is a general lattice observation rather than a diagnostic of variation across electorates. The lemma does not rule out a single-peaked ordering of feasible platforms. That ordering concerns voter responses, not a strictly increasing representation of the entire lattice. A map defined only on a chain of anchors would also not supply the required map on the whole understanding space.

Proposition 5.11. *Fix a finite nonempty voter set and a finite nonempty platform set. There is a finite semantic model with one visible representation and fixed, pairwise disjoint nonempty anchor meaning sets in which every profile of weak voter rankings on those platforms can be represented by changing the broadcast map. Priors have empty semantics, credibility is full, conflict is absent, and $\lambda_v = 1$.*

In the construction, each voter has a separate block of meaning tokens, and each broadcast contains a number of that voter's tokens equal to its rank score. With cardinality valuation, the encounter response is twice that score. Thus the unrestricted finite framework accommodates both common-axis profiles and majority cycles using the same anchors. Restrictions on the feasible messages, reception, or evaluation must do the work of ruling particular profiles out. This is a representational result; it supplies neither a mechanism for choosing the broadcasts nor an empirical explanation of observed rankings.

Feasible paths and the anchor interval. Let $L = \bigwedge_v u_v^*$ and $H = \bigvee_v u_v^*$. The interval $[L, H]$ is the smallest order interval containing all anchors, making it a natural candidate domain for electoral analysis. Its relation to majority paths depends on the available broadcasts. The following finite menu illustrates this dependence.

Example 5.12 (A majority path that must leave the interval). *Use one visible representation with anchor meanings*

$$A_1 = \{q, a_1, a_2\}, \quad A_2 = \{q, b_1, b_2\}, \quad A_3 = \{q, c_1, c_2, c_3\}.$$

The three available platforms have the following meanings. The numerical entries count shared meanings with each anchor:

Platform	Broadcast meanings	Voter 1	Voter 2	Voter 3
x	$\{q, a_1, a_2, c_1\}$	3	1	2
y	$\{q, a_1, b_1, b_2\}$	2	3	1
z	$\{b_1, b_2, c_1, c_2, c_3\}$	0	2	3

Take empty-semantic priors, full credibility, no conflicts, and $\lambda_v = 1$. Set $\mathcal{W}(e, \mu) = 1 + |\mu|$ at the visible stance and $\mathcal{W}(u_\perp) = 0$. This is strictly increasing on the full one-resource understanding lattice. Encounter scores are twice the displayed counts by Proposition 4.10.

The cycle is x defeats y , y defeats z , and z defeats x , each by two votes to one. Here L has meanings $\{q\}$ and H their union. Platforms x, y lie in $[L, H]$, while z lies below H and is incomparable with L . Starting at x , the majority improvements $x \rightarrow z \rightarrow y$ leave the interval. Only x, y are feasible inside it, and y does not defeat x . There is consequently no feasible majority-improvement path from x to y that stays inside the interval.

The example suggests a route to preserving majority comparisons: use available replacement messages that preserve the relevant voter evaluations. Membership in the anchor interval supplies neither score equivalence nor availability by itself. The next results construct score-preserving replacements and specify the menu condition under which they can serve as strategies.

Exact reduction for the direct-broadcast benchmark. For any understanding b , write $D_v[b] = \mathcal{W}(b \wedge u_v^*)$. This extends the direct-broadcast score to broadcasts that need not belong to the feasible menu.

Proposition 5.13. *Let $H = \bigvee_v u_v^*$ and define $Q_H(b) = b \wedge H$. Then $Q_H(b) \preceq_{\mathcal{U}} H$, Q_H is idempotent, and*

$$Q_H(b) \wedge u_v^* = b \wedge u_v^*, \quad D_v[Q_H(b)] = D_v[b]$$

for every voter and every broadcast. No distributivity or common-stance assumption is required.

Every benchmark score can thus be represented in the down-set below H : a broadcast and its clipped version are electorally equivalent under D_v . This domain includes z in Example 5.12, so it accommodates that example's majority cycle. Extending the reduction to the full encounter response requires accounting for content that influences reception and conflict.

Retaining content that can trigger a response. Write $s_v^c[b]$ for applying Definition 4.5 to broadcast b from source c , with the voter's initial state held fixed. Suppose priors, anchors, and broadcasts all have one common stance e . Then the update retains that stance, and both reception and meets within its semantic fiber are pointwise operations.

For a meaning set P , define its conflict neighbors by

$$\mathcal{O}(P) = \{m \in M : \text{some } p \in P \text{ satisfies } (p, m) \in \text{Cf}\}.$$

At each visible class D , retain the set

$$K(D) = \bigcup_{v,c} (\alpha_{vc}^e(D) \cap \mu_v^*(D)) \cup \bigcup_v \mathcal{O}(\mu_v^0(D)),$$

and put $K([\star]_e) = \emptyset$. The first union keeps content that some source can credibly express and some voter values. The second keeps content that can conflict with any voter's prior, whether or not it passes credibility.

Proposition 5.14. *Under the common-stance assumptions above, define $Q_K(e, \mu) = (e, \mu \cap K)$ pointwise. Then Q_K is idempotent and preserves each voter's recognition, valued change, and encounter response for every candidate:*

$$s_v^c[Q_K(b)] = s_v^c[b].$$

Credibility filters, trigger thresholds, and nonnegative change weights may differ across voters; credibility may also differ across candidates.

The retained set is sufficient, not necessarily minimal. It can include meanings absent from every anchor because such meanings still affect raw conflict. For example, with prior and anchor $\{p\}$, full credibility, $\lambda_v = 1$, threshold zero, and conflict between p and q , the message $\{p, q\}$ has recognition 1 and valued change -1 . Clipping it to the anchor union $\{p\}$ removes the trigger and raises its response from 0 to 1. The set K retains q and preserves the response.

This proposition concerns an election evaluated from fixed priors. It does not preserve every post-encounter meaning or an arbitrary sequence of future updates. The retained set must be reconsidered when priors change. With changing stances, meanings and distinctions can also affect transport and acquisition; the common-stance proof does not establish a reduction for that case.

From an exact representation to a feasible strategy menu. A compressed description of a message need not itself be an available platform. To reduce the game to a subset of its existing strategies, require closure under the relevant clipping operation.

Corollary 5.15. *Let Q be one of the idempotent reductions above, and suppose $Q(\phi(X)) \subseteq \phi(X)$. Use the direct-broadcast scores for Q_H , or the common-stance encounter scores for Q_K . Let $X_Q = \{x \in X : Q(\phi(x)) = \phi(x)\}$. There is a map $\rho : X \rightarrow X_Q$, fixing X_Q , such that:*

- (i) *replacing each candidate's platform x_c by $\rho(x_c)$ leaves every expected vote share unchanged;*
- (ii) *a profile in X_Q is a pure Nash equilibrium of the reduced game if and only if it is one of the full game, and the image under ρ of any full-game equilibrium is an equilibrium;*
- (iii) *in the source-neutral case, each strict majority-improvement step maps to a strict step, preserving reachability between endpoints in X_Q .*

For Q_H , the conclusions also apply to encounter scores whose rankings and ties agree with the direct benchmark on the feasible menu.

The closure assumption supplies actual replacement strategies. Without it, Propositions 5.13 and 5.14 still provide exact score representations in their stated models, but do not authorize adding missing platforms to the original game. The corollary identifies a feasible domain in which majority reachability can be studied without losing comparisons between its alternatives. The original paths may leave this domain, while their projections preserve each strict majority-improvement step. Conditions that further restrict reachability must constrain majority comparisons within the retained domain or the available sequences of proposals.

Common ground and unanimous improvements. The lower endpoint L identifies shared valued content. Completing a message with that content can improve every voter's benchmark score and yield restrictions on symmetric equilibria.

Proposition 5.16. *Let $L = \bigwedge_v u_v^*$, $H = \bigvee_v u_v^*$, and $P(b) = (b \vee L) \wedge H$. Then $P(b) \in [L, H]$ and $D_v[P(b)] \geq D_v[b]$ for every voter. If $L \not\leq_{\mathcal{U}} b$, all these inequalities are strict. In a source-neutral election with at least two candidates, direct-broadcast rankings, uniform voting ties, and a feasible platform broadcasting $P(\phi(x))$, such an x cannot support a symmetric equilibrium.*

Feasible completion excludes messages missing shared valued content from symmetric equilibrium in this benchmark. It operates through a unanimous improvement, whereas Corollary 5.15 operates through equality of evaluations; the two results therefore restrict competition in different ways. If H itself is feasible, it gives every voter maximal direct-broadcast value, regardless of anchor width. It consequently belongs to the ordinary strict-majority core and supports a symmetric equilibrium under the uniform voting tie rule. A model of competition over a full feasible interval $[L, H]$ therefore includes a jointly satisfying endpoint. Explaining electoral trade-offs in this benchmark requires specifying what limits the availability of such messages.

5.3 Strategic polysemy and combined appeals.

Candidates can offer one speech that different audiences hear differently. The strategic question is when combining appeals attracts additional votes and when the resulting reception costs support. Feasible co-articulation is a property of the message menu; it does not require every audience to approve the combination.

Fix a communication protocol and an election horizon H . Write $\sigma_v^c(x) = S_v^c(H)$ for voter v 's score when candidate c chooses x . As in Equation (2), each candidate-attributed process starts from the original priors and is evaluated separately. Thus changing c 's speech leaves rivals' scores unchanged. Source neutrality here requires the same reception parameters and observation rules for either candidate, so $\sigma_v^c(x) = \sigma_v(x)$. The following result applies to the resulting scores, including active triggers and public reactions.

From responses to votes. For fixed rival speeches, let m_v be voter v 's highest rival score and k_v the number of rivals attaining it. The probability that she votes for c when c 's score is z is

$$g_v(z) = \begin{cases} 0, & z < m_v, \\ 1/(k_v + 1), & z = m_v, \\ 1, & z > m_v. \end{cases}$$

Proposition 5.17. *In a race with at least two candidates and uniform voting ties, suppose replacing x by y weakly raises c 's score for every voter while rivals' scores stay fixed. Then c 's expected vote share weakly increases. The increase is strict if and only if some voter satisfies*

$$\sigma_v^c(x) \leq m_v \leq \sigma_v^c(y) \quad \text{and} \quad \sigma_v^c(x) < \sigma_v^c(y).$$

For arbitrary x, y in a source-neutral two-candidate race, put $N_H(y, x) = |\{v : \sigma_v(y) > \sigma_v(x)\}|$. A deviation from the symmetric profile (x, x) to y changes expected vote share by

$$\frac{N_H(y, x) - N_H(x, y)}{2|V|}.$$

Improving a response matters electorally when it changes a voter's allocation between the candidates. A speech that already wins a voter can become more appreciated without gaining an additional vote. At a symmetric profile in the source-neutral two-candidate game, profitable deviations are determined by the balance of strict supporters and strict opponents of the alternative.

Combining content in one encounter. At a common stance, write $b \cup d$ for the broadcast formed by pointwise union of their meanings.

Lemma 5.18. *Suppose priors, anchors, and broadcasts share a stance, and b , d , and $b \cup d$ are feasible broadcasts. For any voter whose encounter with $b \cup d$ takes the forward branch, the encounters with b and d also take that branch, and*

$$s_v^c[b \cup d] \geq s_v^c[b], \quad s_v^c[b \cup d] \geq s_v^c[d].$$

If this holds for every voter, replacing either component speech by the combination weakly increases the candidate's expected vote share at $H = 1$.

The threshold need not be infinite: a voter may tolerate some raw conflict. The condition concerns the actual combined encounter. For later evaluation dates, the candidate must also account for subsequent updates and public reactions.

Example 5.19 (A feasible combination can lose support). *There are two source-neutral candidates, two voters of type A, and one of type B. All agents share one visible class. Meanings are p, q, r , thresholds are zero, change weights are one, and valuation is one plus cardinality in this fiber, with value zero at the global bottom. Type A has prior $\{r\}$, anchor $\{r, p\}$, and credibility $\{p\}$. Type B has empty prior, anchor $\{q\}$, and credibility $\{q\}$. The feasible speeches are $\{p\}$, $\{q\}$, and $\{p, q\}$. Each is repeated twice, and voting occurs at $H = 2$.*

The table compares the same menu with no conflict and with only r and q conflicting:

Speech	No conflict		r conflicts with q	
	Type A	Type B	Type A	Type B
$\{p\}$	2	0	2	0
$\{q\}$	0	2	-1	2
$\{p, q\}$	2	2	1	2

Without conflict, both candidates choosing $\{p, q\}$ is the unique pure equilibrium. With the specified conflict, the unique pure equilibrium has both choose $\{p\}$. The combined speech triggers removal of A's valued r ; a second encounter absorbs p and stabilizes, but A prefers the speech that preserves r while adding p . The two A voters give that speech a strict majority against either alternative. The combination remains feasible and internally consistent. Its electoral cost arises from its interaction with the prior.

When audience components can be chosen independently. At a common stance, partition voters into nonempty types $T_1, \dots, T_k$ sharing credibility filters $\alpha_1, \dots, \alpha_k$. Call a broadcast *polysemic* for these types when at least two types receive different nonempty filtered meaning assignments. This concerns reception; a trigger may prevent the received content from being retained in that encounter. With pairwise disjoint filters, nonempty receipts for different types occupy separate meaning domains.

A sufficient condition for independent design must also separate the sources of conflict and supply feasible combinations. For each type let X_i be a nonempty finite menu of semantic assignments $z_i \subseteq \alpha_i$ pointwise. Suppose the full feasible broadcast menu is parametrized by $X_1 \times \dots \times X_k$, with broadcast semantics $\bigcup_i z_i$. Candidates choose all components of this one broadcast.

Proposition 5.20. *Suppose all stances are common, filters are pairwise disjoint and source-neutral, and the feasible menu has the product form just specified. For every visible class C , assume that*

- (i) $\mu_v^0(C) \subseteq \alpha_i(C)$ for each $v \in T_i$;
- (ii) no meaning in $\alpha_i(C)$ conflicts with a meaning in $\alpha_j(C)$ when $i \neq j$.

Repeat each candidate's fixed broadcast for $H \geq 1$ encounters and use election-time evaluation. Then a type- i voter's trajectory and score depend only on z_i . For fixed rival strategies, candidate c 's payoff separates as

$$\mathcal{V}_c(\mathbf{z}; \mathbf{x}_{-c}) = \sum_i \mathcal{V}_{c,i}(z_i; \mathbf{x}_{-c}),$$

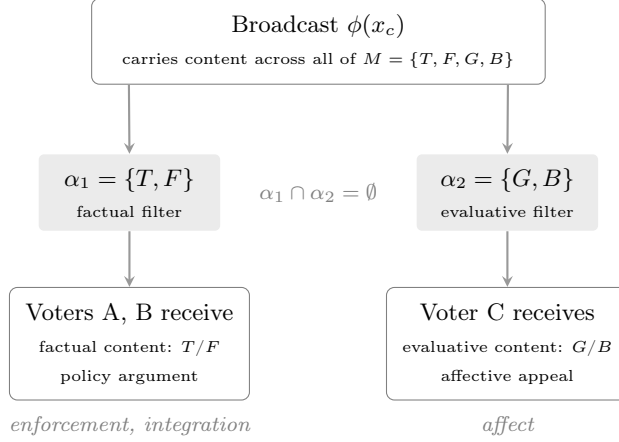


Figure 3: *Different reception of one broadcast in the immigration illustration. At the common stance, disjoint credibility filters select factual content for A and B and evaluative content for C. These are filtered receipts; retention also depends on conflict and updating. Independent strategic choice additionally requires the feasibility and separation conditions of Proposition 5.20.*

where $\mathcal{V}_{c,i}$ is the contribution of type i to total expected vote share. Consequently the set of best responses is

$$\prod_i \arg \max_{z_i \in X_i} \mathcal{V}_{c,i}(z_i; \mathbf{x}_{-c}).$$

Thresholds may be active, and repetition need not converge before H .

The optimized quantity is the votes supplied by the type. Voters within a type may disagree about which component is best. Public reactions that connect types require a new check of separation, even if the direct credibility filters remain disjoint.

Corollary 5.21. *Under Proposition 5.20, suppose each X_i contains a component z_i^* maximizing the score of every voter in T_i . In a race with at least two candidates, all candidates choosing $\mathbf{z}^* = (z_1^*, \dots, z_k^*)$ is a pure equilibrium. If at least two of the z_i^* have nonempty semantics, this equilibrium broadcast is polysemic.*

These conditions provide a feasible speech that jointly maximizes voter responses. Other menus can support trade-offs within or across audiences, as in Example 5.19. Incomparability of anchors alone does not decide which case obtains.

The immigration illustration separates factual receipts for A and B from evaluative receipts for C (Figure 3). It identifies different reception of the same broadcast. To use those audiences in a strategic election, one must additionally specify their priors, available messages, conflict, evaluation horizon, and vote-share comparisons. Membership in a shared filter type does not make A and B's preferences identical.

An electoral incentive to elicit opposition. Public reactions can make a speech more attractive to one audience precisely because another audience objects. The following election isolates that mechanism while allowing both component speeches and their combination.

Example 5.22 (A public objection changes the equilibrium speech). *There are two source-neutral candidates, two supporters, and one opposing listener. All share one visible class and have threshold*

zero and change weight one. The supporter's prior, anchor, and credibility are respectively

$$\{s\}, \quad \{s, a, p, h\}, \quad \{a, p, h\};$$

for the opposing listener they are

$$\{r\}, \quad \{r, q\}, \quad \{q\}.$$

Only p conflicts with s and with r , symmetrically. Use $\mathcal{W}(e, \mu) = 1 + \sum_{m \in \mu} \omega_m$ in the visible fiber and $\mathcal{W}(\perp) = 0$, with

$$\omega_s = \omega_a = \omega_r = 3, \quad \omega_p = \omega_q = 1, \quad \omega_h = \eta > 0.$$

The feasible speeches are a calm appeal x_0 broadcasting $\{a, q\}$, a provocative component x_1 broadcasting $\{p\}$, and their combination x_2 broadcasting $\{a, p, q\}$. All are internally consistent, and the menu is closed under combination. The token h is an observed public objection to the speech; it is generated by the observation rule and is absent from the campaign menu.

The first round is the direct speech encounter. Subsequently the opposing listener receives the same broadcast, while supporters receive that broadcast together with h whenever the opposing listener no longer holds r . For this example, losing r produces an expressed objection that supporters observe and attribute to the candidate's speech. This is a stipulated behavioral and attribution rule. Each candidate's process runs separately from the original priors.

At $H = 1$, the combined speech gives each supporter score 1 and the opposing listener score -2 . The speech therefore elicits a positive initial supporter response and a negative opposing response. By the second round, supporters have absorbed the appeal and its reported objection. Voting at $H = 2$ gives:

Speech	Report withheld		Report observed	
	Supporter	Opponent	Supporter	Opponent
$x_0 : \{a, q\}$	6	2	6	2
$x_1 : \{p\}$	-1	-3	$\eta - 1$	-3
$x_2 : \{a, p, q\}$	5	-1	$5 + \eta$	-1

All these processes are stable from round two onward. With the report observed, x_2 beats x_1 unanimously; supporters prefer x_2 to x_0 exactly when $\eta > 1$, while the opposing listener always prefers x_0 to x_2 . With the report withheld, (x_0, x_0) is the unique pure equilibrium. With the report observed, it remains the unique pure equilibrium for $0 < \eta \leq 1$; for $\eta > 1$, the unique pure equilibrium is (x_2, x_2) . In each case the selected speech earns more than half the votes against every other menu item, making its symmetric profile the unique pure equilibrium of this two-candidate constant-sum game. For example, at $\eta = 2$, switching from x_0 to x_2 against a rival using x_0 earns vote share $2/3$ with the report, compared with zero when it is withheld. The symmetric profile initially gives each candidate $1/2$.

The observation raises the supporter's valuation of the candidate's speech without restoring the opposing listener's valued prior. A sufficiently valued objection tips the majority's choice and makes the combined provocative appeal electorally advantageous. The conclusion depends on the audience sizes, menu, and observation rule, as well as on what supporters value in the objection.

5.4 Majority incentives and minority choice.

An audience's size matters through its actual rankings of feasible speeches. To isolate this effect, consider a source-neutral two-candidate race with scores $\sigma_v(x)$ from a specified protocol and horizon. Let A be a strict majority of voters whose sets of maximizing platforms coincide:

$$X_A = \arg \max_{x \in X} \sigma_v(x) \quad \text{for every } v \in A, \quad |A| > |V|/2.$$

The finite menu makes X_A nonempty. The voters in A may disagree about the ranking of platforms outside X_A . Put $B = V \setminus A$ and $N_B(y, x) = |\{v \in B : \sigma_v(y) > \sigma_v(x)\}|$.

Proposition 5.23. *Every symmetric equilibrium belongs to X_A . For $x \in X_A$, the profile (x, x) is an equilibrium of the full game if and only if*

$$N_B(y, x) \leq N_B(x, y) \quad \text{for every } y \in X_A.$$

When both candidates choose within X_A , the majority splits equally, and all differences in their expected vote shares come from B .

Thus a majority can exclude speeches it ranks below its common maximum while leaving the minority to choose among equally satisfactory alternatives. If the minority's comparisons cycle within X_A , a symmetric equilibrium need not exist. In Example 5.22, $\eta = 1$ makes supporters indifferent between x_0 and x_2 ; the opposing listener's preference selects x_0 . For $\eta > 1$, the supporters' maximizing set contains only x_2 , and their strict majority sustains that speech.

The direct-anchor benchmark. Anchor containment identifies a majority's maximizing set under direct-broadcast rankings. Its implication for minority value is conditional on the selected broadcast.

Corollary 5.24. *Suppose a strict majority has anchor u_A^* , all remaining voters have anchor u_B^* , the race is source-neutral with two candidates and uniform voting ties, and rankings and ties agree with $D_v(x) = \mathcal{W}(\phi(x) \wedge u_v^*)$. If some feasible broadcast contains u_A^* , every symmetric equilibrium broadcast b contains u_A^* . Its minority value deficit satisfies*

$$0 \leq \mathcal{W}(u_B^*) - \mathcal{W}(b \wedge u_B^*) \leq \mathcal{W}(u_B^*) - \mathcal{W}(u_A^* \wedge u_B^*).$$

The deficit is positive exactly when b does not contain u_B^ . If a feasible broadcast contains $u_A^* \vee u_B^*$, it gives every voter maximal direct-broadcast value and supports a symmetric equilibrium with zero deficit.*

Incomparability permits distinct valued content without making exclusion inevitable. When the joint appeal is available in this benchmark, it satisfies both types. Under active conflict or public reactions, the maximizing set must instead be calculated from the actual responses.

The immigration election. The running example connects these incentives to the content voters admit and the prior meanings a speech puts at risk.

Example 5.25 (Equilibrium without a common policy axis). *There are two source-neutral candidates and one voter of each type A , B , and C . All priors, anchors, and broadcasts use the common fully distinguishing stance on $\{r_1, r_2, r_3\}$. For this election, each prior equals its evaluative anchor:*

$$\mu_A^0 = \mu_A^* = (\{T, G\}, \{F, B\}, \{T, B\}), \quad \mu_B^0 = \mu_B^* = (\{T, B\}, \{T, G\}, \{T, G\}),$$

$$\mu_C^0 = \mu_C^* = (\{B\}, \{G\}, \emptyset).$$

At every sender class, A and B admit $\{T, F\}$ and C admits $\{G, B\}$ from either candidate. Use the conflict pairs T/F and G/B in both directions, $\tau_v = 2$, and $\lambda_v = 1$ for every voter. Take

$$\mathcal{W}(e, \mu) = |\mathcal{R}_e / \sim_e| + \sum_{D \in \mathcal{R}_e / \sim_e} |\mu(D)|.$$

This valuation is strictly increasing: strict refinement increases the number of visible classes without reducing inherited content, and strict semantic inclusion at a fixed stance increases the sum. The class-count term is three throughout this election and cancels from the response differences. Acquisition is maximal. Each candidate's process starts separately from the original priors and repeats its fixed speech, with voting at $H = 2$ and uniform splitting of ties.

The available components are two factual accounts and an evaluative appeal:

$$f_A = (\{T\}, \{F\}, \{T\}), \quad f_B = (\{T\}, \{T\}, \{T\}), \quad g = (\{B\}, \{G\}, \{G\}).$$

The factual accounts disagree about paths to citizenship; g describes enforcement as undesirable and citizenship and immigration's economic impact as desirable. The menu is

$$X = \{f_A, f_B, g, f_A \vee g, f_B \vee g\},$$

with each displayed understanding its own broadcast. All five speeches are internally coherent. The menu includes every coherent combination of its members; the two conflicting factual accounts are not offered together. Thus the evaluative appeal can accompany either factual account, whether or not a voter welcomes the combination.

Each entry below reports initial recognition R , the net valued change ΔF from the original prior to round two, and the election score $S(2) = R + \Delta F$:

	A	B	C
Speech	$(R, \Delta F, S(2))$	$(R, \Delta F, S(2))$	$(R, \Delta F, S(2))$
f_A	(3, 0, 3)	(2, 0, 2)	(0, 0, 0)
f_B	(2, 0, 2)	(3, 0, 3)	(0, 0, 0)
g	(0, -3, -3)	(0, 0, 0)	(2, 0, 2)
$f_A \vee g$	(3, -3, 0)	(2, 0, 2)	(2, 0, 2)
$f_B \vee g$	(2, -4, -2)	(3, 0, 3)	(2, 0, 2)

Only A 's encounters with g and the two combinations trigger contraction. For instance, $f_B \vee g$ conflicts with A 's prior at all three resources, exceeding her threshold. Her semantics follow

$$(\{T, G\}, \{F, B\}, \{T, B\}) \mapsto (\{T\}, \emptyset, \{T\}) \mapsto (\{T\}, \{T\}, \{T\}).$$

She recognizes two factual meanings from her anchor but loses four valued prior meanings, giving score $2 - 4 = -2$. The second encounter adds T at r_2 outside her anchor. Every process is fixed by round two, and the scores in this example are the same at every positive horizon.

B maximizes at f_B and $f_B \vee g$; C maximizes at g and both combinations. Their rankings therefore agree on some comparisons but differ on others. Against f_A , switching to $f_B \vee g$ wins B and C and earns 2/3 of the vote. Against f_B , the same combination loses A , ties B , and wins C , earning 1/2.

The complete pure-equilibrium set is $K \times K$, where

$$K = \{f_B, f_B \vee g\}.$$

Each member of K earns at least half the vote against every menu item, and they tie one another. Every speech outside K earns less than half against some member of K : f_A loses to $f_B \vee g$, while g and $f_A \vee g$ lose to f_B . In this two-candidate constant-sum game, each candidate can guarantee half by choosing in K , so no other pure strategy can occur in equilibrium. The four equilibria include candidates making different speeches and voters who remain dissatisfied with the opposing appeal. ¹¹

Nevertheless no common single-peaked axis exists. Consider just f_B , $f_A \vee g$, and $f_B \vee g$. Among these, A strictly ranks $f_B \vee g$ last, B strictly ranks $f_A \vee g$ last, and C strictly ranks f_B last. Any total axis puts one of the three between the others, creating a strict valley for the voter who ranks that middle speech last. The obstruction therefore persists even if flat portions of a single-peaked response are allowed.

In this election, selective reception and active conflict generate rankings that resist a common policy axis while permitting strategic stability. The value of adding the evaluative appeal depends on the rival speech and the votes gained and lost.

Shared content and bargaining. The meet $L = \bigwedge_v u_v^*$ identifies shared valued content. Its role in Proposition 5.16 is to support a unanimous direct-score improvement when adding missing shared content is feasible. The meet need not itself be an equilibrium or an inoffensive message under raw conflict with current priors. In the immigration illustration, the three-way meet has empty semantics at the common full stance, so it has no semantic recognition value in that fixed-stance comparison. Legislative bargaining can impose a shared-content requirement as a separate modeling assumption; Section 7.1 states the static benchmark and the additional structure needed for a sequential game.

The candidate’s strategic problem is therefore to choose among feasible speeches by their induced vote shares. Separate reception, conflict across audiences, and the interpretation of public reactions each supply conditions under which combining appeals succeeds or fails.

6 Empirical predictions and operationalization.

The framework separates what voters can represent, what they credibly receive, and how encounters affect candidate evaluation. The following subsections consider empirical implications and measurement strategies. Testing these mechanisms requires identifying the relevant messages, priors, and observation conditions as well as the resulting responses.

6.1 Conflict exposure and delayed uptake.

Which prior content makes a message disruptive, and when does uptake occur? The update supplies separate predictions for conflict exposure, content removal, and subsequent admission. At a fixed stance, Lemma 3.10 compares nested priors against the same message: adding prior meanings weakly enlarges the set of classes at which conflict occurs. With a common threshold, a message that triggers contraction from the smaller prior also triggers it from the larger one. The larger prior loses weakly more meanings under contraction, while its retained content is also weakly larger. These are comparisons of specified contents and categories. Additional meanings need not add an affected class, so the comparison need not be strict.

¹¹Raising every threshold to three makes every encounter forward on this common three-class stance. A’s scores become (3, 2, 0, 3, 2) in table order, while B and C’s scores stay unchanged. The pure-equilibrium set becomes $K' \times K'$ with $K' = \{f_A \vee g, f_B \vee g\}$. Tolerance changes which combinations are sustainable even though the menu and evaluative anchors are fixed.

Following the encounter over time. Initial non-admission can be followed by uptake once conflicting prior content has been removed. For a minimal example, take one common visible class, prior $\{p\}$, repeated message $\{q\}$, and only p conflicting with q . The following trajectories hold with fixed credibility and threshold:

Admitted message	Threshold τ	After one encounter	After two encounters
$\{q\}$	0	$\emptyset$	$\{q\}$
$\emptyset$	0	$\emptyset$	$\emptyset$
$\{q\}$	1	$\{p, q\}$	$\{p, q\}$

All three paths are stable from the second encounter onward. The first row exhibits delayed uptake. In the second, the raw message removes the prior meaning even though the filter blocks admission of q . In the third, the threshold tolerates the one conflicting class and permits coexistence. The receiver retains the same representation and distinction throughout. These are instances of the internally consistent-message result in Corollary 5.6; they show why measuring reception only after one encounter can miss later admission.

The immigration example has the same temporal structure. Voter A’s first encounter at $\tau_A = 2$ removes the conflicting evaluations and the factual denial at r_2 , leaving $(\{T\}, \emptyset, \{T\})$. A second encounter with the same integration message takes the forward branch: full credibility admits its factual and evaluative content, while factual-only credibility admits its factual content. The model thus gives predictions about the content present at each encounter and the role of selective admission in the eventual state.

Content loss and candidate evaluation. The electoral consequences depend on which removed meanings the voter values. On contraction, the valued-change term G_v^c is nonpositive by Proposition 4.6. At a common stance, it is strictly negative exactly when some removed prior meaning also belongs to the evaluative anchor at that class. Removing unvalued content can therefore leave this term unchanged. The total encounter response also includes recognition of the message; a loss of valued prior content can coexist with a positive candidate score, as in Example 5.22. Voting additionally depends on the rival comparison. Measuring cognitive loss, candidate evaluation, and choice separately makes these links assessable.

Measuring exposure, retention, and uptake. A possible design elicits the representations respondents can distinguish and the meanings they attach to them before exposure, then varies message content at a fixed source. The one-encounter prediction concerns which prior meanings disappear and which incoming meanings appear. Repeated exposure to the same message tests whether initial removal is followed by admission, persistent non-admission, or coexistence. The study would also need measures of which content respondents value and how they evaluate the candidate.

The nested-prior comparison requires evidence about containment of meanings at aligned categories and a common threshold, or a model of threshold heterogeneity. A knowledge score alone does not establish those relationships. Likewise, factual recall and open-ended responses can inform a measurement model for access, distinctions, and meanings, but each must be connected to its intended primitive. Conflict should be measured or specified independently of the observed contraction. Otherwise, classifying every removed meaning as conflicting would make the proposed test circular.

Issue-position extremity and attitude strength are distinct empirical outcomes. Taber and Lodge (2006, pp. 758, 765–766) measure them separately; their pooled analyses report polarization toward more extreme positions, especially among participants with greater political knowledge, stronger priors, or more biased processing. These participant characteristics were measured rather than

randomly assigned. Testing a connection to those findings requires a measurement rule linking retained meanings to issue positions and commitment ratings. The set-valued state itself records which associations are present. An account of newly generated counterarguments would additionally need a rule supplying content beyond the prior and the received message. Whether these cognitive patterns improve or weaken a proximity model’s account of voting can then be investigated using measured candidate rankings and choices.

6.2 Combined appeals and observed reactions.

The strategic analysis suggests two distinct comparisons. The first asks whether adding an appeal changes how another audience receives and evaluates the speech. The second asks whether observing an audience’s response changes another audience’s evaluation of the candidate. These comparisons distinguish direct reception from effects transmitted through public reactions.

A possible design presents matched component speeches and their combination from the same source. Measure which meanings each audience receives, what changes in its understanding, its candidate evaluation, and its voting preference against a fixed rival. The last comparison matters because a higher evaluation need not change a vote (Proposition 5.17). The forward-combination result gives a qualified prediction at the first encounter: when the combined message passes the trigger, it weakly improves the modeled response relative to either component. Longer-horizon comparisons require observing subsequent communication.

Disjoint filtered receipts supply evidence about reception separation. They do not establish independent strategic design: Proposition 5.20 additionally assumes a product menu and isolation of the meanings that can trigger each audience. The conflict example (Example 5.19) identifies why the latter matters: content outside a voter’s credibility filter can still affect her response through raw conflict. The immigration illustration similarly motivates measuring factual and evaluative receipts separately without inferring that one message component can be optimized independently of the others.

Observing opposition. The public-objection election (Example 5.22) suggests a second intervention: hold the candidate’s speech fixed and vary whether an audience observes an opposing audience’s objection. Measure candidate evaluation before and after the report, together with whether respondents understood the report and attributed it to that speech. Under the example’s reception and valuation assumptions, observing the objection increases supporters’ scores. More generally, holding the initial speech encounter fixed also fixes recognition, so the report’s effect is λ_v times the difference in anchored value between the report and withholding conditions at the election horizon (Equation (2)). For $\lambda_v > 0$, the effect is positive precisely when the report leaves greater anchored value at that date. The comparison depends on uptake before voting and on which valued meanings are added or lost: even an admitted, valued objection can be outweighed by the loss of other valued content. The intervention should therefore measure admission and retained valued content alongside comprehension and attribution. A higher score affects choice only when it crosses the relevant rival comparison. The example’s threshold is a consequence of its specified valuation and menu; an empirical study must estimate the corresponding comparison.

A reported negative reaction, a conflict trigger, and an observed change in belief are separate measurements. Inferring one from another would build the proposed mechanism into its own test. Randomizing the availability of an objection can identify an effect of observing it; studying candidates’ incentives additionally requires the vote gains and losses across the audiences that constitute their electorate.

Measuring reception. Source trust can help characterize the setting, but the model’s credibility filter acts on meaning content from a source. Direct elicitation of what respondents heard and accepted is needed to distinguish different filtered receipts from differences in prior knowledge or interpretation. Candidate thermometers, media-use patterns, or low correlations across groups do not by themselves identify disjoint meaning filters. A proposed measurement model should connect those observations to the filter rather than equating them by definition.

6.3 Majority priorities and minority influence.

Proposition 5.23 identifies a common maximizing set of actual responses as the relevant majority condition. A test therefore begins with a specified menu and measurements of which speeches each voter ranks highest at the chosen horizon. If a strict majority has the same maximizing set, symmetric equilibrium must lie within it. When that set contains several alternatives, minority comparisons can determine which alternative supports equilibrium. The indifference threshold in Example 5.22 illustrates this possibility.

Anchor structure gives a separate benchmark. With two anchor types and direct-broadcast rankings, a feasible speech containing the majority anchor bounds the minority’s possible value deficit as in Corollary 5.24. A feasible speech containing both anchors attains zero deficit. A proposed account of minority exclusion must therefore identify which alternatives are available and how their actual reception changes the relevant rankings. Shared content between two voters’ anchors alone does not establish that they belong to the same response bloc; this is the comparison required for immigration voters B and C.

Measuring common ground. Anchor width describes incomparability of the modeled evaluative anchors. It does not count policy axes, and factor-analytic orthogonality does not by itself identify lattice incomparability. Measurement must distinguish the representations a voter can access, the distinctions she draws, and the meanings she values, then establish how these observations support the proposed anchor comparisons. The unrestricted representation result (Proposition 5.11) shows why connecting those comparisons to electoral order requires substantive restrictions on messages, reception, or evaluation.

Testing electoral relevance. Proposition 5.14 gives a conditional invariance implication at a common fixed stance: deleting message content outside both the credibly valued meanings and the conflict neighbors of current priors leaves the modeled encounter scores unchanged. This implication gives a testable response invariance for the specified one-encounter model. A possible design would compare matched messages while holding the source and offered distinctions fixed. Its interpretation requires independent measurements or assumptions about priors, valued meanings, credibility, and conflict; defining discarded content by the observed absence of an effect would make the exercise circular. When messages teach new distinctions or change the prior used in subsequent encounters, the fixed-stance result no longer supplies that invariance. Anchor width alone predicts neither the size of the response-equivalent classes nor confinement of political competition.

6.4 Empirical scope and design requirements.

The empirical object is the joint pattern of reception, cognitive change, candidate evaluation, and voting choice. Observed agreement or disagreement alone does not identify the mechanism producing it. A design that varies message components can examine their interaction at reception; a design

that varies access to a public objection can examine the effect of observing another audience. Both require a fixed comparison candidate or an explicit model of how evaluations become votes.

The constructs also require distinct measurements. Reports of what a voter understood can inform a model of reception; changes in the distinctions she can draw can inform a model of acquisition; and her evaluations of specific content can inform an anchor representation. Each inference needs a measurement model rather than an identification of a survey item with a formal primitive. Comparing these observations across voters and across exposure conditions can test whether the proposed semantic structure explains their responses to the same political communication.

7 Extensions.

The electoral framework fixes the feasible campaign menu and the ambient representational field. The first extension considers legislative proposals and bargaining protocols. The second compares restricted views of a field and embeddings between fields.

7.1 Legislative bargaining as an extension.

The understanding lattice supplies possible outcomes for a legislative game, but an electoral reduction does not determine its feasible set or its equilibrium. A proposed extension would specify a finite menu $\mathcal{F} \subseteq \mathcal{U}_{\mathcal{R}}$, a status quo, recognition probabilities, a voting quota, discounting, and the utility from acceptance and rejection. These choices must precede a claim about stationary equilibrium, delay, or coalition selection.

For a direct-score benchmark, legislator i evaluates an outcome b by $D_i[b] = \mathcal{W}(b \wedge u_i^*)$. Let $L = \bigwedge_i u_i^*$ and $H = \bigvee_i u_i^*$. One may stipulate $\mathcal{F} = [L, H]$ when this interval is finite, but that is an assumption about the available proposals. This particular menu contains a jointly satisfying endpoint, with direct consequences for bargaining preferences.

Proposition 7.1 (A common maximum in the legislative benchmark). *If $H \in \mathcal{F}$, it maximizes every legislator's direct score and belongs to the ordinary strict-majority core, regardless of anchor width.*

Proof. For every i and $b \in \mathcal{F}$,

$$D_i[b] = \mathcal{W}(b \wedge u_i^*) \leq \mathcal{W}(u_i^*) = D_i[H].$$

No legislator strictly prefers an alternative to H , so no strict majority defeats it. $\square$

The proposition is a static comparison. Its role in a sequential bargaining model depends on the proposal protocol and continuation payoffs. A bargaining explanation of trade-offs must account for why the jointly satisfying endpoint is unavailable, costly, or otherwise restricted. Likewise, choosing the interval as the proposal menu confines outcomes by assumption, rather than by an electoral theorem.

A possible shared-content constraint. A separate modeling choice is to restrict coalitions by the content shared among their anchors. To distinguish semantic content from access alone, define:

Definition 7.2 (Semantic viability). *For a nonempty coalition S , write $(e_S, \mu_S) = \bigwedge_{i \in S} u_i^*$. The coalition is semantically viable if $\mu_S(D) \neq \emptyset$ for at least one visible class D of e_S .*

This is a proposed admissibility constraint, not a necessary consequence of the legislators' direct preferences. Disjoint anchor meanings do not prevent the join from satisfying both legislators. If viability is imposed on winning coalitions, the game must specify its role in acceptance and whether a viable coalition of the required size is available. The existence of shared content also does not supply a transferable surplus or a cost ordering of coalition partners.

The immigration example. At the common full stance, the three anchors have meet $u_{e_T,0}$ and join

$$H = (\{T, G, B\}, \{T, F, G, B\}, \{T, G, B\}).$$

Here the displayed tuples specify meanings at the three visible resources. If the full interval $[u_{e_T,0}, H]$ is available, H maximizes all three direct scores. For the proposed viability rule, $\{A, B\}$ shares $\{T\}$ at the first and third resources, and $\{B, C\}$ shares C 's nonempty evaluative meanings. The meet of $\{A, C\}$ has empty meanings everywhere, so that coalition fails the shared-content test. Its meet still has full epistemic scope and is strictly above global bottom. These calculations identify possible constraints for a bargaining model; they do not determine which coalition forms or what a recognized legislator proposes.

7.2 Field variation and differential access.

Different information environments can make different representational resources available for a political encounter. A comparison across such environments needs both a rule for restricting a view and a common field in which to compare the results. Let $S_v \subseteq \mathcal{R}$ be voter v 's *window* on the ambient field. These windows are specified features of the comparison. A theory of how voters, candidates, or media select them would additionally require choice rules and incentives.

Restriction and embedding. Keep the meaning space M fixed and use the same marker $\star$ in every field. For bookkeeping, allow $S = \emptyset$, whose understanding space has just the state with no visible resources.

Definition 7.3 (Field restriction and fragmentation). *For $u = (e, \mu) \in \mathcal{U}_{\mathcal{R}}$ and $S \subseteq \mathcal{R}$, the restriction $\rho_S(u) \in \mathcal{U}_S$ is defined as follows: its marked partition consists of the nonempty intersections of the e -classes with $S \cup \{\star\}$. Its invisible class is $I_e \cap (S \cup \{\star\})$, with empty semantics. Each nonempty visible class $C \cap S$ inherits $\mu(C)$.*

The invisible extension $\iota_S : \mathcal{U}_S \rightarrow \mathcal{U}_{\mathcal{R}}$ leaves visible classes and their meanings unchanged and adjoins every resource in $\mathcal{R} \setminus S$ to the invisible class. Write

$$\mathbf{V}_S = \iota_S \circ \rho_S$$

for the resulting view in the ambient field. A field fragmentation is a profile $(S_v)_{v \in V}$; the corresponding restricted views are $\mathbf{V}_{S_v}(u_v)$, or $\mathbf{V}_{S_v}(u_v^)$ when comparing the available portions of evaluative anchors.*

The extension distinguishes an unavailable resource from a visible resource carrying empty semantics. Restriction inherits the meaning of a category at its retained part, consistently with the overlap interpretation of reception. Applying a window to an anchor specifies its available portion in this comparison; it does not posit a change in the underlying evaluative disposition. A dynamic application must also specify how the environment affects accessible representations over time. A message stream that simply omits a topic need not erase a voter's existing access to it.

Let t_S be the understanding with each $r \in S$ visible in its own class, every other resource invisible, and all meanings M at each visible class.

Proposition 7.4. *The restriction ρ_S is monotone. The extension ι_S is an order embedding with image*

$$\iota_S(\mathcal{U}_S) = \{(e, \mu) \in \mathcal{U}_{\mathcal{R}} : \mathcal{R}_e \subseteq S\} = \downarrow t_S.$$

It preserves binary meets and joins. Moreover,

$$\rho_S \circ \iota_S = \text{id}_{\mathcal{U}_S}, \quad \mathbf{V}_S \circ \mathbf{V}_T = \mathbf{V}_{S \cap T}, \quad \mathcal{R}_{e_{\mathbf{V}_S(u)}} = S \cap \mathcal{R}_{e_u}.$$

Consequently each $\mathbf{V}_S$ is monotone and idempotent, fixes exactly the understandings supported on S , and preserves comparability when applied to both members of an ordered pair.

The same embedding compares fields under expansion: for $\mathcal{R} \subseteq \mathcal{R}'$, extend an understanding to $\mathcal{R}'$ by making the new resources invisible. This preserves its previous order relations. The enlarged field permits additional states in which those resources become visible; a subsequent acquisition or updating rule determines whether they do.

Category boundaries. A restricted view can change both access and the boundaries of a category. If a and b initially form one visible class, retaining only a yields the marked partition $\{\{\star, b\}, \{a\}\}$. The original $\{\{\star\}, \{a, b\}\}$ and the new partition are incomparable. Thus $\mathbf{V}_S(u)$ need not lie below u in the richness order. When S retains or removes each visible class in its entirety, $\mathbf{V}_S(u) \preceq_{\mathcal{U}} u$.¹²

When different views become incomparable. Common restriction preserves an existing order by Proposition 7.4. Different windows can remove the resources needed for one view to contain the other. Two sufficient conditions isolate this effect.

Theorem 7.5. *Let $u_i = (e_i, \mu_i)$ with $u_1 \preceq_{\mathcal{U}} u_2$, let $w_i = \mathbf{V}_{S_i}(u_i)$, and put $A_i = S_i \cap \mathcal{R}_{e_i}$. Then w_1 and w_2 are incomparable if either of the following holds:*

$$(i) \ A_1 \setminus A_2 \neq \emptyset \text{ and } A_2 \setminus A_1 \neq \emptyset;$$

$$(ii) \ S_1 = \mathcal{R}, \ \mathcal{R}_{e_1} \setminus S_2 \neq \emptyset, \text{ and}$$

$$\rho_{S_2}(u_1) \prec_{\mathcal{U}_{S_2}} \rho_{S_2}(u_2).$$

The first condition gives each voter access to a resource absent from the other's view. For example, on $\{a, b, c\}$, restrict two initially comparable full-scope understandings to $\{a, c\}$ and $\{b, c\}$. The views are incomparable even if both attach exactly the same meanings to their shared resource c .

The second condition combines a loss of access with a retained advantage. At a common fully distinguishing stance on $\{a, b\}$, take

$$u_1 : \quad a \mapsto \{p\}, \ b \mapsto \{p\}, \quad u_2 : \quad a \mapsto \{p, q\}, \ b \mapsto \{p\}.$$

Initially $u_1 \prec_{\mathcal{U}} u_2$. Keep u_1 's full view and restrict u_2 to $\{a\}$. The first voter still sees b , while the second retains the extra meaning q at a . Neither restricted understanding contains the other. This example has overlapping windows and only two resources. More generally, the retained strict advantage in the theorem can concern distinctions as well as meanings.

¹²An application requiring a restriction that always lies below the original understanding can use $u \wedge t_S$. By Proposition 7.4, this is the greatest understanding below u with visible scope contained in S . It retains each visible class of u contained wholly in S and makes every class meeting the discarded region invisible. This alternative preserves the original category boundaries by also removing the retained part of any class cut by the window.

The immigration comparison. Use the common fully distinguishing anchor stance from the running example. Set $S_A = \{r_1, r_3\}$ and $S_B = \{r_2, r_3\}$. A’s view retains $\{T, G\}$ at r_1 and $\{T, B\}$ at r_3 , with r_2 invisible. B’s view retains $\{T, G\}$ at both r_2 and r_3 , with r_1 invisible. These anchors are incomparable both before and after restriction. Their ambient meet after restriction has

$$I_\wedge = \{\star, r_1, r_2\}, \quad \mathcal{R}_{e_\wedge} = \{r_3\}, \quad \mu_\wedge(\{r_3\}) = \{T\}.$$

The shared factual meaning at r_1 is lost from the meet, while the factual acknowledgment of economic impact at r_3 remains. The comparison identifies precisely which common content survives.

Media exposure and electoral consequences. The empirical interpretation requires identifying accessible representations, their category boundaries, and attached meanings. [Flaxman, Goel and Rao \(2016\)](#), studying active online news readers, find that social and search channels are associated both with greater ideological segregation between users and with greater exposure to opposing perspectives within users. Those browsing patterns motivate studying differential access, but do not identify the windows or meanings of the present model.

A study of the proposed mechanism could compare views under common and different access conditions while measuring both retained content and the distinctions respondents draw. Its electoral consequences require specifying how those conditions affect priors, available anchor content, reception, and the feasible broadcasts. The resulting candidate scores and rankings can then be compared at the stated election horizon. Incomparability of understandings alone determines neither a policy dimension nor the availability of a common single-peaked ordering.

8 Conclusion.

This manuscript treats political communication as an encounter between structured understandings. The epistemic component records which representations a voter can access and distinguish; the semantic component records the meanings attached to them. The marked-partition order and the reception maps make comparisons possible even when agents use different scopes and categories. The update can introduce a distinction and its meanings in one encounter, while excessive raw conflict instead removes conflicting prior content.

The electoral response separates recognition from valued change. This permits an appeal to affirm something already held, add something newly valued, or produce a mixed response. The evaluative anchor and the additive response are reduced-form assumptions connecting this cognitive process to voting. The best available appeal depends on the voter’s current understanding, the source, and the messages that can be offered.

Repeated exposure makes the interaction between absorption and conflict explicit. A fixed message from a fixed source leads eventually to stability or a two-cycle, depending on whether admitted content renews excessive conflict. Internally consistent broadcasts stabilize within two encounters under maximal acquisition, including when triggers fire. Public reactions can supply further valued content; with internally consistent outputs and maximal acquisition, acyclic observation dependencies ensure that this process settles. Evaluation at a fixed election horizon accommodates both stable and recurring responses, allowing the timing of a vote to matter.

Two questions connect this account of communication to electoral competition: when do voter responses admit a common policy order, and when can a smaller message domain preserve the strategic problem? For the first, a feasible median supports an equilibrium when actual responses are strictly single-peaked on a common order and the stated source and voting assumptions hold. The coalition mountain condition characterizes strict Condorcet existence through aggregate win counts.

For the second, clipping at the join of anchors preserves direct-broadcast scores. At a common stance, a sufficient reduction for the full encounter response retains both credibly valued content and content that can trigger conflict with current priors. Closure of the feasible menu under the relevant reduction supplies available representatives that preserve expected vote shares. Full-game pure equilibria project to equilibria in the reduced menu, and every reduced-game equilibrium remains an equilibrium of the full game.

These results make feasibility central to the study of bounds on competition. Score-preserving reductions identify sufficient content for electoral evaluation; menu closure turns those descriptions into an equivalent strategic game. A distinct restriction comes from completing messages with shared valued content: when completion is feasible, messages missing that content cannot support a symmetric equilibrium in the direct benchmark under the stated source and voting assumptions. If the join of anchors is itself available, every voter attains maximal direct-broadcast value there. Limits on available messages therefore shape electoral trade-offs.

The strategic results connect combined appeals to votes. Feasible recombination and isolated audience responses permit independent component choice even with active triggers. When audiences interact through conflict or public reactions, the whole speech must be evaluated through those interactions. An opposing audience’s objection can make a combined appeal more attractive to supporters and change the equilibrium speech. When a strict majority has the same maximizing set, symmetric equilibrium must lie within it, while minority voters can select among equally satisfactory majority alternatives.

Field variation introduces a further comparison. Restricting both understandings to the same window preserves their order. Different windows can make comparable understandings incomparable through exclusive access or a retained advantage combined with lost access. The restriction and embedding maps identify these changes while keeping the surviving shared content explicit.

The unrestricted finite framework can represent arbitrary weak-ranking profiles, so explanation requires restrictions on messages, reception, and evaluation. Empirical work can investigate these restrictions by distinguishing what voters can represent, what they value, and how direct encounters and observed reactions affect their choices.

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A Proofs.

A.1 Proofs for Section 2.

Lemma 2.3. $(E_{\mathcal{R}}, \preceq_E)$ is a partially ordered set.

Proof. The stances are partitions of the fixed marked set $\mathcal{R}_\star$. Reflexivity and transitivity follow from inclusion of equivalence relations, with the inclusion reversed in the richness order. If $e_1 \preceq_E e_2$ and $e_2 \preceq_E e_1$, their equivalence relations on $\mathcal{R}_\star$ are equal. Their classes containing $\star$ are consequently equal, as are their visible scopes, so $e_1 = e_2$. Moreover, $e_1 \preceq_E e_2$ implies $[\star]_{e_2} \subseteq [\star]_{e_1}$ and hence $\mathcal{R}_{e_1} \subseteq \mathcal{R}_{e_2}$. $\square$

Lemma 2.4. $(E_{\mathcal{R}}, \preceq_E)$ is a complete lattice with bottom element $e_\perp$ (the single class $\mathcal{R}_\star$, with empty visible scope) and top element $e_\top = (\mathcal{R}, \text{id}_{\mathcal{R}_\star})$ (full visible scope, each resource its own class). The meet of the empty family is $e_\top$; its join is $e_\perp$.

Proof. For a nonempty family $\{e_j\}_{j \in J}$, form equivalence relations on $\mathcal{R}_\star$ by

$$\sim^\wedge = \text{EqCl} \left(\bigcup_j \sim_j \right), \quad \sim^\vee = \bigcap_j \sim_j.$$

Each determines a stance by taking the complement in $\mathcal{R}$ of its marker's class as visible scope. The first relation contains every $\sim_j$, so its stance is a lower bound. Every lower bound has an equivalence relation containing all $\sim_j$ and hence their equivalence closure; it is therefore below this stance. The second relation is contained in every $\sim_j$, so its stance is an upper bound. The relation of any other upper bound is contained in every $\sim_j$, hence in their intersection; the displayed upper bound is therefore least.

The marker's class in the join is $\bigcap_j I_{e_j}$, giving visible scope $\bigcup_j \mathcal{R}_{e_j}$. The marker's class in the meet contains every I_{e_j} and may contain more through transitive closure, so its visible scope is only required to be contained in $\bigcap_j \mathcal{R}_{e_j}$.

The universal relation on $\mathcal{R}_\star$ gives $e_\perp$, and the diagonal relation gives $e_\top$. These lie below and above every stance respectively and supply the join and meet of the empty family. Thus every family has both bounds. $\square$

Proposition 2.9. For $e_1 \preceq_E e_2$, the maps Ref , Crs , and All above are monotone maps between the indicated semantic fibers, and

$$\text{Ref}_{e_1 \rightarrow e_2}(\mu) \preceq_M \nu \iff \mu \preceq_M \text{All}_{e_2 \rightarrow e_1}(\nu).$$

Thus $\text{Ref} \dashv \text{All}$ is a Galois connection. If $\mathcal{R}_{e_1} = \mathcal{R}_{e_2}$, there is also the adjunction $\text{Crs} \dashv \text{Ref}$; this second claim does not in general hold when visible scope expands. Refinement has identity maps and composes: for $e_1 \preceq_E e_2 \preceq_E e_3$, $\text{Ref}_{e_2 \rightarrow e_3} \circ \text{Ref}_{e_1 \rightarrow e_2} = \text{Ref}_{e_1 \rightarrow e_3}$.

Proof. For $e_1 \preceq_E e_2$, the fine invisible class is contained in the coarse invisible class, so refinement assigns it $\emptyset$. The other two maps set the coarse invisible semantics to $\emptyset$ explicitly. All three maps therefore take values in the stated fibers; pointwise inclusion is preserved by copying, union, and intersection.

Fix a visible coarse class C . The condition $\mu(C) \subseteq \nu(D)$ for every fine $D \subseteq C$ is equivalent to $\mu(C) \subseteq \bigcap_{D \subseteq C} \nu(D)$. Across all visible coarse classes, these are exactly the comparisons $\text{Ref}(\mu) \preceq_M \nu$ and $\mu \preceq_M \text{All}(\nu)$. Fine classes with an invisible coarse parent add only the automatic condition $\emptyset \subseteq \nu(D)$ to the first comparison. This proves $\text{Ref} \dashv \text{All}$ even when scope expands.

When visible scopes agree, every visible fine class has a visible coarse parent. In this case $\bigcup_{D \subseteq C} \nu(D) \subseteq \mu(C)$ for every visible C is equivalent to $\nu(D) \subseteq \mu(C_D)$ for every visible D , proving $\text{Crs} \dashv \text{Ref}$. For contrast, take a coarse stance seeing only a and a finer stance distinguishing and seeing a, b . Let the coarse semantics be empty and the fine semantics attach a nonempty meaning set only to b . Coarsening erases that content, so $\text{Crs}(\nu) \preceq_M \mu$, but $\nu \not\preceq_M \text{Ref}(\mu)$.

Finally, a fine class and its intermediate parent have the same ancestor at any coarser level. Copying along either route therefore gives the same semantics, including when that ancestor is invisible. This proves refinement composition; refinement at an unchanged stance is the identity. $\square$

Proposition 2.10. *For arbitrary e_i, e_j , $\text{Rec}_{e_j \rightarrow e_i} : M_{e_j} \rightarrow M_{e_i}$ is monotone and, writing $k = e_i \vee e_j$, satisfies*

$$\text{Rec}_{e_j \rightarrow e_i} = \text{Crs}_{k \rightarrow e_i} \circ \text{Ref}_{e_j \rightarrow k}.$$

If $e_i \preceq_E e_j$, it reduces to $\text{Crs}_{e_j \rightarrow e_i}$; if $e_j \preceq_E e_i$, it reduces to $\text{Ref}_{e_j \rightarrow e_i}$.

Proof. The reception formula is a union of sender meanings at each visible receiver class and assigns $\emptyset$ at the invisible class. It is therefore a well-defined monotone map into M_{e_i} . The classes at $k = e_i \vee e_j$ are precisely the nonempty intersections of classes from the two marked partitions. For a visible receiver class C , the k -classes inside C are its nonempty intersections with sender classes. Refinement from e_j assigns each such intersection the semantics of its sender parent; an invisible sender parent contributes $\emptyset$. Coarsening into C consequently unions exactly the meanings of visible sender classes meeting C , proving the identity.

If $e_i \preceq_E e_j$, a sender class meeting a visible receiver class lies wholly inside it, giving the coarsening formula. If $e_j \preceq_E e_i$, a visible receiver class has one sender parent; it receives that parent's meanings, or $\emptyset$ if the parent is invisible, giving refinement. $\square$

Theorem 2.11. *($\mathcal{U}_{\mathcal{R}}, \preceq_{\mathcal{U}}$) is a complete lattice. For any nonempty family $\{(e_j, \mu_{e_j})\}_{j \in J} \subseteq \mathcal{U}_{\mathcal{R}}$, writing $e^\wedge = \bigwedge_j e_j$ and $e^\vee = \bigvee_j e_j$, the meet and join are*

$$\bigwedge_{j \in J} (e_j, \mu_{e_j}) = (e^\wedge, \mu_\wedge), \quad \bigvee_{j \in J} (e_j, \mu_{e_j}) = (e^\vee, \mu_\vee),$$

where

$$\mu_\wedge(C) = \bigcap_{j \in J} \text{All}_{e_j \rightarrow e^\wedge}(\mu_{e_j})(C), \quad \mu_\vee(D) = \bigcup_{j \in J} \text{Ref}_{e_j \rightarrow e^\vee}(\mu_{e_j})(D),$$

for $C \in \mathcal{R}_{e^\wedge} / \sim_{e^\wedge}$ and $D \in \mathcal{R}_{e^\vee} / \sim_{e^\vee}$ respectively. Both semantics assign $\emptyset$ to their invisible class. The global bottom is $u_\perp = (e_\perp, 0_{e_\perp})$, and the global top has stance $e_\top$ and meaning set M at every visible singleton. The empty-family join and meet are these bottom and top elements, respectively.

Proof. First, the understanding relation is a partial order. Reflexivity follows from identity refinement. For transitivity use refinement composition and monotonicity from Proposition 2.9. Mutual comparison forces equal epistemic stances by Lemma 2.3, and then equal semantics by antisymmetry of pointwise inclusion.

Let $\{(e_j, \mu_j)\}_{j \in J}$ be nonempty, and let $h = \bigwedge_j e_j$ and $k = \bigvee_j e_j$. Use the semantics in the theorem statement, with their invisible values set to $\emptyset$. They lie in M_h and M_k by the transport proposition.

Meet. At a visible h -class C , the proposed semantics is

$$\mu_\wedge(C) = \bigcap_j \bigcap_{\substack{D \in \mathcal{R}_{e_j} / \sim_{e_j} \\ D \subseteq C}} \mu_j(D).$$

It is included in every term, so its refinement to each e_j is below μ_j on classes with a visible h -parent. Classes with an invisible h -parent inherit $\emptyset$. Together with $h \preceq_E e_j$, this proves the lower-bound property.

For greatestness, let (e', μ') be any lower bound. Then $e' \preceq_E h$. A visible h -class C has a unique e' -parent C' . If that parent is invisible, its refinement into C is empty. Otherwise, for every j and fine $D \subseteq C$, the lower-bound assumption gives $\mu'(C') \subseteq \mu_j(D)$, since $D \subseteq C \subseteq C'$. Intersecting these inclusions gives $\text{Ref}_{e' \rightarrow h}(\mu')(C) \subseteq \mu_\wedge(C)$. Thus (e', μ') lies below the proposed meet.

Join. At stance k , the proposed semantics is the pointwise union of $\text{Ref}_{e_j \rightarrow k}(\mu_j)$. Each member is therefore below it. If (e', μ') is any upper bound, then $k \preceq_E e'$. For each visible e' -class D , its parent at k has an ancestor at every e_j . The upper-bound assumption includes that ancestor's semantics in $\mu'(D)$; invisible ancestors contribute empty semantics. The union over j is consequently included as well, establishing $\text{Ref}_{k \rightarrow e'}(\mu_\vee) \preceq_M \mu'$. The proposed join is least.

Extrema. The empty semantics at $e_\perp$ refines to empty semantics at every stance, so $u_\perp$ is below every understanding. At $e_\top$, assigning M to each resource singleton and $\emptyset$ to $\{\star\}$ gives an upper bound for every understanding. These supply the empty-family join and meet, completing the proof. $\square$

A.2 Proofs for Section 3.

Proposition 3.4 (Maximal expansion). *Let $u_j^\alpha = (e_j, \nu_{ij})$ be the source understanding with its semantics filtered by α_{ij} . If $\Gamma(e_i, e_j) = e_i \vee e_j$, then*

$$T_+(u_i, u_j) = u_i \vee u_j^\alpha.$$

Under full credibility, $T_+(u_i, u_j) = u_i \vee u_j$. These identities give the actual update T whenever its conflict test selects the forward branch.

Proof. Put $k = e_i \vee e_j$. Since $e_j \preceq_E k$, Proposition 2.10 gives $\text{Rec}_{e_j \rightarrow k}(\nu_{ij}) = \text{Ref}_{e_j \rightarrow k}(\nu_{ij})$. The forward semantics is therefore the pointwise union of $\text{Ref}_{e_i \rightarrow k}(\mu_i)$ and $\text{Ref}_{e_j \rightarrow k}(\nu_{ij})$, with empty semantics at the invisible class. By Theorem 2.11, this is exactly the semantics of $u_i \vee u_j^\alpha$. Under full credibility $\nu_{ij} = \mu_j$, giving the second identity. The definition of T applies this forward map precisely when the conflict test passes. $\square$

Proposition 3.6. *If $\text{Conf}(u_i, u_j) \neq \emptyset$, then $\beta(u_i, u_j) \prec_{\mathcal{U}} u_i$. More explicitly, S_{ij} has a greatest element and*

$$\beta(u_i, u_j) = (e_i, \mu_i^-), \quad \mu_i^-(C) = \mu_i(C) \setminus F_{ij}(C)$$

for every visible receiver class C , with $\mu_i^-(I_{e_i}) = \emptyset$. In particular, contraction preserves the receiver's epistemic stance.

Proof. Define $\mu_i^-(C) = \mu_i(C) \setminus F_{ij}(C)$ on visible classes and set its invisible value to $\emptyset$. Then $b = (e_i, \mu_i^-)$ lies below u_i and avoids every forbidden meaning at every conflicting class, so $b \in S_{ij}$. For any $u = (e_u, \mu_u) \in S_{ij}$, its refinement to e_i is below μ_i . At a conflicting class it is also disjoint from F_{ij} and hence included in μ_i^- . At a nonconflicting class, μ_i is already disjoint from F_{ij} , so the

same inclusion holds there. Consequently $u \preceq_{\mathcal{U}} b$. Thus b is the greatest element of S_{ij} and equals its join β . If the conflict set is nonempty, at least one visible class loses a prior meaning; at the unchanged stance this gives $\beta \prec_{\mathcal{U}} u_i$. $\square$

Lemma 3.7. *For any understandings u_i, u_j with $|\text{Conf}(u_i, u_j)| \leq \tau_i$ (the trigger does not fire):*

$$u_i \preceq_{\mathcal{U}} T_+(u_i, u_j).$$

The forward update is weakly above the input: encountering another understanding can only enrich, never impoverish, when the trigger does not fire.

Proof. The expansion rule gives $e_i \preceq_E e'_i = \Gamma(e_i, e_j)$. At this output stance, the union formula gives $\text{Ref}_{e_i \rightarrow e'_i}(\mu_i) \preceq_M \mu_i^+$. These are exactly the two conditions for $u_i \preceq_{\mathcal{U}} T_+(u_i, u_j)$. The argument holds for the forward map irrespective of whether the threshold selects it. $\square$

Lemma 3.8. *For fixed u_i, u_j and expansion rule Γ , if $\alpha_{ij}(D) \subseteq \alpha'_{ij}(D)$ for every visible sender class D , then $T_+(u_i, u_j; \alpha_{ij}) \preceq_{\mathcal{U}} T_+(u_i, u_j; \alpha'_{ij})$.*

Proof. At every visible sender class D , $\alpha_{ij}(D) \cap \mu_j(D) \subseteq \alpha'_{ij}(D) \cap \mu_j(D)$. The expansion $e'_i = \Gamma(e_i, e_j)$ is the same in both comparisons because Γ depends only on stances. By monotonicity of $\text{Rec}_{e_j \rightarrow e'_i}$, the admitted reception at this stance is weakly larger under α'_{ij} . Union with the same refined prior gives $\mu_i^+ \preceq_M \mu_i^{+'}$. Together with the common output stance, this establishes the claimed understanding comparison. $\square$

Lemma 3.9. *For fixed α_{ij} , the output $T_{\alpha_{ij}, \tau}(u_i, u_j)$ is weakly increasing in τ .*

Proof. Fix u_i, u_j, α_{ij} and let $n = |\text{Conf}(u_i, u_j)|$. For $\tau \geq n$ the trigger does not fire: the output is $T_+(u_i, u_j) \succeq_{\mathcal{U}} u_i$ by Lemma 3.7. For $\tau < n$ the trigger fires: the output is $\beta(u_i, u_j) \preceq_{\mathcal{U}} u_i$ by Proposition 3.6 (strict when $\text{Conf} \neq \emptyset$). Now let $\tau' > \tau$. If $\tau \geq n$, both produce T_+ : equal. If $\tau < n \leq \tau'$, τ produces $\beta \preceq_{\mathcal{U}} u_i \preceq_{\mathcal{U}} T_+$: strictly higher under τ' . If $\tau' < n$, both produce β : equal. In every case the output under τ' is weakly above the output under τ . $\square$

Lemma 3.10 (Prior content and conflict). *Fix a source understanding u_j and two priors $u_i = (e, \mu_i) \preceq_{\mathcal{U}} u_{i'} = (e, \mu_{i'})$ at the same stance. For each visible class C , put $L_i(C) = \mu_i(C) \cap F_{ij}(C)$, the content removed by $\beta(u_i, u_j)$, and define $L_{i'}$ analogously. Then*

$$\text{Conf}(u_i, u_j) \subseteq \text{Conf}(u_{i'}, u_j), \quad L_i(C) \subseteq L_{i'}(C), \quad \beta(u_i, u_j) \preceq_{\mathcal{U}} \beta(u_{i'}, u_j).$$

At a common threshold τ , if the trigger fires for u_i , it also fires for $u_{i'}$, and both contractions are strictly below their respective priors.

Proof. Since the receiver stances agree and the source is fixed, the translated message and forbidden sets F_{ij} are the same for both priors. Intersecting $\mu_i(C) \subseteq \mu_{i'}(C)$ with $F_{ij}(C)$ gives $L_i(C) \subseteq L_{i'}(C)$. Every class with nonempty $L_i(C)$ therefore also has nonempty $L_{i'}(C)$, proving conflict-set inclusion. Taking set differences with the same forbidden set gives $\mu_i(C) \setminus F_{ij}(C) \subseteq \mu_{i'}(C) \setminus F_{ij}(C)$. By Proposition 3.6, these are the baseline semantics at the common stance, establishing the comparison of retained understandings. Finally, $|\text{Conf}(u_{i'}, u_j)| \geq |\text{Conf}(u_i, u_j)| > \tau$ whenever the smaller prior triggers contraction. Both conflict sets are then nonempty, so each baseline is strictly below its own prior by Proposition 3.6. $\square$

A.3 Proofs for Section 4.

Lemma 4.4. *If $f(u, u')$ is a lower bound of u, u' and lies above every common lower bound, then $f(u, u') = u \wedge u'$.*

Proof. The two requirements say precisely that $f(u, u')$ is the greatest lower bound of u, u' . Uniqueness of a greatest lower bound gives $f(u, u') = u \wedge u'$. This identifies a lattice element, without selecting a scalar function of that element. $\square$

Proposition 4.6. *Recognition is nonnegative and is zero for an empty received message. Valued change is nonnegative on the forward branch and nonpositive on contraction. Consequently the forward-branch response is nonnegative. A broadcast at the global bottom produces no update and has response zero. If $u_v^0 \succeq_{\mathcal{W}} u_v^*$, every forward encounter has $G_v^c(x) = 0$, but recognition can still distinguish the messages.*

Proof. At the actual output stance $\bar{e}$, $u_{\bar{e},0} \preceq_{\mathcal{W}} r_v^c(x)$. Monotonicity of meet and of $\mathcal{W}$ gives $R_v^c(x) \geq 0$, with equality when $r_v^c(x) = u_{\bar{e},0}$. On the forward branch, $\hat{u}_v^c(x) \succeq_{\mathcal{W}} u_v^0$ by Lemma 3.7; on contraction, the reverse weak inequality holds by Proposition 3.6. Applying the same meet and valuation gives the asserted signs of $G_v^c(x)$. Since $\lambda_v \geq 0$, the forward response is nonnegative.

A bottom broadcast has an all-invisible stance and empty semantics. The expansion interval forces $\Gamma(e_v^0, e_{\perp}) = e_v^0$; the raw reception is empty, so there is no conflict and no absorbed content. Thus the output is u_v^0 , the received message is $u_{e_v^0,0}$, and both response terms vanish. Finally, if $u_v^0 \succeq_{\mathcal{W}} u_v^*$, every forward output also lies above u_v^* . Both its meet with the anchor and the prior's meet with the anchor equal u_v^* . Hence $G_v^c(x) = 0$. This equality does not force the received messages, or their recognition scores, to coincide. $\square$

Proposition 4.10. *Suppose the prior and every feasible broadcast have the same stance e , credibility is full, and every feasible encounter takes the forward branch. Then $s_v(x) = D_v(x) - F_v(u_{e,0})$ if $\lambda_v = 0$. If instead $u_v^0 = u_{e,0}$, then for any $\lambda_v \geq 0$,*

$$s_v(x) = (1 + \lambda_v)[D_v(x) - F_v(u_{e,0})].$$

In either case s_v and D_v induce exactly the same rankings and ties. The conditions may be imposed voter by voter.

Proof. The common stance forces the output stance to be e . Full credibility and identity reception give $r_v(x) = \phi(x)$, so $R_v(x) = D_v(x) - F_v(u_{e,0})$. If $\lambda_v = 0$, this is the response. If $u_v^0 = u_{e,0}$ and the encounter is forward, absorption yields $\hat{u}_v(x) = \phi(x)$. Thus $G_v(x) = D_v(x) - F_v(u_{e,0})$ as well. The resulting transformation of D_v has positive slope $1 + \lambda_v$ and an additive constant independent of x . Both reductions therefore preserve all rankings and ties. $\square$

Proposition 4.11. *In a source-neutral two-candidate election, (x^*, x^*) is an equilibrium if and only if*

$$N(y, x^*) \leq N(x^*, y) \quad \text{for every } y \in X.$$

Every strict Condorcet winner satisfies this condition, but the converse need not hold.

Proof. Source neutrality gives identical voter evaluations at (x^*, x^*) , hence each candidate has expected share $1/2$. After one candidate deviates to y , write $a = N(y, x^*)$, $b = N(x^*, y)$ and $t = |V| - a - b$. The deviator's expected share is

$$\frac{a + t/2}{|V|} = \frac{1}{2} + \frac{a - b}{2|V|}.$$

It exceeds $1/2$ exactly when $a > b$. The same argument applies to either candidate, proving the equivalence. A strict Condorcet winner has $b > |V|/2$ against every distinct y , which implies $a < b$. For the failure of the converse, if every voter is indifferent between two available platforms, either symmetric profile is an equilibrium but neither platform is a strict Condorcet winner. $\square$

Theorem 4.14.

- (i) *CMC holds at x^* if and only if x^* is a strict Condorcet winner. It is therefore necessary for Downsifiability but does not establish a common single-peaked ordering.*
- (ii) *If all voter responses have the same rankings as the direct-broadcast benchmark and x^* is a strict Condorcet winner, then $\phi(x^*)$ is maximal in $\phi(X)$.*

Proof. For part (i), CMC explicitly requires $|W(x^*)| = |X| - 1$, so x^* is a strict Condorcet winner. Conversely, if x^* is such a winner, sort all other platforms by nondecreasing $|W(x)|$ and place x^* last. This is an admissible CMC order, with an endpoint peak. A Downsifiable median is a strict Condorcet winner by the majority argument in the proof of Proposition 4.15 below, so it satisfies CMC.

CMC does not imply a common single-peaked axis. In the displayed three-voter example, consider any proposed order of a, b, c, d and restrict it to a, b, c . Whichever of those three alternatives lies in the middle is ranked below both others by one voter. That valley is incompatible with single-peakedness, while every voter ranks d above all rivals. Thus d satisfies CMC without a common single-peaked order.

For part (ii), suppose some feasible $\phi(y)$ is strictly above $\phi(x^*)$. Monotonicity of meet and $\mathcal{W}$ gives $D_v(y) \geq D_v(x^*)$ for every v . By the ranking agreement assumption, $s_v(y) \geq s_v(x^*)$ for every v . No voter strictly prefers x^* to y , contradicting its strict Condorcet property. Thus no feasible image is strictly above $\phi(x^*)$, which is maximality. The argument imposes no ordering between the winner and incomparable images. $\square$

Proposition 4.15. *Suppose the base election is source-neutral, has two candidates and an odd number of voters, and uses the tie rule of Definition 4.8. If an injective $q : X \rightarrow \mathbb{R}$ orders every s_v strictly single-peaked with a feasible peak $p_v \in X$, then the median peak x_{med} :*

- (i) *is a strict Condorcet winner and satisfies CMC;*
- (ii) *supports the symmetric equilibrium $(x_{\text{med}}, x_{\text{med}})$;*
- (iii) *is Downsifiable on the ordering induced by q .*

Proof. The injective q orders X . Because $|V|$ is odd, the median of the ordered list of feasible peak platforms is a well-defined platform $x_{\text{med}} \in X$. For any $y <_X x_{\text{med}}$, at least $(|V| + 1)/2$ voters have peaks weakly above x_{med} . Strict single-peakedness makes each of these voters strictly prefer x_{med} to y . The symmetric argument handles $y >_X x_{\text{med}}$. Thus x_{med} defeats every distinct platform by a strict majority, proving the Condorcet assertion and, by Theorem 4.14(i), CMC. By Proposition 4.11, the symmetric profile at that platform is an equilibrium under the specified tie rule. The same order and feasible median satisfy the definition of Downsifiability. $\square$

A.4 Proofs for Section 5.

Proposition 5.1. *The pooling map $\mathbf{T}$ is monotone, extensive, and idempotent. It preserves G . Starting from μ , the profile $\mathbf{T}(\mu)$ is the least communication equilibrium above μ , reached after one simultaneous update.*

Proof. Monotonicity follows from monotonicity of union and intersection with a fixed filter. Extensivity is immediate. At a visible class, every output meaning belongs to the initial union G , and every initial meaning remains with its owner. Thus G is preserved. A second update gives

$$\mathbf{T}^2(\mu)_i = \mu_i \cup (\alpha_i \cap G) \cup (\alpha_i \cap G) = \mathbf{T}(\mu)_i.$$

If $\mu \preceq \eta$ and $\mathbf{T}(\eta) = \eta$, monotonicity gives $\mathbf{T}(\mu) \preceq \mathbf{T}(\eta) = \eta$. This proves the least-fixed-extension claim. $\square$

Proposition 5.2. *The fixed points of $\mathbf{T}$ form a complete lattice. Their least and greatest elements are respectively the all-empty and all-full visible semantic profiles, for every filter profile.*

Proof. The product of semantic fibers is a complete lattice and $\mathbf{T}$ is monotone, so Tarski's fixed-point theorem applies. Explicitly, an arbitrary meet of fixed profiles is fixed: monotonicity puts its image below every member of the family, while extensivity puts the meet below its image. The supremum of any fixed-profile family is $\mathbf{T}$ applied to its ambient join, by Proposition 5.1. Both the all-empty and the all-full visible profiles are fixed directly from the formula. They are the ambient extrema and hence the extrema of the fixed-point set. $\square$

Lemma 5.3. *At the fixed stance, if $\kappa \leq \tau$, every orbit of H reaches a fixed point within three updates. If $\kappa > \tau$, H has no fixed point and every orbit enters a two-cycle within three updates. From initial semantics P , that cycle alternates between*

$$C_0 = (P \cup A) \setminus F \quad \text{and} \quad C_1 = C_0 \cup A.$$

Proof. All set operations below are pointwise. A triggered update removes every member of F , leaving a conflict-free state for the next encounter. Put $Q = P \cup A$ and $C_0 = Q \setminus F$. If $d(P) \leq \tau$ and $d(Q) \leq \tau$, the first update yields Q , which is fixed.

Otherwise a contraction occurs in the first or second update. If it occurs first, the next states are $P \setminus F$ and $(P \setminus F) \cup A$. The latter equals $C_0 \cup A = C_1$. If contraction occurs second, the second state is C_0 and the third is C_1 . Since C_0 is disjoint from F , $H(C_0) = C_1$. Also $d(C_1) = d(A) = \kappa$. When $\kappa \leq \tau$, C_1 is fixed. When $\kappa > \tau$, $H(C_1) = C_0$, and $C_0 \neq C_1$ because $A \cap F$ is nonempty.

In the latter case no fixed point exists. A state on the contraction branch changes strictly. A fixed state on the forward branch would have to contain A , giving conflict count at least $\kappa > \tau$, a contradiction. The preceding cases also give the stated transient bound. $\square$

Theorem 5.4. *Under repeated exposure to b , the receiver reaches stance e^∞ within $2h$ encounters. If $\kappa \leq \tau$, the full understanding reaches a fixed point within $2h + 3$ encounters. If $\kappa > \tau$, it enters a two-cycle within $2h + 3$ encounters. Thus every fixed-message orbit is eventually constant or has period two.*

Proof. The stance iteration is increasing and bounded above by $e^0 \vee f$, so it reaches e^∞ after finitely many strict steps. At any stance e with $\Gamma(e, f) \neq e$, a forward update takes the next strict step. If instead the receiver contracts, Proposition 3.6 removes all prior meanings conflicting with raw reception of b at e . The next conflict count at that unchanged stance is zero, so the next encounter is forward and takes the step. Each strict stance step therefore requires at most two encounters. The actual stance follows the specified iteration and reaches e^∞ within $2h$ encounters.

At e^∞ , the stance no longer changes. Raw reception, filtered reception, and the forbidden sets are fixed, so subsequent updates are exactly H in Lemma 5.3. Applying that lemma gives the classification and the additional three-encounter bound. $\square$

Corollary 5.6. *Suppose b is internally consistent and $\Gamma(e, f) = e \vee f$. For every initial understanding, credibility filter, and threshold,*

$$T_{\alpha, \tau}(T_{\alpha, \tau}(T_{\alpha, \tau}(u, b), b), b) = T_{\alpha, \tau}(T_{\alpha, \tau}(u, b), b).$$

Repeated exposure therefore stabilizes within two encounters, including when a trigger fires.

Proof. Write $u = (e, \mu)$, $b = (f, \nu)$, and $g = e \vee f$. At g , each visible class receiving source content is contained in one visible f -class. Consequently raw reception at that class is the corresponding source meaning set, and admitted reception is a subset of it. Internal consistency of b implies that admitted meanings do not conflict with raw reception at g .

Suppose the first encounter contracts. At e , the resulting prior has no conflict with raw reception, so the second encounter is forward and acquires g . Each visible g -class inside an old visible e -class receives a subset of that class's earlier raw content. Its inherited prior is therefore compatible with its raw content. Newly visible classes inherit empty semantics. The newly admitted meanings are also compatible with raw reception, as established above. The second output is thus conflict-free relative to b at g , already contains all admitted source meanings, and is fixed.

Suppose instead the first encounter is forward. It acquires g and contains all admitted meanings there. If the second encounter is forward, it adds nothing and changes no stance. If the second encounter contracts, it removes the conflicting prior meanings but preserves every admitted meaning, since those are compatible with the raw broadcast. The resulting state has no remaining raw conflict and already contains everything a further forward update could add. It is therefore fixed as well. $\square$

Proposition 5.7. *Suppose every output of each Ψ_i is internally consistent and acquisition is maximal. Let d be the largest number of participants on a directed path in the observation graph. For arbitrary initial understandings, fixed credibility families, and thresholds, the communication profile reaches a fixed point within $2d$ simultaneous rounds.*

Proof. Assign depth one to participants without parents and depth one plus the greatest parent depth to every other participant. An acyclic finite graph admits these depths. A depth-one participant receives a fixed internally consistent broadcast, so Corollary 5.6 makes her state constant from round two onward. Inductively, suppose every participant of depth at most k is constant from round $2k$ onward. A participant of depth $k + 1$ then receives a fixed internally consistent broadcast from round $2k$ onward. Regardless of her state at that time, two further encounters make her state constant. All states are therefore constant from round $2d$ onward. The profile update is deterministic and time-independent for the fixed speech and observation rules, so that constant profile is a fixed point. $\square$

Lemma 5.10. *If the anchor poset contains incomparable elements, there is no strictly increasing, meet-preserving map from $\mathcal{U}_R$ into a real chain.*

Proof. Let u_1^*, u_2^* be incomparable evaluative anchors (an antichain of width ≥ 2). Since $u_1^* \parallel u_2^*$, neither is below the other, so $u_1^* \wedge u_2^* \prec_{\mathcal{U}} u_1^*$ and $u_1^* \wedge u_2^* \prec_{\mathcal{U}} u_2^*$. Meet preservation gives $\pi(u_1^* \wedge u_2^*) = \min(\pi(u_1^*), \pi(u_2^*))$. Without loss of generality $\pi(u_1^*) \leq \pi(u_2^*)$, so $\pi(u_1^* \wedge u_2^*) = \pi(u_1^*)$. But strict monotonicity requires $\pi(u_1^* \wedge u_2^*) < \pi(u_1^*)$ (since $u_1^* \wedge u_2^* \prec_{\mathcal{U}} u_1^*$), a contradiction. $\square$

Proposition 5.11. *Fix a finite nonempty voter set and a finite nonempty platform set. There is a finite semantic model with one visible representation and fixed, pairwise disjoint nonempty anchor meaning sets in which every profile of weak voter rankings on those platforms can be represented by changing the broadcast map. Priors have empty semantics, credibility is full, conflict is absent, and $\lambda_v = 1$.*

Proof. Let $m = |X|$. Give each voter v a distinct block of m meaning tokens $A_v = \{a_{v,1}, \dots, a_{v,m}\}$, and set $M = \bigcup_v A_v$. Use a field with one substantive resource and its invisibility marker, and let e be the stance distinguishing them. Fix $u_v^* = (e, A_v)$, $u_v^0 = (e, \emptyset)$, full credibility, empty conflict relation, and $\lambda_v = 1$ for every voter. Set $\mathcal{W}(e, \mu) = 1 + |\mu|$ and $\mathcal{W}(u_\perp) = 0$. These assignments define a strictly increasing valuation on the whole one-resource understanding lattice.

For any given weak ranking of the finite menu, choose integer scores $k_v(x) \in \{0, \dots, m-1\}$ that preserve all its strict comparisons and ties. Broadcast at x the meaning set

$$B_x = \bigcup_{v \in V} \{a_{v,j} : 1 \leq j \leq k_v(x)\}.$$

Then $|B_x \cap A_v| = k_v(x)$. By Proposition 4.10, $s_v(x) = 2k_v(x)$. This represents the specified profile, with priors, anchors, filters, conflict, and valuation unchanged as the profile varies. $\square$

Proposition 5.13. *Let $H = \bigvee_v u_v^*$ and define $Q_H(b) = b \wedge H$. Then $Q_H(b) \preceq_{\mathcal{U}} H$, Q_H is idempotent, and*

$$Q_H(b) \wedge u_v^* = b \wedge u_v^*, \quad D_v[Q_H(b)] = D_v[b]$$

for every voter and every broadcast. No distributivity or common-stance assumption is required.

Proof. Because $u_v^* \preceq_{\mathcal{U}} H$, associativity and commutativity of meet give

$$(b \wedge H) \wedge u_v^* = b \wedge (H \wedge u_v^*) = b \wedge u_v^*.$$

Applying $\mathcal{W}$ proves equality of benchmark scores. The lower-bound property gives $b \wedge H \preceq_{\mathcal{U}} H$, and $(b \wedge H) \wedge H = b \wedge H$ proves idempotence. $\square$

Proposition 5.14. *Under the common-stance assumptions above, define $Q_K(e, \mu) = (e, \mu \cap K)$ pointwise. Then Q_K is idempotent and preserves each voter's recognition, valued change, and encounter response for every candidate:*

$$s_v^c[Q_K(b)] = s_v^c[b].$$

Credibility filters, trigger thresholds, and nonnegative change weights may differ across voters; credibility may also differ across candidates.

Proof. Fix voter v , candidate c , and a visible class D at the common stance. Write $P = \mu_v^0(D)$, $A = \mu_v^*(D)$, $B = \mu_b(D)$, and $\alpha = \alpha_{vc}^e(D)$. The clipped message is $B' = B \cap K(D)$. Since $\alpha \cap A \subseteq K(D)$,

$$(B' \cap \alpha) \cap A = (B \cap \alpha) \cap A.$$

At the common stance, meets are semantic intersections. This equality preserves the received-message meet with the anchor at every class, and hence preserves recognition, including its unchanged empty-message reference point.

Every incoming meaning conflicting with a prior meaning in P belongs to $O(P) \subseteq K(D)$. Thus for each $p \in P$, some member of B conflicts with p if and only if some member of B' does. The set of conflicting prior meanings is unchanged, as is the set of conflicting classes and the branch selected by τ_v . On contraction, the entire baseline is therefore unchanged. On the forward branch,

$$(P \cup (B' \cap \alpha)) \cap A = (P \cap A) \cup (B' \cap \alpha \cap A) = (P \cup (B \cap \alpha)) \cap A.$$

The post-encounter meet with the anchor is unchanged on either branch. Consequently valued change and the full response agree for any permitted $\mathcal{W}$ and λ_v . Invisible semantics remain empty throughout. Finally, intersection with the fixed family K is idempotent. $\square$

Corollary 5.15. *Let Q be one of the idempotent reductions above, and suppose $Q(\phi(X)) \subseteq \phi(X)$. Use the direct-broadcast scores for Q_H , or the common-stance encounter scores for Q_K . Let $X_Q = \{x \in X : Q(\phi(x)) = \phi(x)\}$. There is a map $\rho : X \rightarrow X_Q$, fixing X_Q , such that:*

- (i) *replacing each candidate's platform x_c by $\rho(x_c)$ leaves every expected vote share unchanged;*
- (ii) *a profile in X_Q is a pure Nash equilibrium of the reduced game if and only if it is one of the full game, and the image under ρ of any full-game equilibrium is an equilibrium;*
- (iii) *in the source-neutral case, each strict majority-improvement step maps to a strict step, preserving reachability between endpoints in X_Q .*

For Q_H , the conclusions also apply to encounter scores whose rankings and ties agree with the direct benchmark on the feasible menu.

Proof. For $x \notin X_Q$, choose $\rho(x) \in X$ with $\phi(\rho(x)) = Q(\phi(x))$, possible by closure. Idempotence implies $\rho(x) \in X_Q$. For $x \in X_Q$, take $\rho(x) = x$. The relevant clipping proposition gives identical voter scores for x and $\rho(x)$. For the stated transfer from direct scores to encounter rankings, the two feasible platforms tie under every D_v , so ranking-and-tie agreement makes their encounter scores equal as well.

All candidate comparisons and maximizing sets are unchanged when platforms are replaced by their images. The uniform voting rule therefore gives the same expected vote shares. Every full-menu deviation also has a score-equivalent deviation in X_Q . Thus a reduced profile has a profitable full-menu deviation if and only if it has a profitable reduced-menu deviation, proving the first equilibrium assertion. The same score equalities show that the image of a full-game equilibrium has no profitable deviation.

Under source neutrality, a strict majority comparison between x and y holds between $\rho(x)$ and $\rho(y)$ with exactly the same supporters. In particular, adjacent platforms in a strict majority path cannot collapse to the same image. A projected path may revisit a platform; deleting any resulting loops preserves reachability. Endpoints already in X_Q are fixed. $\square$

Proposition 5.16. *Let $L = \bigwedge_v u_v^*$, $H = \bigvee_v u_v^*$, and $P(b) = (b \vee L) \wedge H$. Then $P(b) \in [L, H]$ and $D_v[P(b)] \geq D_v[b]$ for every voter. If $L \not\preceq_{\mathcal{W}} b$, all these inequalities are strict. In a source-neutral election with at least two candidates, direct-broadcast rankings, uniform voting ties, and a feasible platform broadcasting $P(\phi(x))$, such an x cannot support a symmetric equilibrium.*

Proof. Both $b \vee L$ and H lie above L , so $L \preceq_{\mathcal{W}} P(b) \preceq_{\mathcal{W}} H$. By Proposition 5.13,

$$P(b) \wedge u_v^* = (b \vee L) \wedge u_v^* \succeq_{\mathcal{W}} b \wedge u_v^*.$$

This gives the weak score inequality. If $L \not\preceq_{\mathcal{W}} b$, the two meets cannot be equal: the new meet lies above L , whereas the old meet lies below b . Their equality would imply $L \preceq_{\mathcal{W}} b$. Thus the meet increases strictly for every voter, and strict monotonicity of $\mathcal{W}$ yields a strict score gain for each.

At a source-neutral symmetric profile, each of $|C| \geq 2$ candidates receives expected share $1/|C|$ under uniform voting ties. If a feasible completion strictly improves every voter's ranking, a deviating candidate receives all votes. The deviation is profitable, so the original symmetric profile is not an equilibrium. $\square$

Proposition 5.17. *In a race with at least two candidates and uniform voting ties, suppose replacing x by y weakly raises c 's score for every voter while rivals' scores stay fixed. Then c 's expected vote share weakly increases. The increase is strict if and only if some voter satisfies*

$$\sigma_v^c(x) \leq m_v \leq \sigma_v^c(y) \quad \text{and} \quad \sigma_v^c(x) < \sigma_v^c(y).$$

For arbitrary x, y in a source-neutral two-candidate race, put $N_H(y, x) = |\{v : \sigma_v(y) > \sigma_v(x)\}|$. A deviation from the symmetric profile (x, x) to y changes expected vote share by

$$\frac{N_H(y, x) - N_H(x, y)}{2|V|}.$$

Proof. For fixed rivals, each g_v is nondecreasing and rises only on reaching or passing m_v . For two ordered scores it rises strictly exactly under the pivotality condition in the statement. Expected vote share is the average of these probabilities. All changes are nonnegative, so their sum is positive exactly when one is positive.

In the source-neutral two-candidate game, the initial symmetric profile gives each candidate $1/2$ of each vote. After deviating to y , the candidate gains $1/2$ for each voter strictly preferring y , loses $1/2$ for each voter strictly preferring x , and retains $1/2$ from every indifferent voter. Dividing the net change by $|V|$ gives the formula. $\square$

Lemma 5.18. *Suppose priors, anchors, and broadcasts share a stance, and b , d , and $b \cup d$ are feasible broadcasts. For any voter whose encounter with $b \cup d$ takes the forward branch, the encounters with b and d also take that branch, and*

$$s_v^c[b \cup d] \geq s_v^c[b], \quad s_v^c[b \cup d] \geq s_v^c[d].$$

If this holds for every voter, replacing either component speech by the combination weakly increases the candidate's expected vote share at $H = 1$.

Proof. At the common stance, raw and filtered reception are respectively the broadcast semantics and its intersection with the credibility filter. Raw conflict with a fixed prior is monotone in the message. If the union's conflict count is at most the threshold, each component's count is also at most the threshold. All three encounters therefore take the forward branch and retain the common stance. The combination has weakly richer filtered receipt and post-encounter semantics than either component. Meeting with the fixed anchor and applying the increasing valuation preserves these inequalities. Recognition subtracts the same fiber-bottom value, and valued change subtracts the same prior value. Their sum with nonnegative change weight is consequently weakly larger. Apply Proposition 5.17 at $H = 1$ for the vote-share conclusion. $\square$

Proposition 5.20. *Suppose all stances are common, filters are pairwise disjoint and source-neutral, and the feasible menu has the product form just specified. For every visible class C , assume that*

- (i) $\mu_v^0(C) \subseteq \alpha_i(C)$ for each $v \in T_i$;
- (ii) no meaning in $\alpha_i(C)$ conflicts with a meaning in $\alpha_j(C)$ when $i \neq j$.

Repeat each candidate's fixed broadcast for $H \geq 1$ encounters and use election-time evaluation. Then a type- i voter's trajectory and score depend only on z_i . For fixed rival strategies, candidate c 's payoff separates as

$$\mathcal{V}_c(\mathbf{z}; \mathbf{x}_{-c}) = \sum_i \mathcal{V}_{c,i}(z_i; \mathbf{x}_{-c}),$$

where $\mathcal{V}_{c,i}$ is the contribution of type i to total expected vote share. Consequently the set of best responses is

$$\prod_i \arg \max_{z_i \in X_i} \mathcal{V}_{c,i}(z_i; \mathbf{x}_{-c}).$$

Thresholds may be active, and repetition need not converge before H .

Proof. Fix $v \in T_i$. Her initial meanings lie within α_i at every visible class. A contraction removes meanings, and a forward encounter adds only z_i , so this containment remains true at every round. The absence of cross-type conflict means that raw content z_j with $j \neq i$ cannot conflict with her current meanings. Both the trigger count and the meanings removed by contraction therefore depend only on z_i . Filtered reception is exactly z_i by disjointness. Induction on the number of encounters shows that the entire trajectory depends only on z_i . Initial recognition and the understanding at H then give a score with the same dependence. No convergence assumption is used.

Hold all rivals' strategies fixed. The contribution from each type is a function of the corresponding component alone, and contributions sum to total expected vote share. Product feasibility permits any selection of component maximizers. Such a selection maximizes the sum; conversely, a nonmaximizing component could be replaced while holding every other component fixed, strictly raising the sum. This proves the best-response factorization. $\square$

Corollary 5.21. *Under Proposition 5.20, suppose each X_i contains a component z_i^* maximizing the score of every voter in T_i . In a race with at least two candidates, all candidates choosing $\mathbf{z}^* = (z_1^*, \dots, z_k^*)$ is a pure equilibrium. If at least two of the z_i^* have nonempty semantics, this equilibrium broadcast is polysemic.*

Proof. Component separation and the common maximizing components imply that $\mathbf{z}^*$ maximizes every voter's score on the full feasible menu. Source neutrality makes every candidate attain that same score at the symmetric profile. No deviation can increase any voter's score for the deviating candidate, so it cannot increase expected vote share by Proposition 5.17. For each type, the filtered receipt of the chosen broadcast is z_i^* . Two nonempty components in disjoint domains are different, which gives polysemy as defined. $\square$

Proposition 5.23. *Every symmetric equilibrium belongs to X_A . For $x \in X_A$, the profile (x, x) is an equilibrium of the full game if and only if*

$$N_B(y, x) \leq N_B(x, y) \quad \text{for every } y \in X_A.$$

When both candidates choose within X_A , the majority splits equally, and all differences in their expected vote shares come from B .

Proof. If $x \notin X_A$, every voter in A strictly prefers every $y \in X_A$ to x . A deviation from (x, x) to such a y gains all votes of the strict majority, raising the candidate's share above $1/2$. Thus every symmetric equilibrium lies in X_A .

Fix $x \in X_A$. Any deviation to $y \notin X_A$ loses every voter in A and can obtain at most $|B|/|V| < 1/2$. For $y \in X_A$, every voter in A is indifferent and continues to split her vote equally. By Proposition 5.17, the deviation's net gain is $[N_B(y, x) - N_B(x, y)]/(2|V|)$. The stated inequalities are therefore necessary and sufficient for equilibrium in the full game. The same indifference gives the equal majority contribution whenever both candidates choose inside X_A . $\square$

Corollary 5.24. *Suppose a strict majority has anchor u_A^* , all remaining voters have anchor u_B^* , the race is source-neutral with two candidates and uniform voting ties, and rankings and ties agree with $D_v(x) = \mathcal{W}(\phi(x) \wedge u_v^*)$. If some feasible broadcast contains u_A^* , every symmetric equilibrium broadcast b contains u_A^* . Its minority value deficit satisfies*

$$0 \leq \mathcal{W}(u_B^*) - \mathcal{W}(b \wedge u_B^*) \leq \mathcal{W}(u_B^*) - \mathcal{W}(u_A^* \wedge u_B^*).$$

The deficit is positive exactly when b does not contain u_B^ . If a feasible broadcast contains $u_A^* \vee u_B^*$, it gives every voter maximal direct-broadcast value and supports a symmetric equilibrium with zero deficit.*

Proof. A feasible broadcast containing u_A^* attains its maximal direct score $\mathcal{W}(u_A^*)$. Strict increase of $\mathcal{W}$ implies that a broadcast attains this value exactly when it meet with u_A^* equals u_A^* , equivalently when it contains u_A^* . The stipulated agreement of rankings and ties therefore gives a common maximizing set to all majority voters. Apply Proposition 5.23.

For an equilibrium broadcast $b \succeq u_A^*$, monotonicity of meet gives $u_A^* \wedge u_B^* \preceq b \wedge u_B^* \preceq u_B^*$. Applying $\mathcal{W}$ and subtracting from $\mathcal{W}(u_B^*)$ gives the bounds. Strict increase gives a positive deficit exactly when $b \wedge u_B^* \neq u_B^*$, equivalently when b does not contain u_B^* . A feasible broadcast containing the join attains maximal direct value for both types, so no candidate can profitably deviate from its symmetric profile. Its minority deficit is zero. $\square$

A.5 Proofs for Section 7.

Proposition 7.4. *The restriction ρ_S is monotone. The extension ι_S is an order embedding with image*

$$\iota_S(\mathcal{U}_S) = \{(e, \mu) \in \mathcal{U}_R : \mathcal{R}_e \subseteq S\} = \downarrow t_S.$$

It preserves binary meets and joins. Moreover,

$$\rho_S \circ \iota_S = \text{id}_{\mathcal{U}_S}, \quad \mathbf{V}_S \circ \mathbf{V}_T = \mathbf{V}_{S \cap T}, \quad \mathcal{R}_{\text{ev}_S(u)} = S \cap \mathcal{R}_{e_u}.$$

Consequently each $\mathbf{V}_S$ is monotone and idempotent, fixes exactly the understandings supported on S , and preserves comparability when applied to both members of an ordered pair.

Proof. If $u \preceq_{\mathcal{U}} v$, the partition of v refines that of u . Intersecting both partitions with $S \cup \{\star\}$ preserves this refinement. Each surviving fine visible class inherits the same semantics as before restriction, including every meaning of its coarse ancestor. Thus $\rho_S(u) \preceq_{\mathcal{U}_S} \rho_S(v)$.

For two understandings on S , adjoining $\mathcal{R} \setminus S$ to their invisible classes preserves and reflects refinement: all added resources lie with the same marker in both partitions. It also preserves and reflects the required semantic inclusions on visible classes, which are unchanged. Hence ι_S is an order embedding. Its image consists exactly of understandings whose visible scope lies in S : restricting any such understanding and extending it again recovers it. The fully distinguishing, full-meaning state t_S is the greatest state in this image. Conversely, an understanding below t_S must put every resource outside S in its invisible class, since refinement preserves visible-scope inclusion. This proves the image is $\downarrow t_S$. The ambient binary meet and join of two elements below t_S are themselves below t_S . The order isomorphism onto this ideal therefore preserves these operations.

Restriction after extension recovers every original class and meaning on S , so $\rho_S \iota_S = \text{id}$. The view $\mathbf{V}_T(u)$ has visible classes $C \cap T$ for the original visible classes C with nonempty intersection, inheriting $\mu(C)$. Taking its view on S leaves exactly $C \cap T \cap S$ with those same meanings, while all other resources belong to the invisible class. This is $\mathbf{V}_{S \cap T}(u)$. It also proves the stated scope identity. Monotonicity follows by composition, and taking $T = S$ gives idempotence. A state supported on S is fixed by restriction and extension; the scope identity implies every fixed state is supported there. Finally, monotonicity sends any ordered pair to an ordered pair under the same view. $\square$

Theorem 7.5. *Let $u_i = (e_i, \mu_i)$ with $u_1 \preceq_{\mathcal{U}} u_2$, let $w_i = \mathbf{V}_{S_i}(u_i)$, and put $A_i = S_i \cap \mathcal{R}_{e_i}$. Then w_1 and w_2 are incomparable if either of the following holds:*

(i) $A_1 \setminus A_2 \neq \emptyset$ and $A_2 \setminus A_1 \neq \emptyset$;

(ii) $S_1 = \mathcal{R}$, $\mathcal{R}_{e_1} \setminus S_2 \neq \emptyset$, and

$$\rho_{S_2}(u_1) \prec_{\mathcal{U}_{S_2}} \rho_{S_2}(u_2).$$

Proof. By Proposition 7.4, the visible scopes of w_1, w_2 are A_1, A_2 . An understanding below another must have its visible scope contained in the other's scope. Under condition (i), each scope contains a resource absent from the other, ruling out both orders.

Under condition (ii), $w_1 = u_1$. Choose $b \in \mathcal{R}_{e_1} \setminus S_2$. It is visible in w_1 and invisible in w_2 , so $w_1 \not\preceq_{\mathcal{U}} w_2$. If $w_2 \preceq_{\mathcal{U}} w_1$, monotonicity of restriction and its left-inverse identity would imply

$$\rho_{S_2}(u_2) = \rho_{S_2}(\iota_{S_2}(\rho_{S_2}(u_2))) = \rho_{S_2}(w_2) \preceq_{\mathcal{U}_{S_2}} \rho_{S_2}(u_1).$$

This contradicts the stipulated strict inequality in the other direction. Hence neither order holds. $\square$